Feynman Equation Results of Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests

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A The Feynman Equations

We consider learning the following 5 Feynman equations [Udrescu and Tegmark, 2020], with details in Table 1. The data corresponding to each Feynman equation is acquired from the Penn Machine Learning Benchmark (`PMLB`; `https://github.com/EpistasisLab/pmlb`) [Olson et al., 2017], containing 10^5 observations.

$$\text{I\_12\_2}: \quad F = \frac{q_1 q_2}{4\pi\epsilon r^2}, \qquad p = 4 \qquad \text{(CL)}$$

$$\text{I\_12\_11}: \quad F = q\left(E_f + vB\sin\theta\right), \qquad p = 5 \qquad \text{(FCE)}$$

$$\text{I\_24\_6}: \quad E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2, \qquad p = 4 \qquad \text{(EHFO)}$$

$$\text{I\_50\_26}: \quad x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)], \qquad p = 4 \qquad \text{(HOQN)}$$

$$\text{II\_36\_38}: \quad f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T}M, \qquad p = 8 \qquad \text{(FMMM)}$$

where, I_12_2 corresponds to Coulomb's law (CL); I_12_11 represents the force on a charged particle in an electromagnetic field (FCE); I_24_6 describes the energy of a harmonic oscillator (EHFO); I_50_26 represents harmonic oscillation with a quadratic nonlinear correction (HOQN); and II_36_38 describes the frictional force on a moving magnetic moment (FMMM).

Table 1: Details on Feynman equations.

Equation	**Output variable**	**Input variables**	**Scientific interpretation**
I_12_2 (CL)	F: electrostatic force	electric charges (q_1, q_2); permittivity (ϵ); distance between charges (r)	Coulomb's law describing the magnitude of the electrostatic force between two point charges. The force is proportional to the product of the charges and inversely proportional to the squared separation distance, with the permittivity controlling the strength of interaction in the medium.
I_12_11 (FCE)	F: Lorentz force	particle charge (q); electric field (E_f); magnetic field (B); particle velocity (v); angle between velocity and magnetic field (θ)	Force acting on a charged particle moving through combined electric and magnetic fields. The term qE_f corresponds to the electric contribution, while $qvB\sin\theta$ captures the magnetic force perpendicular to the velocity–field plane.
I_24_6 (EHFO)	E_n: harmonic oscillator energy	mass (m); angular frequency (ω); natural angular frequency (ω_0); displacement (x)	Energy associated with a harmonic oscillator-like system. The energy scales with mass, squared displacement, and the combined squared frequencies, representing stored mechanical energy in oscillatory motion.
I_50_26 (HOQN)	x: position or displacement	amplitude/initial displacement scale (x_1); angular frequency (ω); time (t); nonlinear correction parameter (α)	Oscillatory displacement with a nonlinear correction term. The first term represents standard cosine oscillation, while $\alpha\cos^2(\omega t)$ introduces a second-order harmonic correction to the motion.

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Table 1: Details on Feynman equations *(continued)*.

Equation	**Output variable**	**Input variables**	**Scientific interpretation**
II_36_38 (FMMM)	f: magnetic response	magnetic moment (mom); magnetic field (H); Boltzmann constant (k_b); temperature (T); polarizability parameter (α); permittivity (ϵ); speed of light (c); mass-like parameter (M)	Magnetic response governed by the balance between field–moment interaction and thermal energy. The first term captures the normalized contribution of the applied magnetic field, while the second term adds a correction involving polarizability, electromagnetic constants, and the material parameter M.

Computational environment. All Feynman equation experiments were implemented in `Python` and executed on a MacBook Air equipped with an Apple M2 processor and 8GB of unified memory.

B Experimental Settings of Competitors

B.1 DSR

We use the `dso.DeepSymbolicRegressor` implementation (https://github.com/dso-org/deep-symbolic-optimization) of Deep Symbolic Regression (DSR) [Petersen et al., 2021] for learning the 5 Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). For each equation, we subsampled $n = 2000$ observations out of 10^5 observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation σ is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and R^2 computed on a 90/10 train/test split. RMSE and R^2 denotes the root mean square error and coefficient of determination, i.e., $\texttt{RMSE} = \sqrt{n^{-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \texttt{RMSE}^2/[n^{-1}\sum_{i=1}^{n}(y_i - \overline{y})^2]$ ($\overline{y} = n^{-1}\sum_{i=1}^{n} y_i$), respectively.

DSR parameterizes a policy over symbolic expressions using an autoregressive recurrent neural network (RNN) which sequentially samples symbolic programs from a prescribed function set. Across all 5 Feynman equations, the operator set is configured as follows.

DSR operator set

```
function_set = [
    "add",      # (x, y) -> x + y
    "mul",      # (x, y) -> x*y
    "neg",      # x -> -x
    "inv",      # x -> 1/x
    "sin",      # x -> sin(x)
    "cos",      # x -> cos(x)
    "exp",      # x -> exp(x)
    "n2",       # x -> x^2
    "n3",       # x -> x^3
]
```

The regression task is run with protected operators enabled. For all equations, the the following DSR configuration is maintained.

DSR settings

```
dsr_settings = {
    "n_samples": 200000,  # total symbolic program sampling budget
    "batch_size": 500,    # samples per RNN policy-gradient update
    "max_length": 20,     # maximum length of each sampled expression
}
```

These settings correspond to approximately 400 RNN policy-gradient updates, unless early stopping occurs. During training, DSR retains the best-reward symbolic program encountered over the search. The final recovered expression is extracted from the fitted estimator through

Extracting the final DSR expression

```
estimator.program_
```

which stores the best symbolic program returned by `dso.DeepSymbolicRegressor`.

B.2 QLattice

We use the `feyn.QLattice` implementation (https://docs.abzu.ai/) of `QLattice` [Broløs et al., 2021] for learning the 5 Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). For each equation, we subsampled $n = 2000$ observations out of 10^5 observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation σ is injected into the target. For each equation and noise level, 5 independent repetitions were performed, with train/test `RMSE` and R^2 computed on a 90/10 train/test split. `RMSE` and R^2 denotes the root mean square error and coefficient of determination, i.e., $\texttt{RMSE} = \sqrt{n^{-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \texttt{RMSE}^2/[n^{-1}\sum_{i=1}^{n}(y_i - \overline{y})^2]$ ($\overline{y} = n^{-1}\sum_{i=1}^{n} y_i$), respectively.

`QLattice` constructs a lattice-based search space over symbolic interactions among input variables and primitive functions, and iteratively generates and scores candidate symbolic models. Candidate formulas are generated through the `auto_run` routine, ranked according to a specified model selection criterion, and the top-ranked model returned by `QLattice` is used as the learned expression. The operator set across all 5 Feynman equations is configured using the primitive functions available in the `feyn` implementation as follows.

QLattice operator set

```
function_names = [
    "inverse",  # z -> 1/z
    "linear",   # z -> a*z + b
    "squared",  # z -> z^2
    "add",      # (z1, z2) -> z1 + z2
    "multiply", # (z1, z2) -> z1 * z2
    "exp",      # z -> exp(z)
    "sin",      # z -> sin(z)
    "cos",      # z-> cos(z)
]
```

For all equations, the `QLattice` search is run with the following configuration.

QLattice settings

```
qlattice_settings = {
    "n_epochs": 25,          # number of search epochs
    "max_complexity": 20,    # complexity of symbolic models
    "criterion": "bic",      # Bayesian information criterion
}
```

The final recovered expression is extracted from the top-ranked `BIC` model using `Python`'s `SymPy` package [Meurer et al., 2017] as

Extracting the final `QLattice` expression

```
# signif -> significant digits in fitted numerical coefficients

best_model = models[0] # top-ranked BIC model
expression = best_model.sympify(signif=6)
```

which converts the selected `QLattice` model into a symbolic expression.

B.3 `SISSO` + +

For sure independence screening and sparsifying operator (`SISSO`) [Ouyang et al., 2018], we use the `SISSORegressor` implementation (https://sissopp_developers.gitlab.io/sissopp/) from `SISSO` + + [Purcell et al., 2022] for learning the 5 Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). For each equation, we subsampled $n = 2000$ observations out of 10^5 observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation σ is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test `RMSE` and R^2 computed on a 90/10 train/test split. `RMSE` and R^2 denotes the root mean square error and coefficient of determination, i.e., $\texttt{RMSE} = \sqrt{n^{-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \texttt{RMSE}^2/[n^{-1}\sum_{i=1}^{n}(y_i - \overline{y})^2]$ ($\overline{y} = n^{-1}\sum_{i=1}^{n} y_i$), respectively.

`SISSO` + + constructs a large symbolic descriptor space (complete enumeration) by recursively applying a prescribed set of mathematical operators to the primary input variables $x_1, \ldots, x_p$. It then performs sure independence screening, `SIS`, followed by sparse descriptor selection through the sparsifying operator, `SO`.

For all equations, the `SISSO`+ + regression task is run with the following common configuration.

`SISSO`++ settings

```
sisso_settings = {
    "n_dim": 3,            # number of selected descriptors
    "max_rung": 3,         # maximum recursive feature-construction depth
    "n_sis_select": 50,    # candidate descriptors retained by SIS
    "n_residual": 20,      # number of residual-based selections during SO
}
```

For `SISSO` + +, we use problem-specific operator sets rather than the common collection of operators across all equations. This is because of the combinatorial growth of the `SISSO` + + descriptor space: as `max_rung` increases, recursively applying a large operator library to the primary variables can generate an extremely large number of candidate descriptors, making the `SIS` and `SO` steps computationally expensive. Therefore, for each Feynman equation, we restrict the operator set to the algebraic and functional components required to represent the corresponding ground-truth expression. This provides a computationally feasible and structurally informed benchmark for `SISSO`+ +, while still allowing the method to search over a rich nonlinear descriptor space generated from the specified operators.

SISSO++ **operator set**

```
# add: +, mult: *, div: /, sq: ^2, sin: sin(), cos: cos(),
sisso_operator_sets = {
    "I_12_2":   ["add", "div", "sq"],           # CL
    "I_12_11":  ["add", "mult", "sin"],         # FCE
    "I_24_6":   ["add", "mult", "sq"],          # EHFO
    "I_50_26":  ["add", "mult", "cos", "sq"],  # HOQN
    "II_36_38": ["add", "mult", "div", "sq"],  # FMMM
}
```

The final recovered expression is taken as the sparse linear combination of selected symbolic descriptors returned by `SISSORegressor`.

B.4 gplearn

We use the `gplearn.genetic.SymbolicRegressor` (https://gplearn.readthedocs.io/) implementation of `gplearn` [Stephens, 2016] for learning the 5 Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). For each equation, we subsampled $n = 2000$ observations out of 10^5 observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation σ is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and R^2 computed on a 90/10 train/test split. RMSE and R^2 denotes the root mean square error and coefficient of determination, i.e., $\texttt{RMSE} = \sqrt{n^{-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \texttt{RMSE}^2/[n^{-1}\sum_{i=1}^{n}(y_i - \overline{y})^2]$ ($\overline{y} = n^{-1}\sum_{i=1}^{n} y_i$), respectively.

`gplearn` evolves symbolic expressions using genetic programming [Koza, 1992]. Starting from an initial population of randomly generated symbolic programs, it repeatedly applies evolutionary operations such as selection, crossover, mutation, and reproduction to improve predictive fit. In our experiments, symbolic programs are scored using RMSE, and a parsimony penalty is included to discourage overly complex expressions.

Across all 5 Feynman equations, the operator set is configured as follows.

gplearn **operator set**

```
function_set = [
    "add",      # (x, y) -> x + y
    "mul",      # (x, y) -> x*y
    "neg",      # x -> -x
    "inv",      # x -> 1/x
    "sin",      # x -> sin(x)
    "cos",      # x -> cos(x)
    "exp",      # x -> exp(x)
    "sq",       # x -> x^2
    "cu",       # x -> x^3
]
```

The unary functions neg, inv, sin, cos, exp, ^{2}, and ^{3} are supplied through custom wrapper `gplearn.functions.make_function`. The inverse operator is protected near zero, and the exponential operator is clipped for numerical stability.

For all equations, the following `gplearn` configurations are maintained.

gplearn settings

```
# metric: fitness metric to select symbolic expression;
# generations: number of evoluationary generations;
# population size: number of symbolic programs maintained in each generation;
# tournament_size: tournament selection pressure used to choose parent programs;
# parsimony_coefficient: expression complexity during fitness evaluation;
# stopping_criteria: early stopping threshold for RMSE;
# p_crossover: subtree crossover probability;
# p_subtree_mutation: subtree mutation probability;
# p_hoist_mutation: hoist mutation probability;
# p_point_mutation: point mutation probability;
# max_samples: fraction of training samples per program.

gplearn_settings = {
    "metric": "rmse",
    "generations": 20,
    "population_size": 2000,
    "tournament_size": 20,
    "parsimony_coefficient": 1e-4,

    # gplearn default settings implemented below
    "stopping_criteria": 0.00,
    "p_crossover": 0.9,
    "p_subtree_mutation": 0.01,
    "p_hoist_mutation": 0.01,
    "p_point_mutation": 0.01,
    "max_samples": 1.0,
}
```

The final recovered expression is extracted from the fitted estimator through

Extracting the final gplearn expression

```
model._program
```

which stores the best symbolic program retained by `gplearn`.

B.5 operon

We use the `pyoperon.sklearn.SymbolicRegressor` implementation (https://github.com/heal-research/operon) of `operon` [Burlacu et al., 2020] for learning the 5 Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). For each equation, we subsampled $n = 2000$ observations out of 10^5 observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation σ is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and R^2 computed on a 90/10 train/test split. RMSE and R^2 denotes the root mean square error and coefficient of determination, i.e., $\texttt{RMSE} = \sqrt{n^{-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \texttt{RMSE}^2/[n^{-1}\sum_{i=1}^{n}(y_i - \overline{y})^2]$ ($\overline{y} = n^{-1}\sum_{i=1}^{n} y_i$), respectively.

operon is a genetic programming-based SR method that evolves expression trees using evolutionary search. Starting from a population of candidate symbolic expressions, operon iteratively applies variation and selection operators to improve predictive fit while searching over a prescribed set of primitive symbols. In our implementation, the symbolic search is performed through the SymbolicRegressor interface from the Python package pyoperon.

Across all 5 Feynman equations, the primitive symbol set is configured as follows.

operon operator set

```
allowed_symbols = [
    "add",       # (x, y) -> x + y
    "mul",       # (x, y) -> x*y
    "sub",       # (x, y) -> x - y
    "div",       # (x, y) -> x/y
    "sin",       # x -> sin(x)
    "cos",       # x -> cos(x)
    "exp",       # x -> exp(x)
    "square",    # x -> x^2
    "variable",  # input variable leaf nodes
    "constant",  # numerical constant leaf nodes
]
```

For all equations, operon is implemented with the following configuration.

operon settings

```
# generations: number of evolutionary generations;
# population_size: number of candidate expressions maintained in the population;
# max_length: maximum length of the evolved expressions

operon_settings = {
    "generations": 2000,
    "population_size": 1000,
    "max_length": 20,
}
```

The operon estimator is initialized as follows.

operon module initialization

```
model = SymbolicRegressor(
    allowed_symbols=allowed_symbols,
    generations=2000,
    population_size=1000,
    max_length=20,
)
```

The final recovered expression is extracted from the fitted estimator using the available `operon` model-string interface viz.,

Extracting the final `operon` expression

```
model.get_model_string(model.model_)
```

whenever the fitted model is stored in `model_`; otherwise, the corresponding best available expression attribute returned by the fitted `SymbolicRegressor` is used.

B.6 `PySR`

We use the `pysr.PySRRegressor` implementation (`https://github.com/MilesCranmer/PySR`) of PySR [Cranmer, 2023] for learning the 5 Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). For each equation, we subsampled $n = 2000$ observations out of 10^5 observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation σ is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test `RMSE` and R^2 computed on a 90/10 train/test split. `RMSE` and R^2 denotes the root mean square error and coefficient of determination, i.e., $\texttt{RMSE} = \sqrt{n^{-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \texttt{RMSE}^2/[n^{-1}\sum_{i=1}^{n}(y_i - \overline{y})^2]$ ($\overline{y} = n^{-1}\sum_{i=1}^{n} y_i$), respectively.

`PySR` performs evolutionary `SR` by searching over expression trees composed of user-specified binary and unary operators. The search maintains multiple evolving populations of candidate expressions and optimizes them using mutation, crossover, selection, and constant optimization. In our experiments, the symbolic search is performed through the `PySRRegressor` interface, with squared-error loss and deterministic serial execution.

Across all 5 Feynman equations, the operator set is configured as follows.

`PySR` operator set

```
binary_operators = [
    "+",        # (x, y) -> x + y
    "*",        # (x, y) -> x*y
]

unary_operators = [
    "neg(x) = -x",           # x -> -x
    "inv(x) = 1 / x",        # x -> 1/x
    "sin",                   # x -> sin(x)
    "cos",                   # x -> cos(x)
    "exp",                   # x -> exp(x)
    "sq(x) = x * x",         # x -> x^2
    "cu(x) = x * x * x",     # x -> x^3
]
```

The custom unary operators neg, inv, square, and cube are additionally mapped to their corresponding `SymPy` [Meurer et al., 2017] expressions to allow symbolic conversion of the final recovered model.

For all equations, PySR is run with the following configuration.

PySR settings

```
# niterations: number of PySR evolutionary iterations;
# populations: number of populations evolved during the search;
# population_size: number of candidate expressions in each population;
# maxsize: maximum expression size.

pysr_settings = {
    "niterations": 200,
    "populations": 10,
    "population_size": 50,
    "maxsize": 20,
    "elementwise_loss": "loss(x, y) = (x - y)^2", # squared-error loss
    "model_selection": "best", # for symbolic expression selection
    "deterministic": True, # to ensure reproducibility
    "parallelism": "serial", # avoid variation owing to parallel execution
}
```

The custom SymPy mappings used for symbolic conversion are as follows.

PySR SymPy mappings for symbolic conversion

```
extra_sympy_mappings = {
    "neg": lambda x: -x,
    "inv": lambda x: 1 / x,
    "sq":  lambda x: x**2,
    "cu":  lambda x: x**3,
}
```

The final recovered expression is extracted from the best PySR model using

Extracting the best PySR model

```
model.get_best()["equation"]
```

whenever available; otherwise, the symbolic expression returned by

Extracting the final SymPy-supported PySR expression

```
model.sympy()
```

is used.

B.7 BMS

We use the BMSRegressor implementation from the AutoRA mdoule (https://autoresearch.github.io/autora/user-guide/theorists/bms/) [Musslick et al., 2024] of Bayesian machine scientist (BMS) [Guimerà et al., 2020] for learning the 5 Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). For each equation, we subsampled $n = 2000$ observations out of 10^5 observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation σ is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and R^2 computed on a 90/10 train/test split. RMSE and R^2 denotes the root mean square error and coefficient of determination, i.e., $\texttt{RMSE} = \sqrt{n^{-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \texttt{RMSE}^2/[n^{-1}\sum_{i=1}^{n}(y_i - \overline{y})^2]$ $(\overline{y} = n^{-1}\sum_{i=1}^{n} y_i)$, respectively.

Across all 5 Feynman equations, the BMS operator set is configured using the primitive operators supported by the BMSRegressor interface. The common operator configuration is as follows.

BMS operator set

```
bms_operator_set = [
    "add",   # (x, y) -> x + y
    "mul",   # (x, y) -> x*y
    "neg",   # x -> -x
    "div",   # (x, y) -> x/y
    "sin",   # x -> sin(x)
    "cos",   # x -> cos(x)
    "exp",   # x -> exp(x)
    "sq",    # x -> x^2
    "cu",    # x -> x^3
]
```

The aforementioned operators are passed through the ops dictionary and a uniform prior weight is assigned to each included operator.

BMS operator dictionary and weights

```
bms_ops = {
    "+": 2,
    "*": 2,
    "/": 2,
    "-": 1,
    "sin": 1,
    "cos": 1,
    "exp": 1,
    "pow2": 1,
    "pow3": 1,
}

bms_prior = {
    "Nopi_+": 1.0,
    "Nopi_*": 1.0,
    "Nopi_/": 1.0,
    "Nopi_-": 1.0,
    "Nopi_sin": 1.0,
    "Nopi_cos": 1.0,
    "Nopi_exp": 1.0,
    "Nopi_pow2": 1.0,
```

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```
    "Nopi_pow3": 1.0,
}
```

For all equations except II_36_38 (FMMM), each BMS chain is run for epochs $= 2000$ MCMC steps. For II_36_38 (FMMM), which is the most structurally complex Feynman equation considered here, we use epochs $= 20000$. BMS is run using 5 independent parallel chains.

BMS settings

```
bms_settings = {
    "n_parallel_chains": 5,
    "epochs": 2000, # 20000 for II_36_38 (FMMM)
    "parallel_chains": True,
}
```

After all chains are completed, the chain with the lowest in-sample RMSE (computed on a 90% train split) is selected, and its symbolic expression is used as the final recovered expression.

Extracting the final BMS expression across parallel chains

```
best_chain = min(
    successful_chains,
    key=lambda chain: chain["chain_train_rmse"],
)

final_expression = best_chain["expression"]
```

B.8 BSR

We use the BSRRegressor implementation from the AutoRA module (https://autoresearch.github.io/autora/user-guide/theorists/bsr/) of Bayesian symbolic regression (BSR) [Jin et al., 2020] for learning the 5 Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). For each equation, we subsampled $n = 2000$ observations out of 10^5 observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation σ is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and R^2 computed on a 90/10 train/test split. RMSE and R^2 denotes the root mean square error and coefficient of determination, i.e., $\texttt{RMSE} = \sqrt{n^{-1}\sum_{i=1}^n (y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \texttt{RMSE}^2/[n^{-1}\sum_{i=1}^n (y_i - \overline{y})^2]$ ($\overline{y} = n^{-1}\sum_{i=1}^n y_i$), respectively.

Across all 5 Feynman equations, the BSR operator set is configured as follows.

BSR operator set

```
bsr_operator_set = [
    "add",    # (x, y) -> x + y
    "mul",    # (x, y) -> x*y
```

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```
    "neg",    # x -> -x
    "inv",    # x -> 1/x
    "sin",    # x -> sin(x)
    "cos",    # x -> cos(x)
    "exp",    # x -> exp(x)
    "sq",     # x -> x^2
    "cu",     # x -> x^3
]
```

The aforementioned operator set is converted to the corresponding BSR prior dictionary.

BSR operator prior dictionary

```
bsr_operator_prior = {
    "+": 1.0,
    "*": 1.0,
    "neg": 1.0,
    "inv": 1.0,
    "sin": 1.0,
    "cos": 1.0,
    "exp": 1.0,
    "pow2": 1.0,
    "pow3": 1.0,
}
```

For all equations except II_36_38 (FMMM), we set `itr_num` = 2000, while for II_36_38 we set `itr_num` = 20000. We use 5 independent restarts for each BSR fit. The main BSR configuration is given below. Note that, the split probability parameters are maintained at their default specifications.

BSR settings

```
# split probability parameters maintained at default settings

bsr_settings = {
    "n_restarts": 5,
    "itr_num": 2000, # 20000 for II_36_38 (FMMM)
    "prior_name": "restricted_ops",
}
```

The number of symbolic trees K is chosen to match the corresponding HierBOSSS configuration described in Section D.1. The equation-specific tree configurations are given below.

BSR K configuration

```
bsr_tree_num = {
    "I_12_2": 4,
    "I_12_11": 3,
    "I_24_6": 4,
```

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```
    "I_50_26": 4,
    "II_36_38": 4,
}
```

Thus, for a given equation, the BSR estimator is initialized as follow.

BSR module initialization

```
model = BSRRegressor(
    tree_num=bsr_tree_num[equation_name],
    itr_num=2000, # 20000 for II_36_38 (FMMM)
)
```

After all restarts are completed, the final recovered expression is taken from the successful restart with the lowest in-sample RMSE (computed on a 90% train set).

Extracting the final BSR expression across restarts

```
best_restart = min(
    successful_restarts,
    key=lambda restart: restart["chain_train_rmse"],
)

final_expression = best_restart["expression"]
```

The final expression is extracted as the additive symbolic model returned by `BSRRegressor`, together with the fitted coefficients of the selected symbolic trees.

C Results of Competitors for Learning Feynman Equations

C.1 Results for I_12_2 (CL)

Table 2: DSR results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$q_2 \{\{r(r+r)(\epsilon+\epsilon)\}^{-1}\}$	0.038	0.044	0.808	0.760	62.7
2	$\left\{q_1^{-1} + (\epsilon r + r)r\right\}^{-1}$	0.049	0.054	0.657	0.679	93.0
3	$q_1 \{r + r\epsilon\}^{-2}$	0.040	0.035	0.806	0.794	83.4
4	$q_2 \{\epsilon r + r\}^{-2}$	0.043	0.040	0.791	0.725	57.7
5	$q_1\left\{q_1 + \epsilon \exp\left[r \exp\{\sin(q_2^{-2})\}\right]\right\}^{-1}$	0.028	0.023	0.898	0.923	85.3
Noise level: $\sigma = 0.15$						
1	$\epsilon q_1 \exp\{-(\epsilon + r)\}$	0.155	0.148	0.223	0.185	95.3
2	$\sin\left[-\left\{-\epsilon \exp(q_2^{-1})(r+r) + \epsilon\right\}^{-1}\right]$	0.159	0.149	0.160	0.116	112.0
3	$q_1\{(r^3 + q_1)\epsilon\}^{-1}$	0.162	0.156	0.185	0.276	109.8
4	$\sin\left[\{q_1[r(\epsilon + q_1)]^{-1}\}^2\right]$	0.155	0.154	0.190	0.214	106.6
5	$\sin\left[\{q_2(\epsilon + r + \exp r)^{-1}\}^2\right]$	0.156	0.145	0.163	0.237	113.3
Noise level: $\sigma = 0.20$						
1	$\{(\exp r + \epsilon)\cos(\epsilon^{-1})\}^{-1}$	0.203	0.206	0.083	0.133	140.3
2	$q_2 \sin\left[\{r^2 + r\epsilon\}^{-2}\right]$	0.205	0.201	0.127	0.185	148.2
3	$\sin\left[\left\{\epsilon r^3 [q_2^2\{\exp(r)\}^{-1}]^{-1} + \epsilon\right\}^{-1}\right]$	0.205	0.208	0.116	0.220	124.7
4	$\sin\left[\epsilon\{r(\epsilon + q_2^{-1})\}^{-2}\right]$	0.200	0.213	0.078	0.111	119.9
5	$q_2\left\{\epsilon^{-1} + \left(r^3 + \exp \epsilon + \cos(q_1 + r)\right)r\right\}^{-1}$	0.208	0.199	0.128	0.076	84.7

Table 3: QLattice results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-2.19\times10^{-4} q_2 + 2.03\times10^{-4} - 0.100(-0.044 q_1 - 0.001)\left[0.065r - 0.211 + (1.077 - 0.002\epsilon)(-1.409 q_2 - 0.021)(0.155r - 1.036)\right] \times \left\{(0.264 - 0.464r)(-0.347\epsilon - 0.010)\right\}^{-1}$	0.003	0.002	0.999	0.999	95.666

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Table 3: QLattice results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
2	$-8.94\times10^{-4} - 2.298\Big[-0.239\epsilon + (1.020 - 0.309q_2)(0.235\epsilon + 0.424) - 0.429\Big] \times (0.015\epsilon + 0.417q_1 + 0.002r - 0.030) \times \Big\{(0.003 - r)^2(\epsilon + 0.567)^2\Big\}^{-1}$	0.001	0.001	1.000	1.000	95.883
3	$-5.11\times10^{-5} - 0.075(-1.224q_1 - 0.020)(0.067q_2 - 0.002) \times \Big[0.024\epsilon - 0.005q_2 + (1.588 - 0.213r)(2.815 - 0.355r) + 0.694\Big] \times \Big\{(0.024 - 0.817\epsilon)(0.428 - 0.808r)\Big\}^{-1}$	0.002	0.002	1.000	1.000	91.454
4	$-4.30\times10^{-5}q_1 + 3.97\times10^{-6} + 0.258(-0.249q_1 - 1.32\times10^{-4})(-0.071q_2 - 0.307) \times (0.214q_2 + 8.66\times10^{-4}) \times \Big\{(0.088 - 0.585r)(-0.195\epsilon - 5.70\times10^{-4}) \times (-0.340q_2 - 1.473)(-0.311r - 0.061)\Big\}^{-1}$	0.000	0.000	1.000	1.000	92.951
5	$-0.231(0.004 - 0.480q_1)(0.038\epsilon - 0.597)(0.484q_2 - 0.022) \times \exp\{2.441\exp(-0.563\epsilon)\} \times \Big[-0.010q_2 + (1.341 - 1.314r)(1.189r + 1.129) - 1.544\Big]^{-1} - 2.15\times10^{-4}$	0.001	0.001	1.000	1.000	93.297
Noise level: $\sigma = 0.15$						
1	$0.373\exp(-0.457\epsilon + 0.346q_1 + 0.261q_2 - 1.107r) + 0.002$	0.150	0.144	0.272	0.226	84.850
2	$1.498\exp\Big[(0.424\epsilon - 0.244q_1 + 1.296) \times (0.264q_2 - 1.283r - 0.635)\Big] + 0.027$	0.153	0.148	0.223	0.120	83.578
3	$1.895(0.233q_1 - 0.080) \times \exp\Big[0.356q_2 - 1.316r + (0.325\epsilon - 2.744)(0.343\epsilon - 0.123)\Big] + 0.003$	0.153	0.157	0.270	0.262	85.915
4	$-2.544(0.090 - 0.207q_1)\exp(-0.492\epsilon + 0.345q_2 - 1.449r) + 0.011$	0.152	0.148	0.230	0.271	81.465
5	$0.408\exp(-0.553\epsilon + 0.303q_1 + 0.370q_2 - 1.306r) + 0.013$	0.150	0.140	0.228	0.293	79.621
Noise level: $\sigma = 0.20$						
1	$0.465\exp(-0.618\epsilon + 0.276q_1 + 0.282q_2 - 1.060r) + 0.014$	0.198	0.197	0.127	0.208	72.570
2	$0.418\exp(-0.468\epsilon + 0.335q_1 + 0.339q_2 - 1.382r) - 0.005$	0.201	0.195	0.164	0.233	78.285
3	$0.458\exp(-0.569\epsilon + 0.298q_1 + 0.371q_2 - 1.358r) + 0.013$	0.201	0.199	0.150	0.289	74.692
4	$-0.277(1.244 - 0.209\epsilon)(0.193q_2 + 0.023) \times \{0.180q_1 - 0.921r - 0.050\}^{-1} + 0.003$	0.197	0.208	0.109	0.152	66.408
5	$0.385\exp(-0.580\epsilon + 0.234q_1 + 0.421q_2 - 1.107r) + 0.006$	0.204	0.198	0.164	0.088	76.088

Table 4: SISSO++ results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-4.642\times10^{-17} + 1.430\times10^{-22}\,\epsilon q_1 q_2^2 (q_2/r)^2 + 2.206\times10^{-19}\,\frac{\epsilon q_2 (q_1/r)}{r^2 (r/q_1)} + 0.080\,\frac{q_1 q_2}{\epsilon r^2}$	0.000	0.000	1.000	1.000	14.700
2	$-5.029\times10^{-17} - 1.126\times10^{-18}\,\frac{q_2^2 (q_1/r)}{\epsilon^2} - 3.676\times10^{-18}\,\frac{q_1/r}{(\epsilon/q_2)\epsilon^2} + 0.080\,\frac{q_1 q_2}{\epsilon r^2}$	0.000	0.000	1.000	1.000	17.523
3	$-4.654\times10^{-17} + 0.080\,\frac{q_1 q_2}{\epsilon r^2} + 6.090\times10^{-20}\,\frac{q_2^2 (q_1 q_2)}{(r\epsilon)(r/q_2)} - 1.016\times10^{-18}\,\frac{(q_1/r)^2}{r^2 \epsilon}$	0.000	0.000	1.000	1.000	16.854
4	$-4.955\times10^{-17} - 2.165\times10^{-17}\,\frac{q_1/r}{(r\epsilon)(r q_2)} + 0.080\,\frac{q_1 q_2}{\epsilon r^2} - 6.040\times10^{-18}\,\frac{(q_1 q_2)/\epsilon^2}{(r/q_2) r^2}$	0.000	0.000	1.000	1.000	17.073
5	$-4.998\times10^{-17} - 3.856\times10^{-19}\,\frac{\epsilon q_1 q_2}{r^2} + 0.080\,\frac{q_1 q_2}{\epsilon r^2} - 6.617\times10^{-19}\left(\frac{q_1 q_2}{r\epsilon}\right)^2$	0.000	0.000	1.000	1.000	17.043
Noise level: $\sigma = 0.15$						
1	$-0.004 + c_1\,\frac{r^5}{\epsilon q_2^2} - 0.009\,\frac{\epsilon r}{q_1^4} + 0.079\,\frac{q_1 q_2}{\epsilon r^2}$	0.147	0.148	0.258	0.238	17.756
2	$0.007 - 0.020\,\frac{q_1^2}{\epsilon^3 q_2^4} + 0.071\,\frac{q_1 q_2}{\epsilon r^2} + 0.009\,\frac{q_1 q_2}{\epsilon^2 r}$	0.148	0.147	0.263	0.273	17.300
3	$0.005 + c_1\,\frac{r^7}{\epsilon} + c_2\,\frac{q_1^4 q_2}{r^3} + 0.073\,\frac{q_1 q_2}{\epsilon r^2}$	0.147	0.150	0.276	0.201	17.438
4	$-0.003 + c_1\,\epsilon^5 q_1 q_2/r + c_2\,q_1 q_2^6/r + 0.084\,\frac{q_1 q_2}{\epsilon r^2}$	0.150	0.146	0.258	0.261	17.967
5	$-0.002 + 0.011\,\frac{q_2^4}{\epsilon^2 q_1^4} + 0.095\,\frac{q_1 q_2}{\epsilon r^2} - 0.066\,\frac{q_2}{\epsilon r^3}$	0.151	0.150	0.254	0.271	17.298
Noise level: $\sigma = 0.20$						
1	$0.003 + c_1\,\frac{r^4 q_1}{\epsilon q_2^2} + 0.019\,\frac{q_1 q_2}{r^3} + 0.077\,\frac{q_1 q_2}{\epsilon^2 r^2}$	0.195	0.198	0.164	0.144	17.098
2	$0.009 - 0.026\,\frac{q_1^2}{\epsilon^3 q_2^4} + 0.068\,\frac{q_1 q_2}{\epsilon r^2} + 0.012\,\frac{q_1 q_2}{\epsilon^2 r}$	0.198	0.196	0.168	0.180	17.476
3	$0.006 + c_1\,\frac{r^7}{\epsilon} + c_2\,\frac{q_1^4 q_2}{r^3} + 0.071\,\frac{q_1 q_2}{\epsilon r^2}$	0.196	0.200	0.176	0.120	17.432

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Table 4: SISSO++ results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
4	$-0.005 + c_1\,\epsilon^5 q_1 q_2/r + c_2\, q_1 q_2^6/r + 0.086\,\frac{q_1 q_2}{\epsilon r^2}$	0.200	0.195	0.165	0.166	17.477
5	$0.002 + 0.016\,\frac{q_2^2}{\epsilon^2 q_1^4} + 0.038\,\frac{q_1^3 q_2}{\epsilon r^2} - 0.004\,\frac{q_1^3 q_2}{\epsilon r^2}$	0.202	0.201	0.163	0.166	18.443

Table 5: gplearn results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$\frac{q_1 q_2^2}{\epsilon\{q_1 q_2^2 r^2 + q_1(r^3+1) + q_2\}}$	0.037	0.045	0.818	0.746	16.505
2	$q_1^2 q_2 \Big\{q_1 [\epsilon\{r + \sin(q_1^{-1}) - 0.509\} + r + \sin(e^{-\epsilon}) - 1.018] + 3\Big\}^{-2}$	0.013	0.013	0.977	0.982	17.639
3	$\Big\{r + \Big[\epsilon + r + \sin(q_1) + \sin(q_2) - \cos\Big\{\frac{r}{r + \{\epsilon + r + 2\sin(q_1) + \sin(q_2)\}^2}\Big\}\Big]^2\Big\}^{-1}$	0.031	0.029	0.886	0.857	17.541
4	$e^{3q_2}\Big\{\epsilon[\{r + \sin(q_1^{-9})\}e^{q_2} + 1]^3\Big\}^{-1}$	0.048	0.043	0.738	0.692	15.868
5	$\Big[\epsilon r^2 + \sin(q_1) + \{q_2 - \sin(q_1^{-1})\}^{-2} + (q_2 - q_2^{-1})^{-2} + q_2^{-1} + q_2^{-2}\Big]^{-1}$	0.030	0.031	0.882	0.860	19.191
Noise level: $\sigma = 0.15$						
1	$q_1^4\{\epsilon(\epsilon^2 + q_1^4 r^2 + q_1^2 r^2 + e^r)\}^{-1}$	0.154	0.151	0.232	0.148	15.586
2	$\{r(\epsilon r + 1)\}^{-1}$	0.161	0.151	0.137	0.092	14.700
3	$q_1 q_2(\epsilon r + e^r)^{-2}$	0.154	0.156	0.263	0.274	15.547
4	$q_1^3 q_2\Big\{r[\epsilon^4 r^3 + q_1^3(q_2 r + 1) + q_1^2(q_2 r + 1)]\Big\}^{-1}$	0.153	0.153	0.213	0.224	16.498
5	$\exp\Big[\frac{r\{-\epsilon q_2^2 - q_1(q_2+2)\}}{q_1 q_2^2}\Big]$	0.152	0.142	0.204	0.270	16.168
Noise level: $\sigma = 0.20$						
1	$q_1^2\{\epsilon(q_1^2 r^2 + q_1 r^2 + 1)\}^{-1}$	0.200	0.202	0.106	0.173	16.173
2	$\sin\Big[\sin\Big\{\sin\big(\sin[\{\epsilon r^3\}^{-1}]\big)\Big\}\Big]$	0.207	0.208	0.112	0.121	15.123
3	$\exp(-0.686/q_2)(\epsilon r^2)^{-1}$	0.206	0.210	0.109	0.205	15.370
4	$\{r^2(\epsilon+1)\}^{-1}$	0.201	0.216	0.066	0.092	14.981
5	$q_2\{\epsilon r(q_2 r + 2)\}^{-1}$	0.209	0.195	0.125	0.111	16.827

Table 6: `operon` results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$0.080\, q_1 q_2 (\epsilon r^2)^{-1}$	0.000	0.000	1.000	1.000	18.776
2	$0.080\, q_1 q_2 (\epsilon r^2)^{-1}$	0.000	0.000	1.000	1.000	18.709
3	$0.080\, q_1 q_2 (\epsilon r^2)^{-1}$	0.000	0.000	1.000	1.000	20.110
4	$0.080\, q_1 q_2 (\epsilon r^2)^{-1}$	0.000	0.000	1.000	1.000	20.938
5	0	0.000	0.000	1.000	1.000	19.030
Noise level: $\sigma = 0.15$						
1	-0.007	0.148	0.145	0.288	0.222	26.672
2	$-0.057(\epsilon r^2)^{-1}\Big[0.035\epsilon r^2 - 0.245 q_1^2 r \sin^2\{1.606 q_1/r\} - 0.042 q_1 q_2 r \sin^2\{1.606 q_1/r\} - q_1 q_2 - 0.171 q_2^2\Big]$	0.152	0.146	0.229	0.144	23.136
3	$0.061(\epsilon r^3)^{-1}\Big[0.132\epsilon r^3 + 0.327 q_1 q_2^2 + 0.432 q_1 q_2 r + q_2 r \sin\{1.248 q_2/r\}\Big]$	0.152	0.156	0.280	0.271	23.952
4	$0.195\{\epsilon q_1 r(r-0.335)\}^{-1}\Big[0.015\epsilon q_1 r^2 - 0.005\epsilon q_1 r + 0.308 q_1^2 q_2 - 0.553 q_1 \cos(0.956\epsilon^2) + \cos(2.539 q_1)\Big]$	0.151	0.152	0.238	0.237	24.718
5	0.014	0.149	0.139	0.235	0.300	22.389
Noise level: $\sigma = 0.20$						
1	$-0.074 e^{-\sin(16.325\epsilon/r)}(\epsilon r^2)^{-1}\Big[0.284\epsilon r^2 e^{\sin(16.325\epsilon/r)}\sin(8.519 q_1 r) - 0.501\epsilon r^2 e^{\sin(16.325\epsilon/r)} + 0.284\epsilon r^2 - q_1 q_2 e^{\sin(16.325\epsilon/r)}\Big]$	0.196	0.199	0.143	0.195	26.691
2	$0.111(\epsilon r^2)^{-1}\Big[-0.073\epsilon q_1^2 - 0.117\epsilon q_1 r + 0.265\epsilon q_1 + 0.202\epsilon r^2 - 0.355\epsilon r + 0.172\epsilon + q_1 q_2 - 0.832 q_2 r + 0.561 q_2\Big]$	0.200	0.194	0.170	0.235	20.375
3	$-0.075 q_2(\epsilon r^2)^{-1}\Big[-q_1 + 0.527 r \sin\{\cos(2.385\epsilon + 2.675 r)\} + 0.527 r \cos(2.375\epsilon + 2.591 q_2)\Big]$	0.200	0.199	0.159	0.285	26.512
4	$0.015\{\epsilon r \cos(1.285/r^2)\}^{-1}\Big[\epsilon r \cos(1.285/r^2) + 0.968 q_1 q_2 e^{\sin(0.572 q_2/r)}\Big]$	0.196	0.211	0.114	0.127	25.906
5	$0.050 e^{-0.310\epsilon r}\{\epsilon r \sin(0.619 r)\}^{-1}\Big[\epsilon q_1 q_2 \sin(0.619 r) + 0.099\epsilon r e^{0.310\epsilon r}\sin(0.619 r) + 0.601 q_2^2\Big]$	0.203	0.196	0.170	0.106	25.677

Table 7: `PySR` results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$0.080\, q_1 q_2 (\epsilon r^2)^{-1}$	0.000	0.000	1.000	1.000	71.351

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Table 7: PySR results for the Feynman equation I_12_2 (CL) $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
2	$0.080\, q_1 q_2 (\epsilon r^2)^{-1}$	0.000	0.000	1.000	1.000	72.965
3	$q_1 q_2 (12.566\, \epsilon r^2)^{-1}$	0.000	0.000	1.000	1.000	73.835
4	$0.080\, q_1 q_2 (\epsilon r^2)^{-1}$	0.000	0.000	1.000	1.000	61.572
5	$0.080\, q_1 q_2 (\epsilon r^2)^{-1}$	0.000	0.000	1.000	1.000	70.044
Noise level: $\sigma = 0.15$						
1	$(\epsilon r^3)^{-1}$	0.160	0.151	0.173	0.156	68.526
2	$\exp\{-1.337r\}$	0.166	0.155	0.077	0.040	68.534
3	$0.082\, q_1 q_2 (\epsilon r^2)^{-1}$	0.152	0.156	0.276	0.274	69.974
4	$\exp\{-1.307r\}$	0.164	0.158	0.099	0.172	83.431
5	$0.163\, r^{-1}$	0.163	0.156	0.079	0.118	73.418
Noise level: $\sigma = 0.20$						
1	$\exp\{-1.314r\}$	0.206	0.210	0.052	0.103	102.550
2	$\exp\{-1.385r\}$	0.211	0.218	0.079	0.036	74.571
3	$\exp\{-1.365r\}$	0.212	0.230	0.057	0.045	68.500
4	$0.178\, r^{-1}$	0.204	0.218	0.046	0.072	70.455
5	$\exp\{-1.348r\}$	0.216	0.196	0.061	0.108	92.480

Table 8: BMS results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set. Here, a_0 denotes a fitted constant learned by BMS.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-\frac{q_1 q_2}{a_0 \epsilon r^2} - \frac{q_2 \sin(a_0)}{a_0^2}$	0.000	0.000	1.000	1.000	231.897
2	$\frac{2 q_1 q_2}{\epsilon r^2 (2a_0 + 1)}$	0.000	0.000	1.000	1.000	198.388
3	$\frac{2e^{2/e} q_1 q_2}{a_0 \epsilon r^2}$	0.000	0.000	1.000	1.000	201.371
4	$\frac{a_0 q_1 q_2}{\epsilon r^2 \{e^{3a_0} + 1\}}$	0.000	0.000	1.000	1.000	218.641
5	$\frac{a_0 q_1 q_2}{\epsilon r^2 (2a_0 + 1)}$	0.000	0.000	1.000	1.000	204.226
Noise level: $\sigma = 0.15$						
1	$\sin\left\{\frac{q_1 q_2}{a_0 \epsilon r^2}\right\}$	0.149	0.144	0.281	0.232	349.022
2	$-\frac{q_1 q_2}{\epsilon r^2 (a_0 + r)}$	0.153	0.146	0.221	0.144	365.649

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Table 8: BMS results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set. Here, a_0 denotes a fitted constant learned by BMS *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	$\frac{a_0 q_1 (a_0 r + q_2)}{\epsilon r^2}$	0.152	0.156	0.275	0.274	363.815
4	$a_0 q_1 q_2 (\epsilon r^2)^{-1}$	0.152	0.148	0.228	0.275	363.012
5	$-\frac{q_1 q_2}{a_0 \epsilon r^2}$	0.149	0.139	0.232	0.305	365.000
Noise level: $\sigma = 0.20$						
1	$\sin\left\{a_0\{q_1 q_2 + \cos(a_0 q_2)\}(\epsilon r^2)^{-1}\right\}$	0.197	0.196	0.132	0.221	406.794
2	$a_0 q_1 q_2 (\epsilon r^3)^{-1}$	0.201	0.195	0.164	0.229	391.551
3	$a_0 q_1 q_2 \sin\{a_0/(2r)\}(\epsilon r)^{-1}$	0.201	0.199	0.148	0.286	421.309
4	$\sin\left\{q_1 q_2 \cos^3(a_0)(\epsilon r^2)^{-1}\right\}$	0.197	0.211	0.104	0.131	393.521
5	$q_1 q_2 \left\{a_0^2 \epsilon r[\epsilon + r(2a_0 + q_1)]\right\}^{-1}$	0.204	0.195	0.166	0.112	382.448

Table 9: BSR results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$0.141 - 0.066\exp\{\cos(q_1)\} + 0.001\exp(q_1) - 0.002\sin\{(\epsilon + q_2)^3\} - 0.007 q_1(1+\epsilon)$	0.081	0.081	0.153	0.191	189.495
2	$2.105 - 2.008\cos\{\exp(-\epsilon)\} + 0.076\cos(r) - 0.014\cos(\epsilon) - 0.004 q_1 r$	0.069	0.079	0.332	0.325	188.710
3	$-0.013 - 5.302\times10^{-5}\exp(q_1) + 4.900\times10^{-4}\exp(q_2) + 3.541\times10^{-5}\frac{\cos(q_1)}{\sin(q_1^2)} + 0.019 q_1$	0.086	0.073	0.095	0.093	188.960
4	$0.052 - 0.019\{\epsilon + \exp[\cos(\epsilon)]\} + 6.265\times10^{-4}\exp(q_1) + 0.019 q_2 + 0.001\sin(q_1^2)$	0.086	0.071	0.171	0.154	188.388
5	$0.013 + 0.005\{\cos(q_2^{-1}) + q_2 + r\} - 0.026 q_2^{-1} + 0.077 r^{-1} + 8.526\times10^{-6}(-r q_2)^3$	0.075	0.069	0.260	0.288	189.140
Noise level: $\sigma = 0.15$						
1	$9.919\times10^{-11} + 2.028\times10^{-10} r + 2.141\times10^{-10} q_1 r^{-1} + 7.839\times10^{-12}\sin(\epsilon^2) - 6.104\times10^{-11}\sin^{-2}(q_1)$	0.186	0.168	-0.117	-0.047	190.750
2	$-0.308 - 0.009\cos(\epsilon^2) - 0.435\exp(-r) - 0.008\cos\{\sin(r)\} + 0.271\exp(r^{-1})$	0.165	0.153	0.088	0.062	188.704
3	$0.051 + 0.030\exp\{\sin(\epsilon)\} + 0.096\cos(r) + 9.383\times10^{-4}\sin(\epsilon^2) - 0.027\cos(q_1)$	0.171	0.168	0.090	0.151	188.453
4	$0.023 + 4.914\times10^{-5} r\exp(q_2) + 0.058\sin\{\sin(r)\} - 6.124\times10^{-5}\sin^{-1}(q_1) + 6.006\times10^{-4}\exp(q_1)$	0.167	0.170	0.063	0.046	189.518

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Table 9: BSR results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
5	*numerical overflow*	—	—	—	—	—
Noise level: $\sigma = 0.20$						
1	$9.680\times10^{-18} + 3.878\times10^{-12}\exp(q_2 r) - 4.213\times10^{-14}(-\epsilon q_1)^9 + 2.551\times10^{-11}\exp(\epsilon q_2/r) - 2.554\times10^{-17}\{\cos(q_1) - r\}$	0.220	0.230	-0.076	-0.080	188.188
2	$6.596\times10^{-13} - 2.350\times10^{-13}\sin(q_2) - 3.259\times10^{-13}\cos(q_1) + 6.368\times10^{-13}\exp\{\exp(r)\} + 2.641\times10^{-12}\epsilon r$	0.223	0.233	-0.032	-0.095	187.441
3	$0.068 - 0.132 q_1^{-1} - 0.010\sin(-\epsilon^3) + 0.050\cos\{\cos(q_1)\} + 0.005\sin\{\cos(q_1^2)\}$	0.217	0.234	0.014	0.014	188.789
4	$0.114 - 0.034\cos\{\cos(q_1)\} - 0.027\cos^2\{\sin(q_2^2)\} + 5.686\times10^{-4} q_1^3 - 0.057\sin^2\{\cos(\epsilon) + q_2\}$	0.206	0.225	0.019	0.010	190.897
5	$0.042 + 0.288 r^{-1} - 0.026\epsilon^2 + 3.220\times10^{-5}\exp\{\cos(r q_1) + r\} + 0.004\epsilon^3$	0.213	0.197	0.092	0.095	194.460

C.2 Results for I_12_11 (FCE)

Table 10: DSR results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$v\{-B - E_f\sin(\theta)\}$	16.714	15.133	0.590	0.596	48.5
2	$v\{q^2\sin(\theta) + B\} + \theta^{-3}$	14.464	11.720	0.681	0.788	46.3
3	$q + \sin(\theta)v\left\{B\left[(\exp(\theta) + E_f)^{-1} + q\right] + B\right\} + B$	9.827	9.812	0.852	0.829	44.4
4	$Bvq\sin^3(\theta)$	12.989	12.120	0.729	0.799	62.4
5	$\{q + E_f\sin(\theta)\}^2$	16.952	20.242	0.566	0.443	42.0
Noise level: $\sigma = 0.25$						
1	$B\sin(\theta)(q + v)$	15.523	14.612	0.642	0.667	42.6
2	$Bqv\left[\{\exp(B)v + \theta\}^{-1} + \sin^3(\theta)\right] + \theta$	11.446	11.847	0.814	0.800	42.1
3	$v\left[E_f B\sin(\theta) + \sin\{-\exp(-E_f)\}^3 + E_f\right]$	14.015	16.898	0.709	0.600	44.1
4	$q^2 B\sin(\theta) + v$	15.082	13.882	0.644	0.764	41.7
5	$B\sin(\theta)\exp\{\exp[\cos(v^{-1})]\}$	15.955	15.260	0.620	0.603	42.5
Noise level: $\sigma = 1.00$						
1	$vq\sin(\theta)\{B - \cos(-\theta)\}$	11.291	11.289	0.804	0.806	42.6
2	$v\{E_f + Bq\sin(\theta)\}$	5.422	5.420	0.955	0.951	42.0
3	$(q + v)\sin(\theta)\{q - \sin(v)\}$	14.773	15.797	0.681	0.668	42.7
4	$\{v + qB\sin(\theta)\}q\cos(B^{-1})$	13.950	15.722	0.723	0.604	42.6
5	$Bvq\sin(\theta)$	10.228	10.272	0.840	0.843	42.6

Table 11: QLattice results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-48.303(0.126q + 5.576\times10^{-4}) \times \left[-0.163E_f + (0.165 - 0.393\theta)(1.908 - 0.326\theta) \times (4.395 - 1.400\theta)(-0.625B - 0.002)\right] \times (-4.409\times10^{-5}E_f - 1.210)(0.156v + 0.002) - 0.062$	0.531	0.522	1.000	1.000	96.089
2	$-38.383(1.643 - 0.294\theta)(0.060q + 1.766\times10^{-5}) \times (-0.072\theta - 0.758)\left[0.602E_f + (2.118 - 0.672\theta)(0.107E_f + 3.200) \times (0.241\theta - 0.086)(0.459v - 0.170) \times (0.936B - 0.215E_f + 0.276) - 0.106\right] + 0.867$	1.091	1.096	0.998	0.998	94.457

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Table 11: `QLattice` results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	$19.478q + 42.662(0.536 - 0.027E_f)(-0.354q - 0.002) \times \Big[0.018\theta + (0.554 - 0.177\theta)(0.838B - 0.044) \times (0.451v - 0.029)\{-0.088E_f + (0.530\theta - 0.548) \times (0.559\theta - 2.919) - 0.602\} + 2.340\Big] + 0.143$	0.550	0.447	1.000	1.000	90.534
4	$2.996q - 18.907(0.006 - 0.290B)(2.827 - 0.901\theta) \times (3.622 - 0.621\theta)(0.242q + 0.009) \times (0.629\theta - 0.263)(0.282v - 0.001) - 29.826(1 - 0.060E_f)^2 + 19.714$	1.433	1.548	0.997	0.997	91.767
5	$20.585(4.180 - 1.332\theta)(2.972\times10^{-4}E_f - 0.351) \times (0.060q + 0.001)\Big[0.317B + (0.450 - 2.152B)(1.003 - 0.171\theta) \times (1.197\theta - 0.474)(0.533v + 0.031) - 0.828\Big] + 1.113(-0.568E_f - 0.559q + 1)^2 + 1.825$	0.899	0.996	0.999	0.999	92.527
Noise level: $\sigma = 0.25$						
1	$-37.575(0.002 - 0.258B)(1.145q + 0.005) \times (0.178\theta - 0.558)(0.243\theta - 1.422) \times (-0.326v - 0.001)(0.003B - 0.843\theta + 0.339) - 37.575(0.053 - 1.318q)(0.156 - 4.545\times10^{-4}v) \times (0.132E_f - 0.009) + 0.246$	0.429	0.441	1.000	1.000	91.071
2	$20.395(0.025q - 0.003)\Big[2.120E_f + (0.017 - 1.983v)(0.757 - 0.130\theta) \times (1.015 - 0.001E_f)(1.442 - 0.015q) \times (2.870 - 0.911\theta)(-1.397B - 0.004) \times (0.586\theta - 0.244) - 0.782\Big] + 0.684$	0.635	0.657	0.999	0.999	93.601
3	$-0.021E_f + 9.681q - 17.240(-0.219q - 0.002) \times \Big[0.266E_f + (0.138 - 0.326\theta)(3.221 - 0.552\theta) \times (3.269 - 1.042\theta)(0.385B + 5.972\times10^{-4}) \times (-0.485v - 0.001) - 2.566\Big] - 0.096$	0.462	0.502	1.000	1.000	91.798
4	$34.556(0.013 - 0.048\theta)(-0.007E_f - 3.883) \times (0.318q - 0.007)\Big[0.653\theta + (2.967 - 0.508\theta) \times \{0.144E_f + (-0.328B - 0.063)(0.201\theta - 0.625) \times (1.753v + 0.146) + 1.108\} - 3.621\Big] + 0.273$	1.070	1.086	0.998	0.999	91.377
5	$38.879\Big[-0.048E_f + (0.392 - 0.950\theta)(1.024 - 0.326\theta) \times (0.382B + 0.001)(0.102v - 8.045\times10^{-4}) \times (-0.525\theta + 7.335\times10^{-4}v + 3.070) - 6.423\times10^{-4}\Big] \times (-0.532q + 0.002v - 0.009) - 0.018$	0.451	0.374	1.000	1.000	92.678
Noise level: $\sigma = 1.00$						
1	$77.446(0.084q - 8.236\times10^{-4}) \times \Big[0.156E_f - 0.001q + (0.183 - 0.438\theta)(2.442 - 0.778\theta) \times (0.389B - 0.002)(0.776v + 0.005) \times (-0.001q + 0.200\theta - 1.169) - 0.005\Big] + 0.143$	1.233	1.253	0.998	0.998	95.471
2	$22.135(0.105 - 1.077q)\Big[-0.043E_f + (0.078 - 0.193\theta)(3.925 - 0.669\theta) \times (0.289\theta - 0.909)(0.141v + 0.002) \times (-1.071B + 0.043q - 0.150) + 0.004\Big] + 0.471$	1.132	1.113	0.998	0.998	92.821

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Table 11: `QLattice` results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	$2.899E_f + 25.179(0.144q + 0.002) \times \big[1.058(0.037 - 0.091\theta)(2.247 - 0.385\theta) \times (-1.059B - 0.025)\{-0.649\theta + (3.902 - 1.244\theta) \times (0.739v - 0.523) + 2.026\} + 0.828\big] - 8.668$	1.686	1.682	0.996	0.996	96.179
4	$-29.274(3.813{\times}10^{-4} - 0.085q) \times \big[0.406E_f + (9.843{\times}10^{-4} - 0.419B)(0.281 - 0.683\theta) \times (0.951 - 0.303\theta)(1.904 - 0.325\theta) \times (1.872v - 0.002) - 0.009\big] - 0.152$	1.129	1.165	0.998	0.998	96.701
5	$-21.119(2.281{\times}10^{-4} - 0.033q) \times \big[1.391E_f - 0.525\theta + (0.099 - 0.589v)(0.130 - 0.819B) \times (0.527 - 1.152\theta)(2.975 - 0.514\theta) \times (5.584{\times}10^{-4}q + 0.835)(0.845\theta - 2.657) + 1.597\big] - 0.143$	1.238	1.140	0.998	0.998	86.482

Table 12: `SISSO` + + results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-9.001{\times}10^{-16} - 1.354{\times}10^{-16}(v + B + \sin B)(\sin v + v) + E_f q + Bvq\sin(\theta)$	0.000	0.000	1.000	1.000	42.554
2	$8.925{\times}10^{-16} + \{\sin(v) + E_f q + E_f q + v\} - \{E_f q + v + \sin(v)\} + Bvq\sin(\theta)$	0.000	0.000	1.000	1.000	44.682
3	$-8.076{\times}10^{-17} - 2.549{\times}10^{-16}\{\sin(E_f) + \sin(q) + E_f\} + E_f q + Bvq\sin(\theta)$	0.000	0.000	1.000	1.000	45.950
4	$4.641{\times}10^{-16} - 1.679{\times}10^{-16}(\theta + v + vB)q\sin(\theta) + E_f q + Bvq\sin(\theta)$	0.000	0.000	1.000	1.000	41.782
5	$4.508{\times}10^{-16} + 2.420{\times}10^{-16}\{\sin(q) + E_f + \sin(E_f)\} + E_f q + Bvq\sin(\theta)$	0.000	0.000	1.000	1.000	43.372
Noise level: $\sigma = 0.25$						
1	$0.019 + 0.998\,E_f q + 0.994\,Bvq\sin(\theta) + 0.004\,q\sin(\theta)(vB + v)$	0.245	0.248	1.000	1.000	42.689
2	$0.011 + 1.000\,E_f q + 0.996\,Bvq\sin(\theta) + 0.004\,v\sin(\theta)\{\sin(B) + Bq\}$	0.248	0.246	1.000	1.000	42.115
3	$0.094 + 1.026\,E_f q - 0.026\{\sin(B) + E_f q + B\} + 1.000\,Bvq\sin(\theta)$	0.245	0.249	1.000	1.000	42.134
4	$0.031 - 0.006\{\sin(B) + \sin(E_f) + B + q + \sin(E_f)\} + 1.001\,E_f q + 1.000\,Bvq\sin(\theta)$	0.250	0.244	1.000	1.000	43.311
5	$0.012 + 0.003\,(v + B + \sin q)\sin(E_f)\sin(q) + 0.999\,E_f q + 1.000\,Bvq\sin(\theta)$	0.252	0.251	1.000	1.000	43.518
Noise level: $\sigma = 1.00$						

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Table 12: `SISSO++` results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
1	$0.076 + 0.991\,E_f q + 0.987\,Bvq\sin(\theta) + 0.008\,v\sin(\theta)(Bq + E_f + q)$	0.978	0.994	0.998	0.998	43.563
2	$0.043 + 0.999\,E_f q + 0.982\,Bvq\sin(\theta) + 0.017\,v\sin(\theta)\{\sin(B) + Bq\}$	0.992	0.985	0.999	0.999	42.256
3	$0.375 + 1.102\{\sin(B) + E_f q + E_f q + B\} - 1.205\{\sin(B) + E_f q + B\} + 1.001\,Bvq\sin(\theta)$	0.981	0.995	0.999	0.998	46.512
4	$0.124 - 0.024\{\sin(B) + \sin(E_f) + B + q + \sin(E_f)\} + 1.002\,E_f q + 0.999\,Bvq\sin(\theta)$	1.001	0.975	0.999	0.998	42.650
5	$0.003 + 0.912\,E_f q + 0.044\{\sin(B) + E_f q + \sin(E_f) + E_f q\} + 1.001\,Bvq\sin(\theta)$	1.010	1.005	0.998	0.998	43.966

Table 13: `gplearn` results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$q\{E_f + Bv\sin(\theta)\}$	0.000	0.000	1.000	1.000	15.034
2	$qE_f + q^2\sin(\theta) + \sin^3(\theta) + v^2\sin\{\sin[\sin(\theta)]\} + B^2\sin(\theta)$	8.008	7.412	0.902	0.915	17.532
3	$q\{E_f + Bv\sin(\theta)\}$	0.000	0.000	1.000	1.000	15.353
4	$q\{E_f + Bv\sin(\theta)\}$	0.000	0.000	1.000	1.000	17.214
5	$qv\{1 + B\sin(\theta)\}$	5.166	5.210	0.960	0.963	16.575
Noise level: $\sigma = 0.25$						
1	$qE_f\exp\{\sin[\sin\{\sin(\theta)\}]\} + 2Bv\sin(\theta) + qB\sin\{\sin[\sin(\sin\{\sin(\theta)\})]\}$	8.225	7.592	0.899	0.910	17.947
2	$q + E_f + q\sin(\theta)\{q + 2v - \theta\} + B^2\sin(\theta) + \exp(0.374^3)$	7.488	7.239	0.921	0.925	16.998
3	$q\{E_f + Bv\sin(\theta)\}$	0.254	0.260	1.000	1.000	15.842
4	$(E_f + q)\{0.571^{-1} + \sin(\theta)\} + 0.877\,qBv\sin\{\sin(\theta)\}$	4.251	4.668	0.972	0.973	17.093
5	$qE_f + qBv\sin\{\sin(\theta)\} + v\sin(\theta)$	1.896	1.733	0.995	0.995	16.505
Noise level: $\sigma = 1.00$						
1	$q\{E_f + Bv\sin(\theta)\}$	0.988	0.978	0.999	0.999	17.098
2	$2q + E_f + B^2\sin(\theta) + \sin\{q - \exp[B^2\sin(\theta)]\} - 0.384 + 2qv\sin(\theta)$	6.973	7.579	0.925	0.905	19.384
3	$qE_f + \sin(\theta)\big[q^2 + B^2 + v^2 + \sin(\theta) - 0.239 - 2\exp\{\sin(\theta)\}\big]$	7.699	8.219	0.913	0.910	15.741

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Table 13: `gplearn` results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
4	$qE_f + v\Big[qB\sin(\theta) + \exp\{-(0.569^{-1})^3\} + \exp\{-\exp(0.569^{-1})\} + \exp\{-(0.569^{-1})^3\}\Big]$	0.985	1.052	0.999	0.998	16.375
5	$q\{E_f + Bv\sin(\theta)\}$	1.021	0.973	0.998	0.999	15.157

Table 14: `operon` results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-0.324 + 1.497\Big[2.035q + \Big\{-0.548qv\sin(1.002\theta) - 0.023E_f\Big\}(-1.218B) - e^{0.711}\,0.459q\sin(1.021E_f)\Big]$	0.816	0.864	0.999	0.999	27.316
2	$4.572 + 7216.561\Big[\{0.462q\sin^2(0.022E_f) + 0.022B\sin(0.996\theta)\} \times 0.022q\{0.273v + 0.022E_f\sin[\sin(0.871\theta)]\}\Big]$	2.459	2.396	0.991	0.991	29.131
3	$9.072 + 3.994(-0.481E_f)\Big[\cos(0.521q) - \Big\{0.520qv\sin(\theta) - 0.177qB^{-1}\sin(1.029E_f)\Big\}\Big]$	1.535	1.552	0.996	0.996	26.635
4	$-8.787 - 0.866\Big[-3.317E_f - 3.563q + 2.948q\sin(1.001\theta)\{-1.297v + \exp(-3.417E_f + 0.678q\sin(0.976\theta))\}\ 0.302B\Big]$	1.274	1.362	0.997	0.997	27.056
5	$0.028 + 0.496\Big[2.020qE_f + 1.822B\sin(\theta)\{1.115qv\}\Big]$	0.049	0.046	1.000	1.000	22.940
Noise level: $\sigma = 0.25$						
1	$6.992 - 13.761\Big[(0.256q/1.414E_f)^2 + \cos(-0.129E_f) - 0.122E_f + 0.093qBv\sin(-1.001\theta) - 0.256q\Big]$	1.025	1.023	0.998	0.998	25.396
2	$2.510 - 0.303\Big[-1.782q + 2.586B - 0.535qE_f^2 - 2.303B\{0.359q + 1.430qv\sin(1.002\theta)\}\Big]$	0.753	0.816	0.999	0.999	25.028
3	$0.178 + 2.419\Big[\{0.179qv\sin(1.007\theta) + c_1\,qE_fB^{-1}\}\,2.303B\Big]$	0.448	0.443	1.000	1.000	26.952
4	$-8.880 + 3.419\Big[1.214\,qBv\sin(1.002\theta) + 0.874E_f + 0.878q\Big]$	1.344	1.311	0.997	0.998	22.984
5	$-0.002 + 0.279\Big[1.790qE_f + 1.790qE_f + 0.012E_f + 3.598qBv\sin(\theta)\Big]$	0.249	0.231	1.000	1.000	23.109
Noise level: $\sigma = 1.00$						
1	$0.056 - 180.230(0.806q)\Big[2.276(0.073B)(0.073v)\sin(-0.995\theta) - 0.007E_f - (0.073B)(0.073v)\sin(-0.988\theta)\Big]$	0.986	0.985	0.999	0.999	24.154
2	$-0.155 - 0.011\Big[\Big\{(0.327v/-1.160)[(-0.972 - 1.873B)\sin(-0.993\theta) + \sin(-0.964\theta)] - 0.525E_f\Big\}\{0.693q/(-0.063)^2\}\Big]$	1.013	0.984	0.998	0.998	23.321

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Table 14: `operon` results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	$3.797 + 0.085\Big[\{3.382B\sin(1.036\theta) + 0.442E_f - \sin[\sin(1.409\theta)]\}1.533v - 1.230v + 0.807qE_f\Big]$	1.641	1.707	0.996	0.996	26.583
4	$-1.497 + 1.487\Big[0.369\{0.660E_f + \cos[\sin(1.027\theta)]\} + 0.837q\{0.800Bv\sin(1.003\theta) + 0.720E_f + 0.305\}\Big]$	1.003	1.075	0.999	0.998	25.101
5	$-5.545 + 8.714\Big[-0.373q\exp(-0.434E_f) - 0.115qBv\sin(-1.003\theta) - 0.216E_f + 0.459q\Big]$	1.319	1.266	0.997	0.998	24.734

Table 15: `PySR` results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$q\{E_f + Bv\sin(\theta)\}$	0.000	0.000	1.000	1.000	136.217
2	$q\{E_f + Bv\sin(\theta)\}$	0.000	0.000	1.000	1.000	135.984
3	$q\{E_f + Bv\sin(\theta)\}$	0.000	0.000	1.000	1.000	122.059
4	$q\{E_f + Bv\sin(\theta)\}$	0.000	0.000	1.000	1.000	115.471
5	$q\{E_f + Bv\sin(\theta)\}$	0.000	0.000	1.000	1.000	126.500
Noise level: $\sigma = 0.25$						
1	$q\{E_f + Bv\sin(\theta)\}$	0.248	0.237	1.000	1.000	119.976
2	$q\{E_f + Bv\sin(\theta)\}$	0.255	0.244	1.000	1.000	129.114
3	$q\{E_f + Bv\sin(\theta)\}$	0.254	0.260	1.000	1.000	128.247
4	$q\{E_f + Bv\sin(\theta)\}$	0.253	0.247	1.000	1.000	124.288
5	$q\{E_f + Bv\sin(\theta)\}$	0.249	0.231	1.000	1.000	105.516
Noise level: $\sigma = 1.00$						
1	$q\{E_f + Bv\sin(\theta)\}$	0.988	0.978	0.999	0.999	124.517
2	$q\{E_f + Bv\sin(\theta)\}$	1.006	0.959	0.998	0.998	115.130
3	$q\{E_f + Bv\sin(\theta)\}$	1.007	0.996	0.999	0.999	112.441
4	$q\{E_f + Bv\sin(\theta)\}$	0.987	1.056	0.999	0.998	117.745
5	$q\{E_f + Bv\sin(\theta)\}$	1.021	0.973	0.998	0.999	146.730

Table 16: BMS results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set. Here, a_0 denotes a fitted constant learned by BMS.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$qE_f + Bqv\sin(\theta)$	0.000	0.000	1.000	1.000	267.921
2	$q\{E_f + Bv\sin(\theta)\}$	0.000	0.000	1.000	1.000	257.230
3	$qE_f + a_0Bqv\sin(\theta)$	0.000	0.000	1.000	1.000	318.374
4	$q\{E_f + Bv\sin(\theta)\} + \sin^3(a_0)$	0.000	0.000	1.000	1.000	282.034
5	$qE_f - a_0Bqv\sin(\theta)$	0.000	0.000	1.000	1.000	235.786
Noise level: $\sigma = 0.25$						
1	$qE_f + Bqv\sin(\theta)$	0.248	0.237	1.000	1.000	294.344
2	$qE_f + B\{qv + 2a_0\}\sin(\theta)$	0.255	0.244	1.000	1.000	314.851
3	$qE_f - Bqv\sin\{-(\theta + 2a_0)\}$	0.254	0.260	1.000	1.000	281.084
4	$qE_f + Bqv\sin(\theta)$	0.253	0.247	1.000	1.000	303.930
5	$qE_f + Bq(v + a_0\theta)\sin(\theta)$	0.249	0.232	1.000	1.000	272.353
Noise level: $\sigma = 1.00$						
1	$qE_f + Bqv\sin(\theta)$	0.988	0.978	0.999	0.999	285.738
2	$qE_f + Bv\{q + a_0e^{\theta}\}\sin(\theta)$	1.003	0.962	0.998	0.998	302.691
3	$q\Big[E_f - Bv\sin(\theta)\Big\{-\exp\{a_0(B+\theta)\}\Big\}^9\Big]$	1.007	0.996	0.999	0.999	344.995
4	*numerical overflow*	—	—	—	—	—
5	$qE_f + Bqv\sin(\theta)$	1.021	0.973	0.998	0.999	246.953

Table 17: BSR results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-25.320 - 1.303\sin(v) + 0.022(B^3 - v) + 87.165\,\theta^{-1}$	19.583	17.165	0.437	0.481	164.419
2	$33.811 - 0.067(2\theta)^3 - 1.468B + 0.055e^{B}$	16.695	17.040	0.575	0.551	167.938
3	$20.823 + 2.685E_f - 0.464\{e^{\theta} + \cos(\theta) + v\} + 0.536\,\theta\sin(v)$	17.821	16.277	0.512	0.531	165.756
4	$17.839 + 0.142\{B + e^{\cos(B)}\} - 0.447e^{\theta} + 2.890E_f$	18.087	19.149	0.474	0.499	165.655
5	$-29.316 + 0.775\{v + \cos(e^{q\theta})\} + 1.235B + 87.299\,\theta^{-1}$	18.864	20.446	0.462	0.431	164.489
Noise level: $\sigma = 0.25$						

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Table 17: BSR results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
1	$19.906 - 0.970(v + q\theta) + 19.892\sin\{\sin(e^{v}) + \theta\} - 1.615\cos(E_f)$	18.838	19.449	0.472	0.410	167.556
2	$3.495{\times}10^{-10} + 2.548{\times}10^{-10}e^{-\cos^2(B)} + 4.434{\times}10^{-9}E_f^2 - 8.757{\times}10^{-9}e^{v\theta}$	28.094	28.733	-0.119	-0.176	165.173
3	$6.308 + 0.105E_f^3 - 0.069B\cos(B) - 1.104\sin(B)$	25.714	26.146	0.022	0.041	165.598
4	$25.442 - 0.395B^2\theta - 0.144\sin(v) - 5.693v^{-1}$	23.227	25.609	0.157	0.195	165.248
5	$6.565 + 0.088E_f^3 + 0.039(B^3 + B) - 5.808\sin(q)$	25.314	24.442	0.044	-0.018	164.711
Noise level: $\sigma = 1.00$						
1	$9.736 + 0.775\cos^2(E_f) - 9.109q^{-2} + 0.061e^{q}$	25.246	25.885	0.022	-0.020	167.802
2	$11.635 + 0.048v^3 + 0.525\sin\{E_f e^{1/q}\}\sin(B) - 3.670\cos\{\cos(q)\}$	25.429	24.696	0.005	-0.012	164.314
3	$-5.198 + 2.083(B + \theta) + 29.568\sin(\theta) + 3.449\sin(B)$	15.892	17.252	0.631	0.604	164.415
4	$1.150{\times}10^{-7} - 2.940{\times}10^{-4}\theta^6 + 5.812{\times}10^{-8}\sin^2\{\cos(B) + E_f\} + 1.265{\times}10^{-7}\{v + \exp(B + v)\}^2$	28.476	28.072	-0.156	-0.264	165.759
5	$32.862 - 2.512q\theta - 0.009\{\sin(E_f B^2)\}^{-1} + 0.054B^2$	22.195	22.518	0.247	0.246	164.758

C.3 Results for I_24_6 (EHFO)

Table 18: `DSR` results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$\omega_0^2 + m\omega x^2$	5.183	4.936	0.863	0.882	31.6
2	$x\{x + m\omega\omega_0\}$	4.917	4.782	0.886	0.889	31.2
3	$x\{x + m\omega\omega_0\}$	4.697	4.721	0.889	0.899	30.6
4	$mx^2\{\omega + \cos(\omega_0 + x)\}$	4.701	4.793	0.885	0.890	33.6
5	$x\{m\omega\omega_0 + x\}$	4.733	4.727	0.889	0.899	32.0
Noise level: $\sigma = 0.15$						
1	$x^2 + m\omega\omega_0 x^2$	4.752	4.784	0.893	0.885	33.2
2	$\omega + \{-\omega_0 + \omega + m\omega_0\}x^2$	4.245	4.385	0.914	0.904	32.1
3	$x\{m\omega\omega_0 + x\}$	4.867	4.877	0.889	0.884	34.9
4	$x^2 + m\omega\omega_0 x^2$	4.709	4.732	0.897	0.886	31.2
5	$m\{-\omega\omega_0^2\cos(x) + \omega + x + m\}$	4.773	4.405	0.894	0.913	32.2
Noise level: $\sigma = 0.25$						
1	$x\{m\omega\omega_0 + x\}$	4.845	5.300	0.888	0.869	34.7
2	$m\omega\omega_0 x + x^2$	4.862	4.900	0.890	0.864	33.2
3	$m\omega\omega_0 x + x^2$	4.936	4.922	0.883	0.886	30.3
4	$x\{x + m\omega\omega_0\}$	4.892	4.577	0.888	0.894	29.8
5	$m\omega\omega_0 x + x^2$	5.088	4.995	0.888	0.889	30.2

Table 19: `QLattice` results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-8.917(-0.036m - 2.068\times10^{-4})(0.177x + 0.454)(1.553x - 0.855) \times \Big[1.874(0.050 - \omega)^2 + (0.193 - 0.006m)(2.987 - 0.980\omega) + (0.679\omega_0 - 0.346)\{1.481x + (0.062m + 2.235)(1.097\omega_0 + 0.301) - 2.960\}\Big] + 0.417$	0.292	0.285	1.000	1.000	95.182
2	$-13.136(-0.408m - 0.004)\Big[-0.009\omega + (2.563 - 2.036x) \times(-0.565x - 0.729) + 1.886\Big]\exp\{(2.029 - 0.474\omega_0)(0.462\omega - 1.980)\} - 0.061$	0.284	0.280	1.000	1.000	99.424

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Table 19: `QLattice` results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	$-0.051\omega_0 - 17.463(-0.802m - 0.022)(0.794x + 0.099) \times \exp\Big[-(0.784 - 0.121x)(-0.004\omega - 0.356)(2.746x - 5.284) - (1.255 - 0.295\omega_0)(3.185 - 0.745\omega)\Big] - 0.139$	0.286	0.300	1.000	1.000	93.490
4	$0.039m - 11.667(0.969 - 2.099x) \times \Big[0.250(0.469m + 0.018)(-0.667x - 0.544) \times \{-0.475\omega_0 + 0.565(2.908 - 0.676\omega_0)(-0.080\omega - 0.296) \times (0.710\omega - 2.618)e^{0.449\omega_0} - 1.202\} - 0.018\Big] + 0.288$	0.253	0.271	1.000	1.000	95.968
5	$-10.519(1.361 - 2.571x) \times \Big[-0.008m + (0.130m + 7.335\times10^{-4})\{0.698(-0.363x - 1)^2 + 0.809\}\{1.834\exp[(1.871 - 0.419\omega_0)(0.441\omega - 1.976)] - 0.033\} - 0.005\Big] + 0.300$	0.275	0.270	1.000	1.000	94.381
Noise level: $\sigma = 0.15$						
1	$2.459(-6.498\times10^{-5}m - 1)^2(0.122m + 1.847\times10^{-4}) \times (-0.008\omega + 1.825x - 0.955) \times \Big[1.256\omega_0 + 0.720(\omega - 0.068)^2 - 2.947 + (0.547 - 0.123\omega_0)^{-1}\Big] \times e^{0.295x} - 0.017$	0.251	0.265	1.000	1.000	94.493
2	$-26.030(0.638m + 0.005)(1.145x - 0.241) \times \Big[-0.324x - 0.786 + (1.472 - 0.009\omega_0)^{-1}\Big] \times \exp\{(1.745 - 0.405\omega)(0.538\omega_0 - 2.303)\} - 0.107$	0.318	0.310	1.000	1.000	92.749
3	$0.497(0.002 - 0.197m)(1.393x - 0.658) \times \Big[3.234(0.045 - \omega)^2 + 2.792(\omega_0 - 0.060)^2(-0.025x - 1)^2\Big] \times (0.032\omega - 0.520x - 0.617) + 0.166$	0.263	0.262	1.000	1.000	97.141
4	$1.316(0.001 - 0.674m)(0.466 - 1.261x)(0.426x + 0.440) \times \exp\Big[0.532\omega_0 + 0.052x + (1.300 - 0.695\omega)(0.309\omega_0 - 1.341)\Big] - 0.081$	0.362	0.350	0.999	0.999	94.907
5	$-18.166(0.271 - 1.498x)(-0.570m - 0.003)(-0.460x - 0.112) \times \exp\Big[-0.033\omega + (1.093 - 0.246\omega_0)(0.888\omega - 3.807)\Big] - 0.077$	0.322	0.295	1.000	1.000	94.036
Noise level: $\sigma = 0.25$						
1	$-33.528(0.760 - 1.956x)(0.058x + 0.037)(1.545m + 0.017\omega_0 - 0.059) \times \exp\Big[(1.933 - 0.456\omega)(0.003m + 1.022)(0.473\omega_0 - 2.023)\Big] + 0.078$	0.393	0.445	0.999	0.999	98.325
2	$-5.035(0.748 - 1.850x)(0.158m - 0.002)(0.188x + 0.120) \times \exp\Big[0.523\omega + 0.439\omega_0 + (0.145\omega - 0.335)(0.343\omega_0 - 2.334)(1.201\omega_0 - 2.243) \times (0.004x + 0.836)\Big] + 0.028$	0.357	0.333	0.999	0.999	93.343
3	$-8.165(0.114 - 0.484x) \times \Big[(0.694 - 0.530m)(0.339x + 0.122) + (-0.197x - 0.061)\Big\{-0.906m + (0.012 - 3.019m) \times \big[0.103(\omega + 0.025)^2 + 0.103(-\omega_0 - 0.027)^2\big] + 1.223\Big\}\Big] - 0.015$	0.285	0.295	1.000	1.000	94.746
4	$20.732(0.382 - 1.114x)(0.386m - 0.001)(-0.394x - 0.199) \times \Big[(-0.191\omega - 0.095)\{-0.375(-0.002m - 0.554)(-0.027\omega - 2.015) \times (0.809\omega - 1.322) - 0.441\} + 0.069(-\omega_0 - 0.015)^2\Big] + 0.108$	0.277	0.294	1.000	1.000	95.126

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Table 19: `QLattice` results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
5	$19.375\big[0.575m + (5.514\times10^{-4} - 0.106m)(4.037 - 1.571x) \times(0.185\omega + 0.947) + 5.200\times10^{-5}\big](0.359\omega + 1.133x - 0.748) \times \exp\{(2.362 - 0.544\omega)(0.397\omega_0 - 1.491)\} - 0.003$	0.412	0.393	0.999	0.999	93.102

Table 20: `SISSO++` results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-5.394\times10^{-16} - 7.822\times10^{-18}(x\omega)(xm)(\omega_0 + \omega)(\omega_0 m) - 5.118\times10^{-18}(x\omega)\omega^2(x + \omega + x^2) + 0.250\,mx^2(\omega_0^2 + \omega^2)$	0.000	0.000	1.000	1.000	20.426
2	$3.502\times10^{-16} + \omega^4 x^2 + 0.250\,mx^2(\omega_0^2 + \omega^2) - 6.855\times10^{-18}(x + \omega_0 + \omega_0^2)(x\omega)(xm)$	0.000	0.000	1.000	1.000	19.014
3	$3.471\times10^{-16} + 6.981\times10^{-18}x^2(\omega_0 m)(\omega^2 + x) + 0.250\,mx^2(\omega_0^2 + \omega^2) + 1.516\times10^{-17}(x + \omega_0)^2(xm)\omega$	0.000	0.000	1.000	1.000	19.793
4	$2.494\times10^{-16} - 1.995\times10^{-17}(x + m)\omega_0^2 x^2 - 6.534\times10^{-17}(x\omega + \omega_0)(xm)(\omega_0 + \omega) + 0.250\,mx^2(\omega_0^2 + \omega^2)$	0.000	0.000	1.000	1.000	19.437
5	$4.628\times10^{-16} - 7.202\times10^{-18}(x\omega_0 + \omega + m)(x\omega)(\omega m) - 4.890\times10^{-17}(x\omega_0 + xm)(x\omega + \omega_0 + m) + 0.250\,mx^2(\omega_0^2 + \omega^2)$	0.000	0.000	1.000	1.000	19.006
Noise level: $\sigma = 0.15$						
1	$0.017 + 0.250\,mx^2(\omega_0^2 + \omega^2) + 0.001(x\omega_0)m(x + \omega)^2 - 0.003(x\omega_0)m(x\omega + x + \omega)$	0.146	0.149	1.000	1.000	19.172
2	$0.014 - 1.842\times10^{-5}(\omega_0\omega)^2(\omega m)^2 + 0.249\,mx^2(\omega_0^2 + \omega^2) + 2.906\times10^{-4}x^2(\omega m)(\omega_0^2 + \omega + m)$	0.149	0.148	1.000	1.000	18.933
3	$0.007 + 0.250\,mx^2(\omega_0^2 + \omega^2) + 6.386\times10^{-4}(x + \omega_0)(x\omega)(xm + \omega_0 m) - 6.527\times10^{-4}(x^2 + x + m)(xm)(\omega_0\omega)$	0.147	0.150	1.000	1.000	18.841
4	$-0.001 + 7.489\times10^{-4}(\omega_0\omega)\omega_0^2(\omega_0 + \omega + \omega_0^2) - 7.301\times10^{-4}(\omega_0 + \omega + \omega^2)\omega_0^4 + 0.250\,mx^2(\omega_0^2 + \omega^2)$	0.150	0.146	1.000	1.000	19.038
5	$-0.003 + 8.834\times10^{-6}\omega_0^4(\omega_0 m)^2 - 7.401\times10^{-5}(x\omega_0)\omega_0^2(x^2 + \omega_0) + 0.250\,mx^2(\omega_0^2 + \omega^2)$	0.151	0.150	1.000	1.000	19.092
Noise level: $\sigma = 0.25$						
1	$-0.012 - 2.034\times10^{-4}x^4\omega^4 + 3.247\times10^{-4}(x\omega)(x + \omega)x^2\omega^2 + 0.250\,mx^2(\omega_0^2 + \omega^2)$	0.244	0.249	1.000	1.000	19.302
2	$0.023 - 3.070\times10^{-5}(\omega_0\omega)^2(\omega m)^2 + 0.249\,mx^2(\omega_0^2 + \omega^2) + 4.844\times10^{-4}x^2(\omega m)(\omega_0^2 + \omega + m)$	0.248	0.246	1.000	1.000	18.769

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Table 20: `SISSO` ++ results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	$0.011 + 0.250\, mx^2(\omega_0^2 + \omega^2) + 0.001(x + \omega_0)(x\omega)(x + \omega_0)m - 0.001(x\omega)(\omega_0 m)(x^2 + x + m)$	0.245	0.250	1.000	1.000	18.868
4	$-0.001 + 0.001(\omega_0\omega)\omega_0^2(\omega_0 + \omega + \omega_0^2) - 0.001(\omega_0 + \omega + \omega^2)\omega_0^4 + 0.250\, mx^2(\omega_0^2 + \omega^2)$	0.250	0.243	1.000	1.000	19.545
5	$-0.005 + 1.472{\times}10^{-5}(\omega_0^2)(\omega_0 m)^2 - 1.233{\times}10^{-4}(x\omega_0)\omega_0^2(x^2 + \omega_0) + 0.250\, mx^2(\omega_0^2 + \omega^2)$	0.252	0.251	1.000	1.000	19.332

Table 21: `gplearn` results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$m\omega_0\{x^2 + \cos[\cos(x) + \omega x]\} + \cos\{x\cos(\omega - 0.615)\} + \cos(\omega + x) + \omega(\omega - 0.615)$	2.953	3.633	0.956	0.936	16.194
2	$m\omega x^2 + x^4\omega_0^8\sin^8(0.286) + m^2x^2\omega_0^6\sin^6(0.286)$	3.393	3.114	0.946	0.953	21.492
3	$e^{-1/\omega_0}\Big[e^{-2/\omega_0}\{mx^2(\omega + 1)e^{-1/\omega_0} + x(0.842 + m\omega x)\} + x(\omega_0^{-1} + m\omega x)\Big]$	2.182	2.154	0.976	0.979	17.269
4	$m\omega\omega_0 x + x^2$	4.729	4.574	0.884	0.900	14.323
5	$m\omega\omega_0 + m + x^3 - 0.081 + \cos(m\omega) + \cos(x\omega) + \cos(m\omega_0)$	6.596	7.069	0.784	0.775	14.708
Noise level: $\sigma = 0.15$						
1	$x(\omega_0^2 - \omega_0) - x + m\omega x^2$	3.220	3.448	0.951	0.940	14.527
2	$\omega_0^2 + m\omega x^2$	5.192	4.815	0.872	0.885	14.160
3	$m\omega\omega_0 x + x^2 + \cos(mx) + 4\cos(2x)$	3.289	3.232	0.949	0.949	14.774
4	$m\omega\omega_0 x + x^2$	4.709	4.732	0.897	0.886	14.162
5	$\sin(0.678^2)\cos(\sin\omega)\Big[m\omega_0 x^2\cos\{\cos[\sin(m\sin\omega\cos(\sin\omega))]\} + \omega x^2\Big] + m\omega_0 x^2\cos(\sin\omega)$	2.464	2.428	0.972	0.974	16.442
Noise level: $\sigma = 0.25$						
1	$x^2 + \omega\omega_0(mx - 2x^{-3})$	4.011	4.487	0.923	0.906	15.331
2	$m\omega x^2 + 2\omega_0^2 - \sin^3(\omega_0) - \sin^3(m) - 6.454$	3.390	3.271	0.946	0.939	15.304
3	$\omega^2 + (x - 0.123)^2 + m\omega_0(x - 0.123)^2 - \dfrac{\omega^2 + m\omega_0(x - 0.123)^2 + \omega^2\omega_0}{me^x} - \dfrac{me^x}{\omega_0\omega^2}$	3.449	3.497	0.943	0.943	16.284
4	$x^2 + m\omega\omega_0 x$	4.892	4.577	0.888	0.894	14.208
5	$\cos(x + \omega) + 2\omega^2 + m\omega_0 x^2 + e^{-e^x} + 2\{\sin(-0.286)\}^{-1} + \exp\{\cos(mx)\}$	3.168	3.523	0.957	0.945	17.446

Table 22: `operon` results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2+\omega_0^2)x^2$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-3.546 + 0.531\left[0.770x \cdot 0.606\omega\{(0.569x)^2 + 1.004\omega + (0.756\omega_0)(-1.298x)(-1.391m)\} + (0.756\omega_0)(-1.298x)(-1.633m)\right]$	1.699	1.991	0.985	0.981	23.375
2	$-0.541 + 1.069\left[\frac{2.718x}{\sin\{\sin(0.749\omega_0)\}} \frac{0.833m}{\sin(0.749\omega)} \times \sin\{2.416\omega_0 \exp(-0.993)(-0.164\omega)\}(-1.230x)\right]$	1.544	1.533	0.989	0.989	29.189
3	$0.607 - 3.798\left[\sin(-1.594\omega_0)(0.318m) + 0.318m(0.650\omega_0)(-1.010x)\right] \times \left[\exp(-1.344\omega) + \sin(-1.483\omega_0) + (-0.663\omega)(-1.010x)\right]$	0.842	0.853	0.996	0.997	28.206
4	$-0.633 - 0.510\left[(-0.442\omega)(1.043x)\{(-2.308m)(-0.442\omega)\} + (0.320\omega)(2.116\omega_0)(1.045x)\{\cos(-1.177\omega) + 1.750x\}(-0.846m)\right]$	1.515	1.557	0.988	0.988	24.353
5	$0.485 - 2.089\left[e^{1.002\omega}(0.707)(-0.290x)e^{0.300\omega_0} + \sin\{e^{0.477\omega_0}\}\right] \times \{(0.434m)(2.718x)e^{-0.581\omega}\}$	0.672	0.704	0.998	0.998	29.436
Noise level: $\sigma = 0.15$						
1	$0.011 - 0.089\left[-0.066\omega_0 + \{(-1.043\omega_0 - 1.101\omega)(-1.911x)\}^2 + \{(-1.047\omega_0 + 0.990\omega)(-2.012x)\}^2\right](-0.333m)$	0.149	0.143	1.000	1.000	22.110
2	$-0.002 - 0.027(1.805m)\left[(3.824\omega_0)(0.509x)(-1.326x)(1.997\omega_0) - \{(1.649\omega)(-1.377x)\}^2\right]$	0.153	0.146	1.000	1.000	20.319
3	$-1.079 + 0.201(0.693\omega)\left[(1.777x)^2 + (-0.123\omega_0)\{(-1.754x)^2 - 0.693\omega\}(0.424\omega)\right] \times (0.060\omega + 0.961m)e^{0.559\omega_0}$	1.632	1.553	0.987	0.988	23.922
4	$-0.068 + 0.737\left[\{-2.053\omega(0.122x) - (-0.270\omega_0)^4 + 0.136\omega - (0.307x)^2\} \times (-0.874m)\{(1.210x)(1.305\omega_0) + 2.053\omega\}\right]$	1.099	1.097	0.994	0.994	22.964
5	$-0.077 + 22.227\left[(2.456 - 1.226\omega_0 - 1.949\omega)e^{(-0.193\omega_0)^2}(0.506x) \times (-0.148m)\right]\left[0.506x\cos\{\cos(e^{-0.887\omega})\}\right]$	0.853	0.812	0.997	0.997	28.447
Noise level: $\sigma = 0.25$						
1	$-2.085 - 0.156\left[\frac{1.014\omega}{1.123\omega_0} - 0.693\omega_0 - (-0.877x)(1.287\omega_0)\right] \times \left[-0.958x + 1.661m\{(-0.958x)(1.446\omega) - 1.414\omega_0\}\right]$	1.002	1.192	0.995	0.993	23.710
2	$-0.341 + 7.613\left[(0.493x)(0.313m)(0.493x) \times \{0.619\omega_0 - \sin(1.106\omega_0) - [(0.493x - 1.712\omega) + (0.592x - 0.493x)(-2.037\omega_0)]\}\right]$	0.879	0.946	0.996	0.995	25.662
3	$-1.330 + 1.419\left[\frac{(-2.871x)(-0.505\omega)}{(1.145\omega_0)^2} - \{(0.188\omega_0)(1.425) + 1.218\omega\}(0.336x)(-2.572x)(0.858m)(0.327\omega_0)\right]$	1.490	1.698	0.989	0.986	22.994
4	$0.248 - 0.281\left[\{(0.485x - 1.914x) - 3.145\omega\}(0.485x)(0.707m)(1.875\omega) - \{(-0.809\omega_0)(0.784x)\}^2(2.190m) + 2.074\omega(2.190m)\right]$	0.737	0.838	0.997	0.996	24.293
5	$-0.216 - 0.260\left[\{(-3.487x)(0.711\omega_0) - (0.116x)(1.614\omega)(5.721\omega)\} \times (0.116x)\{0.456 + 0.483 + (-0.238\omega_0)^2\}(6.283m)\right]$	0.778	0.724	0.997	0.998	22.522

Table 23: PySR results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$0.250\, mx^2(\omega^2 + \omega_0^2)$	0.000	0.000	1.000	1.000	66.755
2	$mx^2\{\omega + \omega_0 - 1.095 + \sin[\cos(\omega + 1.120)]\}$	0.792	0.791	0.997	0.997	85.093
3	$mx^2\Big\{0.328 - 0.529(\omega + \omega_0 - 0.164\omega\omega_0)\Big\}^2$	0.545	0.615	0.999	0.998	63.298
4	$mx^2\{\omega_0 + \sin(-0.780\omega_0) + \omega + \sin(-0.684\omega - 0.202)\}$	0.176	0.170	1.000	1.000	74.764
5	$mx^2\{\omega + 0.252\omega_0^2 - 0.925\}$	0.759	0.836	0.997	0.997	59.788
Noise level: $\sigma = 0.15$						
1	$0.250\, mx^2(\omega_0^2 + \omega^2)$	0.149	0.143	1.000	1.000	63.275
2	$m\Big[x^2\{\omega - 0.946 + 0.251\omega_0^2\} + \cos\{\omega\sin(\omega)\}\Big]$	0.440	0.401	0.999	0.999	73.962
3	$x\Big[mx\{e^{0.422\omega} + \omega_0 - 2.257\} + \cos\{\omega_0\sin(\omega_0)\}\Big]$	0.383	0.373	0.999	0.999	77.065
4	$mx^2\{-1.246\sin(\omega_0 + 0.687) + \omega + 0.628\}$	0.826	0.796	0.997	0.997	86.585
5	$mx\{x - 0.387\sin(\omega_0)\sin(\omega)\}\{\omega + \omega_0 - 1.648\}$	0.688	0.684	0.998	0.998	82.477
Noise level: $\sigma = 0.25$						
1	$\sin(\omega^2) + mx^2\{0.250\omega_0^2 + \omega - 0.936\}$	0.657	0.767	0.998	0.997	70.417
2	$m\Big[(\omega_0 - 2.003)^2 + x^2(\omega_0 + \omega - 1.897)\Big]$	0.939	0.987	0.996	0.994	64.486
3	$mx^2\{\omega - 57.814\cos(-0.093\omega_0) + 56.898\}$	0.772	0.728	0.997	0.998	81.822
4	$mx^2\{\omega_0 + 0.261(\omega - 1.998)^2 + \omega - 1.918\}$	0.798	0.768	0.997	0.997	65.787
5	$0.250\, mx^2(\omega^2 + \omega_0^2)$	0.255	0.243	1.000	1.000	57.047

Table 24: BMS results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set. Here, a_0 denotes a fitted constant learned by BMS.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$m\Big[a_0^2(a_0 + 1)^2x^2(a_0 + \omega_0 + \omega) - x\Big]$	1.286	1.268	0.992	0.992	202.400
2	$a_0^4mx^2(\omega + a_0)\{\exp[-\sin(\omega_0)\sin(\omega)] + \omega_0\}$	1.238	1.160	0.993	0.993	245.711
3	$mx^2\{2a_0 + \omega_0 + \omega\}$	1.103	1.143	0.994	0.994	241.044
4	$mx^2\{\omega + \omega_0 + 2a_0\}$	1.083	1.171	0.994	0.993	241.607
5	$\Big[\omega_0 + x\{2\omega + \omega_0 a_0\cos^2(a_0\omega)\}\Big] \times x\, a_0^6 m(\omega_0 + a_0) + \cos(\omega + x) + \cos(a_0 + m)$	1.286	1.355	0.992	0.992	248.992
Noise level: $\sigma = 0.15$						
1	$mx^2(a_0 + \omega_0 + \omega) + (\omega_0 + a_0)^2$	0.946	0.950	0.996	0.995	208.908

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Table 24: BMS results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set. Here, a_0 denotes a fitted constant learned by BMS *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
2	$\{(\omega+\omega_0)^2 + a_0 m\omega_0\}m\sin(a_0)x^2 + \{a_0(a_0+\omega^2)+\cos(\omega_0)\}^2$	0.819	0.566	0.997	0.998	253.143
3	$x\{a_0 + (a_0 + a_0^3 + \omega_0)mx + a_0^2\} + m\omega x^2$	1.103	1.221	0.994	0.993	247.547
4	$mx^2\{\cos(a_0+\omega_0)+\omega\} + x$	1.502	1.437	0.989	0.989	229.507
5	$mx^2(a_0+\omega_0+\omega) + \exp(a_0)$	1.106	1.027	0.994	0.995	224.604
Noise level: $\sigma = 0.25$						
1	$x\Big[mx(\omega+a_0)a_0(a_0+\omega_0) + \sin\{\omega_0+\omega_0+a_0+\sin(a_0\omega_0+\omega)\}\Big] + a_0^2\omega_0$	1.229	1.433	0.993	0.990	251.926
2	$m\omega_0 x^2 + \omega(\omega+a_0) - \{mx\cos(\omega) + \sin(x)\omega + \omega + x^2\}$	1.707	1.667	0.986	0.984	233.893
3	$a_0\Big[\{a_0 x(\exp[\sin(\sin a_0)]^3 + m)(\omega+a_0+\omega_0)\}^2 + x\cos\{\omega_0+\omega_0\sin(\omega)\}\Big]$	0.970	0.901	0.995	0.996	246.667
4	$m\{\omega x+\omega_0\}x^2 a_0^2\omega_0 e^{a_0} + mx(a_0+\omega)$	1.563	1.585	0.989	0.987	239.007
5	$-a_0 mx(\omega+\omega_0)\{a_0[a_0\omega\omega_0]^2 - x\}x(\omega+\omega_0) + a_0$	0.983	0.906	0.996	0.996	241.769

Table 25: BSR results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-8.483 + 3.470e^{\cos(m)} - 0.249e^m + 0.514\sin\{(\omega_0+\omega)^3\} + 4.413(m+\omega_0 m)$	11.979	12.930	0.268	0.191	190.583
2	$77.459 - 88.506\cos(e^{-\omega_0}) - 8.088\cos(x) - 12.942\cos(\omega_0) + 5.242mx$	7.071	6.964	0.763	0.764	195.174
3	$-5.975 + 0.200e^m + 1.005e^\omega + 0.009\dfrac{\cos(m)}{\sin(m^2)} + 7.105m$	12.327	12.784	0.238	0.263	200.450
4	$-59.296 + 18.947\{\omega_0 + e^{\cos(\omega_0)}\} + 1.090e^m + 8.011\omega - 1.337\sin(m^2)$	11.193	10.677	0.349	0.453	200.248
5	$-21.076 + 6.629\{\cos(\omega^{-1}) + \omega + x\} + 16.613\omega^{-1} - 11.968x^{-1} - 0.050(-x\omega)^3$	8.496	9.116	0.641	0.625	195.475
Noise level: $\sigma = 0.15$						
1	$-43.096 + 24.557x + 11.088mx^{-1} - 0.697\sin(\omega_0^2) + 0.111\sin^{-2}(m)$	9.273	9.513	0.591	0.546	198.356
2	$27.426 + 2.462\cos(\omega_0^2) + 67.775e^{-x} + 49.295\cos\{\sin(x)\} - 30.300e^{x^{-1}}$	10.210	10.241	0.505	0.478	191.194
3	$29.326 - 9.605e^{\sin(\omega_0)} - 21.107\cos(x) - 1.870\sin(\omega_0^2) - 11.162\cos(m)$	7.887	7.834	0.707	0.702	191.329
4	$18.336 + 0.667xe^\omega - 30.475\sin\{\sin(x)\} - 2.175\sin(m)^{-1} + 1.496e^m$	7.578	7.591	0.732	0.707	191.405

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Table 25: BSR results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
5	$-30.878 + 0.729\cos(\omega) - 5.018m^{-1} - 6.903(\omega + \omega_0)$ $+8.028(x + m + \omega\omega_0 + x)$	6.226	5.786	0.820	0.850	190.559
Noise level: $\sigma = 0.25$						
1	$2.171\times10^{-7} + 1.295\times10^{-4}e^{\omega x} - 3.129\times10^{-7}\{(-\omega_0 m)^3\}^3$ $+ 1.016\times10^{-6}e^{x^{-1}\omega_0\omega} - 6.119\times10^{-7}\{\cos(m) - x\}$	22.016	22.501	-1.321	-1.363	190.690
2	$1.211\times10^{-7} + 8.496\times10^{-8}\sin(\omega) - 6.173\times10^{-8}\cos(m)$ $+1.594\times10^{-7}e^{e^x} + 5.889\times10^{-7}\omega_0 x$	22.210	21.097	-1.300	-1.519	190.801
3	$45.128 - 16.608m^{-1} + 1.523\sin(-\omega_0^3) - 20.747\cos\{\cos(m)\}$ $-2.033\sin\{\cos(m^2)\}$	13.339	13.665	0.146	0.122	192.065
4	$-9.015 + 19.457\cos\{\cos(m)\} + 1.670\cos^2\{\sin(\omega^2)\} + 1.134m^3$ $-0.678\sin^2\{\cos(\omega_0) + \omega\}$	13.416	13.236	0.155	0.117	205.891
5	$21.838 - 32.250x^{-1} + 1.481\omega_0^2 + 0.577\exp\{\cos(xm) + x\} + 0.269\omega_0^3$	9.159	8.730	0.638	0.660	191.737

C.4 Results for I_50_26 (HOQN)

Table 26: DSR results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$\{\exp[\cos(t\omega + \exp(x_1)^{-1})]\}^2$	1.566	1.378	0.395	0.525	44.8
2	$\{\exp[\cos(\omega t)] - (t^{-1})^3\alpha^{-1}\}^2$	1.455	1.454	0.506	0.461	46.5
3	$x_1\exp[\cos(\omega t)]$	1.430	1.514	0.471	0.400	43.8
4	$\exp[\cos(\omega t)] + (\omega t)^{-1}$	1.654	1.778	0.311	0.209	45.0
5	$\alpha\exp[\cos\{-[-(\omega t)^{-1}]^{-1}\}]$	1.494	1.518	0.472	0.354	44.3
Noise level: $\sigma = 0.15$						
1	$x_1\exp[\omega\sin^2(\omega)\cos(\omega + t)]$	1.607	1.659	0.300	0.357	47.1
2	$\{\exp[\cos(t - \omega)]\}^2$	1.496	1.348	0.456	0.437	44.7
3	$\{\exp[\cos(-\omega - t)]\}^2$	1.481	1.393	0.457	0.612	44.8
4	$\{\sin[\exp\{\exp(t)^{-1}\}] + \cos(\omega t)\}^3$	1.462	1.526	0.449	0.290	46.4
5	$x_1\exp[\cos(-t\omega)]$	1.494	1.449	0.436	0.474	45.8
Noise level: $\sigma = 0.18$						
1	$\{\exp[\cos(-\omega t)]\}^2$	1.477	1.370	0.458	0.461	44.9
2	$t\alpha\cos^2(t\omega)$	1.485	1.568	0.438	0.463	47.7
3	$\exp[\cos\{t((\exp(x_1)t)^{-3} + \omega)\}]$	1.664	1.825	0.283	0.292	47.1
4	$\alpha\exp[\{\exp[\exp(x_1^3)] + \cos(t\omega)\}^3]$	1.577	1.497	0.333	0.518	45.3
5	$x_1\exp[\cos(t\omega)]$	1.447	1.482	0.431	0.429	45.0

Table 27: QLattice results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set. Row-specific constants are introduced inside the learned-expression column for compactness.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$0.530 + \frac{1.358}{A_1}$, $A_1 = -0.700\omega - 0.708t - 0.143x_1 + 88.824(1 - 0.203\omega - 0.201t)^2 + 4.059$	1.186	1.041	0.653	0.729	85.414
2	$-0.330t + 0.313x_1 + 1.256(0.105 - 0.211x_1) \times(3.152 - 2.507\alpha) - 0.186 + \frac{1.256}{A_2}$, $A_2 = 43.822B_2^2 + 0.199$, $B_2 = 0.353\omega + 0.162t - 0.404(0.505\omega + 0.743) \times e^{-0.403t} - 1$	1.043	1.058	0.747	0.714	85.057

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Table 27: QLattice results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set. Row-specific constants are introduced inside the learned-expression column for compactness *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	$-5.665(-0.118\alpha - 0.237)\exp(A_3 B_3) + 0.518$, $A_3 = -0.239\alpha - 0.966x_1 + (2.127 - 1.290\omega) \times(2.701 - 2.085t) - 1.151$, $B_3 = -0.042\alpha - 1.596\omega - 0.264(t + 0.491)^2 + 6.067$	1.281	1.367	0.576	0.511	87.361
4	$-0.114(3.120 - 4.436\alpha)(0.776x_1 + 1)^2 e^{-0.126\omega} A_4{}^{-1} +0.463$, $A_4 = (0.697\alpha + 0.174)B_4{}^2 + 0.357e^{-0.046\omega}$, $B_4 = 134.737\{-0.197t + (0.209\omega - 1)^2 + 0.269\}$	1.058	1.221	0.718	0.627	84.035
5	$2.335\exp(A_5 B_5 C_5) + 0.817$, $A_5 = 0.278\omega + 1.233$, $B_5 = 0.889(1 - 0.945\omega)^2 + (0.722t - 2.922) \times(0.229x_1 + 2.571) + 0.944$, $C_5 = -0.202\alpha - 2.093\omega - 2.157t - 0.230x_1 + 11.011$	1.113	1.140	0.707	0.636	84.498
Noise level: $\sigma = 0.15$						
1	$0.449(1.220t - 1.359)A_1 + 0.978$, $A_1 = 1.354\alpha + (3.986 - 2.332t)(-0.168x_1 - 0.238)e^{B_1} -3.123$, $B_1 = (2.781\omega - 5.056)\{-0.671(2.312t - 4.288)C_1 +1.066\}$, $C_1 = 1.269\omega + 0.738t - 0.206x_1 - 4.366$	1.176	1.103	0.625	0.715	83.788
2	$2.071(0.678t - 1.130)A_2 + 0.703$, $A_2 = 0.311x_1 + e^{B_2 C_2 D_2} - 1.348$, $B_2 = 1.960 - 1.127\omega$, $C_2 = 1.994\omega - 2.320$, $D_2 = \{1.173t + (0.916 - 0.013\alpha)(1.327\omega - 3.304) -3.591\}\{1.537t + (2.852 - 0.607x_1)(0.139\alpha - 0.772) -1.898\}$	1.173	1.090	0.665	0.632	89.515
3	$1.224\exp(A_3) + 0.767$, $A_3 = 0.738x_1 + 20.096(3.050 - 0.741\omega)(-0.192\omega -0.063)B_3{}^2 e^{-0.070\omega}$, $B_3 = -0.005\alpha - 0.457(1.221\omega - 1.627) \times(1.351t - 1.883) + 1$	1.138	1.207	0.679	0.709	97.835
4	$0.990\alpha + 5.697\exp(A_4) - 1.152$, $A_4 = 42.077(2.792 - 0.532t)(-0.026\omega - 2.553) \times (-0.553t - 1.258)(0.452x_1 - 1.699)(1 - 0.201\omega -0.197t)^2$	1.020	1.004	0.732	0.693	80.437
5	$4.614(0.098\alpha - 1)^2 A_5{}^2 + 0.023$, $A_5 = -0.386\alpha - 0.265x_1 + 0.849 - e^{-45.106 B_5{}^2}$, $B_5 = 0.418\omega + 0.268(1.747 - 0.661t)(-0.178\alpha -2.005) - 1$	0.944	0.956	0.775	0.771	79.372
Noise level: $\sigma = 0.18$						
1	$1.029\alpha + 5.267\exp(A_1 B_1 C_1) - 1.342$, $A_1 = -0.314\alpha - 0.862$, $B_1 = -0.024\alpha + (3.639 - 2.339t)(1.788\omega - 3.272) +2.433$, $C_1 = -1.586\omega - 2.031t + 9.235$	1.129	0.998	0.683	0.714	80.967

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Table 27: `QLattice` results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set. Row-specific constants are introduced inside the learned-expression column for compactness *(continued)*.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
2	$1.031\alpha - 2.827(0.227\omega - 2.374)\exp(A_2B_2C_2) - 1.257$, $A_2 = 0.461 - 2.930t$, $B_2 = 0.842\omega + 1.024t - 4.755$, $C_2 = 1.203t + (0.127 - 0.594t)(2.409 - 1.268\omega) - 3.949$	1.120	1.182	0.681	0.694	83.521
3	$0.331 - \frac{5.052}{(1.130x_1 - 4.640)A_3}$, $A_3 = (-1.929\omega - 0.248t + 4.478)B_3 + 101.679e^{-1.687t}$, $B_3 = 0.556\omega + 1.051t + 0.691(1 - 0.713t)^2 \times(\omega - 0.893)^2(0.147t - 2.259) - 0.897$	1.189	1.207	0.634	0.691	83.611
4	$0.994\alpha - 0.044t + 0.728x_1 + 1.085e^{A_4B_4} - 2.569$, $A_4 = -0.269\omega - 2.001$, $B_4 = 0.076\omega + 100.427(1 - 0.200\omega - 0.196t)^2 - 0.810$	1.018	0.980	0.723	0.793	95.180
5	$3.635(0.264 - 0.269t)(1.068x_1 - 0.401)A_5 + 0.368$, $A_5 = -0.429\alpha - 0.742\omega - 1.490e^{B_5} + 3.223$, $B_5 = 1.619(1 - 0.702\omega)^2(2.258 - 1.258t)(5.321 - 2.166\omega) \times(6.294 - 2.535t)$	1.234	1.300	0.586	0.561	83.509

Table 28: `SISSO` ++ results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-1.698 + 0.034(\alpha + t + \omega + x_1)(\alpha x_1 + \alpha) + 3.251\cos^2(t\omega) + 0.196\cos(t\omega)(tx_1\omega)$	0.585	0.585	0.914	0.903	26.002
2	$-1.476 + 0.044\{(\alpha + t)(\alpha x_1) + (\alpha + \omega)(\alpha x_1)\} + 3.232\cos^2(t\omega) + 0.196\cos(t\omega)(\omega x_1 t)$	0.589	0.595	0.910	0.910	27.093
3	$-2.374 + 0.306\{\alpha x_1 + \alpha + x_1 + \cos(t + \omega)\} + 3.338\cos^2(t\omega) + 0.193\cos(t\omega)(\omega x_1 t)$	0.585	0.566	0.904	0.910	25.880
4	$-1.755 + 0.035(\alpha + t + \alpha + \omega)(\alpha x_1 + x_1) + 3.268\cos^2(t\omega) + 0.191\cos(t\omega)(tx_1\omega)$	0.596	0.576	0.905	0.910	26.835
5	$-2.511 + 0.211\{\alpha + t + \alpha x_1 + \alpha x_1 + \cos(t)\} + 3.347\cos^2(t\omega) + 0.206\cos(t\omega)(tx_1\omega)$	0.593	0.604	0.910	0.907	26.050
Noise level: $\sigma = 0.15$						
1	$-1.704 + 0.034(\alpha + \omega + t + x_1)(\alpha x_1 + \alpha) + 3.249\cos^2(t\omega) + 0.195\cos(t\omega)(\omega x_1 t)$	0.606	0.608	0.908	0.897	25.956
2	$-1.491 + 0.051(\alpha + t + t + \omega)(\alpha x_1 + \cos t) + 3.242\cos^2(t\omega) + 0.200\cos(t\omega)(t\omega x_1)$	0.604	0.600	0.906	0.908	27.228
3	$-2.367 + 0.306\{\alpha x_1 + \alpha + x_1 + \cos(t + \omega)\} + 3.341\cos^2(t\omega) + 0.192\cos(t\omega)(\omega x_1 t)$	0.605	0.592	0.898	0.902	25.630

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Table 28: SISSO++ results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
4	$-1.759 + 0.035(\alpha + t + \alpha + \omega)(\alpha x_1 + x_1) + 3.267\cos^2(t\omega) + 0.191\cos(t\omega)(tx_1\omega)$	0.613	0.595	0.900	0.904	26.012
5	$-2.518 + 0.213\{\alpha + t + \alpha x_1 + \alpha x_1 + \cos(t)\} + 3.331\cos^2(t\omega) + 0.206\cos(t\omega)(t\omega x_1)$	0.614	0.622	0.904	0.902	25.768
Noise level: $\sigma = 0.18$						
1	$-1.705 + 0.034(\alpha x_1 + \alpha)(\alpha + x_1 + t + \omega) + 3.249\cos^2(t\omega) + 0.195\cos(t\omega)(\omega x_1 t)$	0.615	0.617	0.905	0.894	25.894
2	$-1.491 + 0.051(\alpha + t + t + \omega)(\alpha x_1 + \cos t) + 3.242\cos^2(t\omega) + 0.200\cos(t\omega)(tx_1\omega)$	0.611	0.607	0.904	0.907	25.341
3	$-2.365 + 0.306\{\alpha x_1 + \alpha + x_1 + \cos(t + \omega)\} + 3.341\cos^2(t\omega) + 0.192\cos(t\omega)(tx_1\omega)$	0.613	0.602	0.895	0.899	25.675
4	$-1.760 + 0.035(\alpha + t + \alpha + \omega)(\alpha x_1 + x_1) + 3.267\cos^2(t\omega) + 0.191\cos(t\omega)(\omega x_1 t)$	0.621	0.603	0.898	0.902	25.893
5	$-2.519 + 0.213\{\alpha + t + \alpha x_1 + \alpha x_1 + \cos(t)\} + 3.328\cos^2(t\omega) + 0.206\cos(t\omega)(tx_1\omega)$	0.622	0.630	0.901	0.900	27.191

Table 29: gplearn results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$\{\alpha\cos^2(\omega t) + x_1[0.145\alpha x_1\cos^2(\omega t) + 0.145\omega\{0.145t + 0.145\cos[0.145\omega t\{\alpha x_1\cos^2(\omega t) + \sin(\omega t)\}]\} \times \{x_1\cos^2(\omega t) + \cos^2(t)\}\cos(\omega t) + 0.145t + 0.021x_1 + 0.290\cos(\omega t)]\}\cos\{\sin(\omega t)\} + \cos(\omega t)$	0.350	0.326	0.970	0.973	20.960
2	$\{0.125\alpha + 0.125\omega^2(0.488t - 0.488\sin\omega) \times \cos[\omega(0.488t - 0.488\sin\omega) - \cos\{\omega\cos(\sin t)\} + \cos(\sin t)] \times \cos[-t + \sin\omega + \cos\{\omega\cos(\sin t)\} + \cos\{-t + \sin\omega + \cos x_1 + \cos[\omega(0.488t - 0.488\sin\omega)\cos\{-t + \sin\omega + \cos x_1 + \cos(\omega\cos(\sin t))\}]\}]\} \times \{\omega + 0.125tx_1[-\omega\cos\alpha + \omega - \cos\{\cos[t - \sin\omega - \cos(\omega\cos(\sin t)) + \cos(\omega + t + \sin\{0.016x_1^2[\alpha - \omega(t - \sin\omega)\sin\omega \times \cos(-t + \sin\omega + \cos x_1 + \cos(\omega\cos(\sin t)))]^2 - \cos(\sin t)\})]\}] + x_1\}$	1.172	1.269	0.680	0.589	20.866
3	$\alpha x_1\cos^2(\omega t) + 2\cos(\omega t)$	0.413	0.419	0.956	0.954	16.811
4	$\alpha\{e^{\sin(e^{\cos[\sin(e^{\cos\alpha})]}\cos(\omega t))} + \cos(\sin x_1)\}\cos^2(\omega t)$	0.730	0.719	0.866	0.870	19.996
5	$0.281\alpha\omega t x_1\cos^2(\omega t) \times \cos\{\sin[\cos(\sin\{\cos[\sin(\omega t)\cos(\sin\{\sin[\sin(0.281\alpha\omega t x_1 \times \cos^2(\omega t)\cos(\sin(\omega t))\cos(\sin(\cos(\omega t))))]\}) + \sin(\cos(\omega t))]\})]\} + \sin(\cos(\omega t))$	0.411	0.389	0.960	0.958	17.816
Noise level: $\sigma = 0.15$						

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Table 29: `gplearn` results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
1	$0.391x_1\{2\alpha\cos^2(\omega t) - \sin(0.868x_1 - \cos(\omega t)) + \cos^4(\omega t) + 3\cos(\omega t) + \cos(\cos x_1)\}$	0.319	0.324	0.972	0.976	19.991
2	$\alpha + \sin\{(\omega + t)\cos(\omega + t)\} + \cos\{\omega + t + \sin[(\alpha + \omega + \cos\{\alpha + \sin(\omega + t)\})\cos(\omega + t)]\} + \cos\{t + x_1 + \sin[(\alpha + \omega)\cos(\omega + t)]\} + \cos\{\sin(\omega t)\}$	0.889	0.790	0.808	0.807	18.550
3	$\alpha + 0.284x_1 + \sin^2(1.004/t) + \sin^2\{[\alpha - \sin(\cos\omega) + \cos^2 t - 0.615] \times \sin^2(\cos\omega)\} + \sin^2(\cos\omega) + \cos^2 t - 0.615 - \frac{1.004}{x_1} - \frac{2.004}{t}$	1.593	1.864	0.371	0.306	18.179
4	$2\alpha\cos^2(\omega t) + 2\cos(\omega t)$	0.725	0.707	0.865	0.847	16.677
5	$-0.395\{-0.395\omega\cos\omega + [-0.395\omega\cos\omega + t + \{-0.395\omega\cos\omega + (0.823\alpha + t - 0.665e^{1/t})\cos(\sin x_1)\} \times \cos\{\sin(1.992t + \sin[e^{-e^{-(0.992t - 0.665e^{1.040/\cos\omega} + 1.504e^{-1/t})^2}}] + \cos^2\omega)\}\} - 0.395[-0.395\omega\cos\omega + \{0.823\alpha + t + [-0.395\omega\cos\omega + (0.823\alpha + 1.992t)\cos(\sin x_1)] \times \cos\{\sin(1.992t + \sin[e^{-e^{-(-0.395\omega\cos\omega + 0.992t - 0.665e^{1/t})^2}}] + \cos^2\omega)\}\} + 0.992\cos\omega]\cos(\sin x_1)\} \times \cos\omega\cos\{\sin(1.992t + \sin[e^{-e^{-0.992t}}] + \cos^2\omega)\} \times \cos(\sin x_1)\cos\omega\cos\{\sin(1.992t + \sin[e^{-e^{-0.992t}}] + \cos^2\omega)\} + (0.823\alpha + t - 0.665e^{1/t})\cos(\sin x_1)\} \times \cos\{\sin[-0.395\omega\cos\omega + 1.992t + \cos^2\omega]\} \times \cos\{\sin[-0.395\omega\cos\omega + 1.992t + \cos^2\omega - e^{-e^{1/t}}/\{\alpha\cos(\sin x_1)\}]\}$	1.214	1.192	0.628	0.643	29.848
Noise level: $\sigma = 0.18$						
1	$x_1\Big\{\alpha + \sin\Big[(\alpha + \cos\omega)\cos^3(\omega t) + \sin^3\{\sin^3[(\alpha + \cos(\omega + 0.528x_1))\sin^2((\alpha + \cos(\omega + 0.528x_1)) \times \cos^2(\omega t) - \omega^{-3}) - t^{-3}] + \cos^2 x_1\cos(\omega t)\}\Big]\Big\}\cos^2(\omega t) + \cos(\omega t)$	0.524	0.533	0.932	0.919	20.623
2	$\alpha + \cos(\omega t) + 2\cos(2\omega t) + \cos\{\omega t\cos[e^{-\omega\{\alpha + t + 2\cos(\omega t) + 2\cos(2\omega t)\}}]\}$	0.829	0.904	0.825	0.821	18.325
3	$0.434\alpha^2\cos^2(\omega t) - \omega\{-0.032\alpha^2 x_1^2\cos^2(\omega t) + 0.226\alpha - 0.259x_1 + 0.470\cos(\omega^{-1})\cos(\omega - 0.409) - 0.475\cos(\omega t)\} \times \cos\Big\{\sin\Big(\sin\big[\omega\cos(\omega + t)\cos(\sin t) - \cos\{\cos[t + \{0.434\alpha^2\cos^2(\omega t) - \omega(-0.249\alpha - 0.470\cos(\omega - 0.409)\cos(\cos(\omega + t)) - 0.475\cos(t + \cos(t + e^{-\omega - t}))) \times \cos(\sin[\sin\{\omega\cos(0.259\alpha x_1)\}])\}^{-1}]\}\big]\Big)\Big\}$	0.695	0.775	0.875	0.872	24.314
4	$0.376\alpha\omega^2 e^{-\cos[t\cos\{\sin(\omega t)\}]}\sin(0.427x_1) \times \sin\{\sin[0.376\omega t\sin(e^{1/x_1}\sin\{\sin[\cos(x_1^{-1})]\})]\}\cos\{\sin(\omega t)\}$	0.820	0.762	0.820	0.875	18.661

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Table 29: gplearn results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
5	$\alpha + \Big[\{0.263\omega + 0.263t + 0.263\cos^2(0.263\omega + 0.263t + 0.263\sin[\cos(0.572\omega^2)]) + 0.263\cos\{\cos[\cos(0.572\omega^2\cos t)]\} - 0.263x_1^{-1}\}^3 + \sin(\omega^2\cos t) - \cos t - \cos^2(0.572x_1) - \cos(0.572\omega^2\cos t) - \cos^2\{0.572\cos[(\omega + \cos\{(0.263\omega + 0.263t + 0.263\sin(\omega^2\cos t) - 0.076)^3\}) \times (-0.150\omega^2\cos t + 0.263t - 0.076)\cos(0.572\omega^2\cos t)] + 0.572x_1^{-1}\}\Big] \times \sin\{\sin(\omega^2\cos t)\} - \sin x_1 - \cos(0.572\omega^2\cos t)$	1.062	1.071	0.693	0.702	22.754

Table 30: operon results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$0.232\omega t\Big[0.726\alpha x_1\cos\{\sin(2.580\omega)\} + 1.011x_1\sin(4.924t)\sin\{\sin(3.889\omega)\}\Big] - 0.500$	1.117	1.076	0.692	0.710	30.061
2	$-0.972\Big[0.248\omega t(-0.693\alpha - 0.910x_1) + 1.537\Big] \times \Big[0.224\alpha + 0.118x_1 - \sin(1.767\omega t)\Big] + 0.372$	0.520	0.580	0.937	0.914	22.804
3	$0.003\omega^2(2.752\alpha + 1.266tx_1)^2 \times \Big[0.211t\sin(1.773\omega t) - \sin(0.512\omega + 0.729)\Big]^2 - 0.253$	0.639	0.657	0.895	0.887	25.191
4	$0.134\alpha x_1(1.540 - 1.160\omega t) \times \Big[-\sin\{1.094\omega(0.157 - 1.684t)\} - 0.994\Big] - 0.165$	0.531	0.606	0.929	0.908	22.250
5	$-0.049\alpha t x_1\Big[1.613\omega t\sin(1.773\omega t) - 5.275\omega + 0.792x_1\Big] - 0.527$	0.604	0.549	0.914	0.915	23.483
Noise level: $\sigma = 0.15$						
1	$-0.442\omega t\Big[-0.302\alpha x_1 - \sin(1.837\omega t - 28.722)\Big] - 0.692$	0.804	0.816	0.825	0.844	24.823
2	$1.297\Big[0.653\alpha + \sin\{\sin(1.223\omega t)\}\Big] \times \Big[0.277\omega x_1 + \sin\{\sin(2.304\omega t)\}\Big] - 0.009$	0.769	0.704	0.856	0.847	29.172
3	$4.241\Big[0.236\alpha x_1 + \dfrac{0.233x_1}{\cos(1.013\omega t)}\Big]\cos^2(1.013\omega t) - 0.007$	0.208	0.205	0.989	0.992	24.984
4	$2.148\Big[0.761\alpha\sin\{\sin[\sin(0.037\omega^2t^2)]\} - 0.047x_1\Big] \times \Big[-0.693x_1\sin(1.748\omega t) + 0.698x_1\Big] - 0.224$	0.483	0.458	0.940	0.936	25.884
5	$0.169x_1\Big[5.934\alpha + \dfrac{5.908}{\cos(1.006\omega t)}\Big]\cos^2(1.003\omega t) - 0.011$	0.155	0.143	0.994	0.995	26.387
Noise level: $\sigma = 0.18$						
1	$2.183x_1\Big[0.434\omega - e^{-0.681\alpha}\Big]\Big[0.441t - e^{-0.681\alpha}\Big] \times \Big[e^{e^{-1.557\alpha}} + \cos(2.018\omega t)\Big] - 0.173$	0.633	0.591	0.901	0.900	29.021
2	$4.712\Big[0.306\omega x_1\cos(1.230t) + \sin(1.091t)\Big] \times \sin^2\{0.449t^2\cos^2(0.979\omega)\}\cos(1.299t) + 0.853$	1.259	1.324	0.596	0.617	29.952

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Table 30: `operon` results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	$0.576\alpha\omega x_1\left[0.126t + \cos(1.016\omega t)\right] \times \sin(0.379t)\cos(1.016\omega t) - 0.075$	0.511	0.503	0.932	0.946	28.517
4	$0.120\alpha\omega t x_1 e^{-\sin\{\sin(1.757\omega t)\}}\sin(0.309\omega t) - 0.511$	0.625	0.631	0.895	0.914	28.719
5	$-0.774x_1\left[1.383\alpha - \frac{1.481\alpha}{\cos\{\cos(1.003\omega t)\}} + 0.354\right] + 1.954\cos(1.001\omega t) + 0.534$	0.458	0.447	0.943	0.948	25.782

Table 31: `PySR` results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$x_1\{\cos(\omega t) + \alpha\cos^2(\omega t)\}$	0.000	0.000	1.000	1.000	100.024
2	$x_1\{\cos(\omega t) + \alpha\cos^2(\omega t)\}$	0.000	0.000	1.000	1.000	120.296
3	$x_1\cos(\omega t)\{\alpha\cos(\omega t) + 1.000\}$	0.000	0.000	1.000	1.000	127.832
4	$x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.000	0.000	1.000	1.000	108.169
5	$0.247\left[\{x_1 + \alpha\}\{\cos(\omega t) + 0.119\} + \alpha^{-1}\right]^2 - 0.310$	0.219	0.229	0.989	0.985	87.682
Noise level: $\sigma = 0.15$						
1	$x_1\{\cos(\omega t) + \alpha\cos^2(\omega t)\}$	0.149	0.142	0.994	0.995	99.897
2	$x_1\{\cos(\omega t) + \alpha\cos^2(\omega t)\}$	0.153	0.146	0.994	0.993	102.262
3	$x_1\cos(\omega t)\{\alpha\cos(\omega t) + 0.999\}$	0.152	0.156	0.994	0.995	105.729
4	$x_1\{\cos(\omega t) + \alpha\cos^2(\omega t)\}$	0.152	0.148	0.994	0.993	107.485
5	$x_1\{\cos(\omega t) + \alpha\cos^2(\omega t)\}$	0.149	0.139	0.994	0.995	101.043
Noise level: $\sigma = 0.18$						
1	$0.242\left[\{x_1 + \alpha\}\left\{\frac{1}{2\alpha} + \cos(\omega t)\right\}\right]^2 - 0.283$	0.258	0.246	0.983	0.983	92.106
2	$x_1\{\cos(\omega t) + \alpha\cos^2(\omega t)\}$	0.181	0.173	0.992	0.993	98.596
3	$x_1\cos(\omega t)\{\alpha\cos(\omega t) + 1.003\}$	0.181	0.179	0.991	0.993	119.619
4	$x_1\{\cos(\omega t) + \alpha\cos^2(\omega t)\}$	0.178	0.190	0.992	0.992	99.745
5	$x_1\left[\{\{\cos(\omega t) - 0.232\}(0.385\alpha + 0.628) + 0.687\}^2 - 0.139\right]$	0.210	0.208	0.988	0.989	91.471

Table 32: BMS results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set. Here, a_0 denotes a fitted constant learned by BMS.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-\omega^2 a_0^2(x_1 + a_0 - x_1 t)\{\alpha a_0 + \alpha\cos[\sin(\omega t)]\}$	0.785	0.827	0.848	0.829	268.458
2	$\{2\alpha + \omega t + \sin\alpha + x_1 + a_0^3\}\cos[\sin(\omega t)] + 2a_0$	0.999	1.053	0.767	0.717	280.043
3	$\sin\Big(\omega\big[a_0 t^2 + a_0\sin(2a_0)(t + a_0)^3(\alpha + \omega + x_1) \times \cos\{-2\omega t\}\big]\Big)\{a_0 + \sin(\alpha + 2\omega) + \alpha x_1 + x_1\}$	0.857	0.802	0.810	0.832	231.521
4	$a_0^2\sin(a_0)\,\omega t\, x_1 \times \{-\cos\alpha + \sin(2\omega t + a_0 x_1 + a_0^2) + \sin(a_0^3) + a_0\}$	0.978	0.970	0.759	0.764	281.771
5	$\exp\Big[a_0^2\{3\omega\}\cos(\omega t)\cos t\sin(\omega + t) \times \{x_1 + a_0[\sin(\sin a_0) + a_0]\,[\alpha(\omega^2 + \alpha + a_0) + \omega]\}\Big]^3$	1.089	1.122	0.719	0.647	308.486
Noise level: $\sigma = 0.15$						
1	$a_0(t + \omega)\Big[-a_0^2(t^2 + \omega x_1\alpha + a_0\omega x_1\sin\{a_0\alpha(t^2 - x_1)\}) \times \cos(\omega t)\Big]^2$	0.739	0.701	0.852	0.885	286.765
2	$\omega\Big[t x_1\{\cos(a_0^3) + \sin(2a_0)\}\{\cos(\omega t) + a_0 + \cos(2\omega t) + a_0^3 + \alpha\} + \alpha\exp\{-\exp[a_0 + \sin(a_0) + \exp(t + a_0)]\}\Big]$	0.693	0.641	0.883	0.873	326.618
3	$[\{t + \cos[\sin(\omega t)](\alpha + \omega + \alpha + t) + x_1 + \omega\}\sin a_0 + a_0]^2$	0.819	0.893	0.834	0.841	302.188
4	$\omega a_0(\alpha + t)\Big[\alpha + x_1\{\sin a_0 + \sin[\sin(a_0^3)] + 2\alpha\} + \omega e^t\{a_0\omega e^{x_1}\cos(\omega t) + a_0^2 e^{-a_0}\}\Big]$	1.242	1.250	0.602	0.524	247.684
5	$-\{\omega + a_0\omega(x_1 + \omega t)\{a_0 + \cos(2\omega t)\} + a_0\sin(-\alpha)(2\alpha)\} \times \cos[\sin\{\sin(\omega t)\}]\alpha(-a_0 x_1 t)$	0.764	0.808	0.852	0.836	309.985
Noise level: $\sigma = 0.18$						
1	$a_0 x_1(t + \alpha)\Big[a_0 + \omega(\omega + a_0)(\alpha - a_0 x_1)\Big] \times \Big[a_0\{a_0^3\alpha a_0 - \omega\alpha t^3\}\cos^2\{\cos(\omega t)\} + t\Big]$	1.273	1.239	0.597	0.559	290.545
2	$a_0\alpha\omega t x_1\{a_0 + \cos(2\omega t) + \cos[a_0 + \sin(\omega + \cos(-x_1\sin(a_0 + a_0\cos a_0 + a_0 x_1 + t)))]\}$	0.727	0.657	0.865	0.906	331.377
3	$\Big[a_0\cos(e^{\sin t}) + \{\cos(\omega t) + a_0\}e^{\cos(a_0 x_1)} \times x_1(a_0 + a_0\sin a_0)a_0\{[\cos(-\omega\sin a_0) + \cos a_0] \times [\sin x_1 + \alpha] + \omega\} + a_0\Big]^2$	0.521	0.521	0.930	0.942	317.026
4	$\Big[x_1 a_0\{a_0 e^{e^{\cos[\omega(t + a_0 x_1)]}} e^{\omega}(\sin a_0 + a_0)(a_0 + e^t)\}^3 + \alpha - a_0\Big] \times \omega t a_0\{(\cos t + x_1)\alpha + t\}$	1.139	0.956	0.652	0.803	269.761
5	$x_1\sin^2(a_0)\Big[\sin(\cos\omega) + \{\alpha + a_0 + \alpha\omega^2(2a_0\cos(\omega t))^2\}t\Big] + a_0 x_1$	0.774	0.854	0.837	0.811	291.594

Table 33: BSR results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-1.947 + 0.754e^{\cos x_1} - 0.055e^{x_1} - 0.017\sin\{(t+\omega)^3\} + 0.562x_1(1+t)$	1.842	1.883	0.163	0.112	191.627
2	$35.478 - 37.326\cos(e^{-t}) - 0.388\cos\alpha - 2.534\cos t + 0.457\alpha x_1$	1.734	1.727	0.299	0.239	190.812
3	$-0.554 + 0.051e^{x_1} + 0.111e^{\omega} + 0.001\dfrac{\cos x_1}{\sin(x_1^2)} + 0.309x_1$	1.837	1.853	0.127	0.101	193.801
4	$-7.632 + 2.206\{t + e^{\cos t}\} + 0.089e^{x_1} + 1.114\omega - 0.213\sin(x_1^2)$	1.759	1.863	0.220	0.131	190.078
5	$-1.243 + 0.698\{\cos(\omega^{-1}) + \omega + \alpha\} - 0.315\omega^{-1} - 0.986\alpha^{-1} + 0.002(\alpha\omega)^3$	1.806	1.676	0.229	0.213	191.675
Noise level: $\sigma = 0.15$						
1	$-2.506 + 1.531\alpha + 0.871\dfrac{x_1}{\alpha} - 0.303\sin(t^2) + 0.021\sin^{-2}(x_1)$	1.775	1.960	0.146	0.102	193.315
2	$3.383 + 0.669\cos(t^2) + 3.919e^{-\alpha} + 2.082\cos(\sin\alpha) - 2.159e^{1/\alpha}$	1.886	1.719	0.134	0.085	191.531
3	$4.018 - 1.416e^{\sin t} - 1.264\cos\alpha - 0.447\sin(t^2) - 0.969\cos x_1$	1.711	1.962	0.275	0.231	191.691
4	$0.553 + 0.065\alpha e^{\omega} - 1.222\sin(\sin\alpha) - 0.080\sin^{-1}(x_1) + 0.093e^{x_1}$	1.747	1.677	0.214	0.143	191.321
5	$-2.954 - 0.051\cos\omega - 0.922x_1^{-1} - 0.002(\omega + t) + 0.502(x_1 + \omega t + 2\alpha)$	1.621	1.622	0.336	0.340	190.397
Noise level: $\sigma = 0.18$						
1	$2.020\times10^{-8} + 7.707\times10^{-6}e^{\omega\alpha} - 2.830\times10^{-8}(-tx_1)^9 + 5.019\times10^{-7}e^{\omega t/\alpha} - 5.383\times10^{-8}\{\cos x_1 - \alpha\}$	2.357	2.195	-0.379	-0.384	190.049
2	$1.025\times10^{-8} + 6.823\times10^{-9}\sin\omega - 5.033\times10^{-9}\cos x_1 + 9.683\times10^{-9}e^{e^{\alpha}} + 4.875\times10^{-8}t\alpha$	2.449	2.642	-0.527	-0.526	191.367
3	$3.473 - 1.635x_1^{-1} - 0.024\sin(-t^3) - 1.294\cos(\cos x_1) - 0.054\sin\{\cos(x_1^2)\}$	1.915	2.082	0.050	0.080	192.903
4	$-0.640 + 1.373\cos(\cos x_1) - 0.346\cos^2\{\sin(\omega^2)\} + 0.085x_1^3 + 0.573\sin^2\{\cos t + \omega\}$	1.871	2.064	0.063	0.083	204.162
5	$0.116 - 1.696\alpha^{-1} + 0.965t^2 + 0.037e^{\cos(\alpha x_1)+\alpha} - 0.227t^3$	1.706	1.770	0.209	0.186	191.171

C.5 Results for II_36_38 (FMMM)

Table 34: DSR results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$T^{-1}\exp\{k_b^{-1} + \exp[-\text{mom}^{-1}]\}$	0.731	0.842	0.561	0.537	41.900
2	$\exp\{\sin[\text{mom}(Tk_b)^{-1}]\}$	0.794	0.731	0.456	0.450	40.600
3	$\text{mom}\{T^{-1} + \sin[(k_b^2)^{-1}]\}$	0.668	0.650	0.625	0.635	41.300
4	$(\alpha + \text{mom mom})k_b^{-1}T^{-1}$	0.710	0.678	0.570	0.580	41.100
5	$H\{k_b\cos(T^{-1})\}^{-1}$	0.949	0.808	0.417	0.450	42.200
Noise level: $\sigma = 0.15$						
1	$\sin(T^{-1})\{H + \text{mom}\sin(k_b^{-1})\}$	0.778	0.857	0.544	0.476	46.500
2	$\{\text{mom}^{-1}\{k_b^{-1}\exp[-(-T)^{-1}]\}^{-1}\}^{-1}$	0.710	0.699	0.566	0.582	58.100
3	$\{\text{mom}^2 + \sin(\text{mom})\}(k_b T)^{-1}$	0.655	0.634	0.645	0.653	50.600
4	$-[-\{k_b^{-1} + \exp(\sin[(-\exp\{(\exp k_b)^{-1}\})^{-1}]\epsilon)\}\text{mom}]$	0.837	0.877	0.415	0.394	84.700
5	$\text{mom} + \cos[-\{(-k_b^3)^{-1} + T\}]$	0.851	0.739	0.477	0.538	56.700
Noise level: $\sigma = 0.20$						
1	$(\alpha + c + \text{mom}^2)(T + k_b T)^{-1}$	0.714	0.677	0.590	0.648	57.800
2	$\text{mom} - \{k_b H^{-1} - T^{-1}\}$	0.750	0.759	0.516	0.524	82.300
3	$\exp(k_b^{-1})T^{-1}\text{mom}$	0.721	0.730	0.591	0.615	54.600
4	$\text{mom} + \sin(T + k_b)$	0.840	0.863	0.466	0.433	71.800
5	$\exp\{\text{mom}(k_b T)^{-1}\}$	0.718	0.732	0.579	0.580	63.400

Table 35: QLattice results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$0.207\alpha + 0.257\Big[0.648H + (0.366T - 2.270)(0.065\epsilon - 0.701) \times \{-0.869T - 0.578c - 1.064k_b + 0.897\text{mom} + (0.868 - 0.162\epsilon) \times (0.483M - 0.091) + 2.964\} + 0.238\Big]^2 + 0.132$	0.270	0.262	0.940	0.955	89.144
2	$0.662\exp\Big[(0.527 - 0.058\text{mom})(0.065\epsilon - 1.061) \times \{-0.395T + (1.643 - 0.424k_b)(-0.077M - 0.555) \times (-0.141\alpha - 1.002) + 1.349\}\Big](-0.471H + 0.373c - 1.003\text{mom} - 0.293) - 0.725$	0.278	0.342	0.933	0.879	116.290

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Table 35: `QLattice` results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom}H}{k_bT} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T}H$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
3	$13.534(1.128-0.227T)(2.260-0.422T) \times \exp\Big[0.151M+(0.134k_b+0.226)(0.879H+0.364\alpha-0.709c -0.388\epsilon+1.255\text{mom}-8.372)\Big]+0.239$	0.274	0.299	0.937	0.923	133.543
4	$0.966\exp\Big[-0.171\epsilon-0.585k_b+0.494\text{mom}+(1.393-0.013T) \times(0.253H-0.448)+0.223(-0.391M-0.187) \times(1.435c-4.172)+2.201(0.283T-1)^2\Big] +0.038$	0.277	0.263	0.934	0.937	108.359
5	$-3.622(0.056\alpha+0.209)(0.480k_b-1.923)(0.167\text{mom}-0.040) \times\Big[-0.520k_b+(0.672T-2.312)\{1.485T+(0.881M-0.402) \times(0.669c-1.872)-2.753\}+2.860\Big]\exp(0.383H-0.191\epsilon)+0.250$	0.312	0.264	0.937	0.941	121.090
Noise level: $\sigma=0.15$						
1	$-0.195k_b-1.809\Big[-0.715H+(0.124-0.546\alpha)(1.655-0.579c) \times(0.553M-0.110)+0.207\Big] \times(-0.680\epsilon+2.558\text{mom}+0.449)\exp(-0.653T)\exp(-0.537k_b)+0.649$	0.284	0.347	0.939	0.914	112.151
2	$6.936\exp\Big[0.170\alpha+(0.173M-0.176\epsilon-1.618) \times\{-0.247H+0.655T+(0.074T+0.201)(-0.711T+1.289k_b -0.911)+(-0.110c-0.202)(0.897\text{mom}-3.770)-0.341\}\Big]+0.280$	0.309	0.291	0.918	0.928	118.903
3	$1.498(2.233-0.538k_b)(0.130T-0.498) \times(0.141\epsilon-1.200)(1.421\text{mom}-0.068) \times\Big[0.566H+(0.234-0.045k_b)(3.692-1.241c) \times(1.074M+0.961\alpha-1.967)+0.011\Big]+0.012$	0.307	0.310	0.922	0.917	110.390
4	$2.162(0.055H+0.055)(0.401k_b-1.489)(-0.630T+1.209\text{mom}+1.071) \times\Big[0.901T-0.171\epsilon+2.238(1.291c-3.960)(0.421M+0.412\alpha-0.628) \times\exp(-0.699\epsilon)-3.276\Big]+0.107$	0.342	0.341	0.902	0.909	104.237
5	$3.706\Big[-0.173H-0.074M-0.069\alpha+0.078\epsilon +0.635(0.104-0.340\text{mom})(1.711-0.346c) +0.635(3.327-0.741k_b)(0.201T-0.920)+1\Big]^2+0.577$	0.354	0.317	0.910	0.915	105.278
Noise level: $\sigma=0.20$						
1	$5.648\Big[-0.136H+0.215T+0.117c+0.210k_b +0.349(-0.088M-0.260)\{1.171\text{mom}+(2.583-0.729\epsilon) \times(0.307\alpha-0.088)-1.731\}-1\Big]^2+0.566$	0.348	0.316	0.903	0.923	162.774
2	$4.933(1-0.066\epsilon)^2(0.147H-2.571)(1.322\text{mom}-0.333) \times(-0.800H-0.260\alpha+0.477)\exp(-0.337c)\exp(-0.666T-0.646k_b) +0.212$	0.339	0.308	0.901	0.922	129.613
3	$28.951\exp\Big[0.145M-0.641T+(0.847-0.110H) \times(-0.293\epsilon-1.027k_b+0.894\text{mom}-3.160) +(-0.171\alpha-0.065)(0.718c-2.117)\Big]+0.156$	0.308	0.344	0.925	0.915	121.915
4	$-11.655(0.108-0.398\text{mom})(0.320-0.067c)(0.154M+0.596) \times(0.198\alpha+0.693)(0.300\epsilon-1.940)(0.718k_b-3.043) \times(-0.149H+0.243T+0.093k_b-1)^2+0.381$	0.385	0.429	0.888	0.860	93.531

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Table 35: `QLattice` results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
5	$-26.832(-1.192H - 0.810) \times \exp\Big[-0.572T - 0.276c - 0.150\epsilon - 0.586k_b + (1.373 - 0.347\text{mom})(0.209M - 1.849)\Big] + 0.033$	0.340	0.352	0.905	0.903	95.179

Table 36: `SISSO++` results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-0.487 + (0.374)\{(M\,\text{mom})/(c\epsilon)\} + (1.450)\{(\alpha/k_b)/(cT)\} + (0.995)\{(H\,\text{mom})/(Tk_b)\}$	0.197	0.207	0.970	0.961	17.056
2	$-0.459 + (0.368)\{(\alpha\,\text{mom})/(c\epsilon)\} + (1.400)\{(M/k_b)/(cT)\} + (0.993)\{(H\,\text{mom})/(Tk_b)\}$	0.173	0.173	0.975	0.971	17.246
3	$-0.467 + (0.385)\{(M\alpha)/(c\epsilon)\} + (0.959)\{(H\,\text{mom})/(Tk_b)\} + (1.492)\{(\text{mom}/T)/(ck_b)\}$	0.194	0.172	0.972	0.974	17.072
4	$-0.451 + (0.381)\{(M\alpha)/(c\epsilon)\} + (0.958)\{(H\,\text{mom})/(Tk_b)\} + (1.445)\{(\text{mom}/c)/(Tk_b)\}$	0.195	0.226	0.966	0.965	17.246
5	$-0.480 + (0.394)\{(M\alpha)/(c\epsilon)\} + (0.961)\{(H\,\text{mom})/(Tk_b)\} + (1.501)\{(\text{mom}/T)/(ck_b)\}$	0.192	0.197	0.970	0.971	17.013
Noise level: $\sigma = 0.15$						
1	$-0.493 + (0.376)\{(M\,\text{mom})/(c\epsilon)\} + (1.439)\{(\alpha/k_b)/(cT)\} + (1.001)\{(H\,\text{mom})/(Tk_b)\}$	0.248	0.258	0.953	0.942	16.934
2	$-0.466 + (0.372)\{(\alpha\,\text{mom})/(c\epsilon)\} + (1.418)\{(M/c)/(Tk_b)\} + (0.993)\{(H\,\text{mom})/(Tk_b)\}$	0.227	0.229	0.958	0.951	17.099
3	$-0.465 + (0.388)\{(M\alpha)/(c\epsilon)\} + (0.959)\{(H\,\text{mom})/(Tk_b)\} + (1.498)\{(\text{mom}/c)/(Tk_b)\}$	0.243	0.235	0.956	0.953	16.735

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Table 36: SISSO++ results for the Feynman equation II_36_38 (FMMM): $f = \frac{\mathrm{mom} H}{k_b T} + \frac{\mathrm{mom}\alpha}{\epsilon c^2 k_b T} H$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
4	-0.464 $+ (0.384)\{(M\alpha)/(c\epsilon)\}$ $+ (0.960)\{(H\,\mathrm{mom})/(Tk_b)\}$ $+ (1.465)\{(\mathrm{mom}/k_b)/(cT)\}$	0.242	0.265	0.949	0.952	16.932
5	-0.483 $+ (0.396)\{(M\alpha)/(c\epsilon)\}$ $+ (0.959)\{(H\,\mathrm{mom})/(Tk_b)\}$ $+ (1.513)\{(\mathrm{mom}/c)/(Tk_b)\}$	0.244	0.245	0.952	0.955	16.868
Noise level: $\sigma = 0.20$						
1	-0.495 $+ (0.376)\{(M\,\mathrm{mom})/(c\epsilon)\}$ $+ (1.435)\{(\alpha/c)/(Tk_b)\}$ $+ (1.003)\{(H\,\mathrm{mom})/(Tk_b)\}$	0.281	0.290	0.941	0.927	16.772
2	-0.469 $+ (0.373)\{(\alpha\,\mathrm{mom})/(c\epsilon)\}$ $+ (1.424)\{(M/c)/(Tk_b)\}$ $+ (0.994)\{(H\,\mathrm{mom})/(Tk_b)\}$	0.262	0.263	0.945	0.936	16.868
3	-0.464 $+ (0.389)\{(M\alpha)/(c\epsilon)\}$ $+ (0.959)\{(H\,\mathrm{mom})/(Tk_b)\}$ $+ (1.499)\{(\mathrm{mom}/k_b)/(cT)\}$	0.276	0.272	0.944	0.939	16.968
4	-0.469 $+ (0.385)\{(M\alpha)/(c\epsilon)\}$ $+ (0.961)\{(H\,\mathrm{mom})/(Tk_b)\}$ $+ (1.471)\{(\mathrm{mom}/c)/(Tk_b)\}$	0.275	0.294	0.936	0.942	16.960
5	-0.484 $+ (0.397)\{(M\alpha)/(c\epsilon)\}$ $+ (0.959)\{(H\,\mathrm{mom})/(Tk_b)\}$ $+ (1.517)\{(\mathrm{mom}/T)/(ck_b)\}$	0.278	0.278	0.939	0.943	16.558

Table 37: gplearn results for the Feynman equation II_36_38 (FMMM): $f = \frac{\mathrm{mom} H}{k_b T} + \frac{\mathrm{mom}\alpha}{\epsilon c^2 k_b T} H$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$\{\mathrm{mom} + \sin(e^{\sin(\cos^3 k_b)}) + \cos(k_b) - \cos[H + \sin(\cos c)]\} \times \sin(e^{\cos T})$	0.497	0.550	0.797	0.802	19.230
2	$\frac{H}{T} + \sin(k_b^{-4}) + \frac{-\frac{H}{T} - \sin(k_b^{-2}) + k_b^{-2} e^{e^{\sin(\cos T)} + k_b^{-1}\cos(\mathrm{mom})} \cos(\mathrm{mom})}{k_b} \times \frac{\cos(\mathrm{mom})}{k_b}$	0.485	0.558	0.797	0.679	20.785

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Table 37: **gplearn** results for the Feynman equation II_36_38 (FMMM): $f = \frac{\mathrm{mom}H}{k_b T} + \frac{\mathrm{mom}\alpha}{\epsilon c^2 k_b T} H$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	$T^{-1}\Big[\{\mathrm{mom}k_b^{-1} + (0.286 + \mathrm{mom} + \cos k_b + \cos(H/k_b)\cos(0.286 + k_b + H/k_b))\cos(\sin H)\}\cos(\sin H) + 0.286 + \mathrm{mom}k_b^{-1}\Big]$	0.484	0.464	0.803	0.814	19.168
4	$\frac{H\,\mathrm{mom}}{T k_b^2} + \frac{H}{T c k_b} + \frac{\mathrm{mom}}{T c}$	0.360	0.324	0.889	0.904	19.868
5	$\mathrm{mom} + \cos(k_b) - \cos\{H + \cos(T)\}$	0.760	0.615	0.626	0.682	17.906
Noise level: $\sigma = 0.15$						
1	$c^{-1} + T^{-1}\Big[\frac{H(\mathrm{mom} - 0.859)}{k_b} + T^{-1}\Big\{\frac{H\,\mathrm{mom}\cos(k_b^{-1})}{k_b^2} - 0.859 + \frac{2}{k_b}\Big\}\Big]$	0.443	0.590	0.852	0.752	18.665
2	$\cos\{\sin[\sin A_1]\} - \frac{c - \mathrm{mom} + k_b^{-1}(T - \mathrm{mom} - 1.799\sin k_b)\sin k_b}{k_b} + T^{-1} - H^{-3}$ $A_3 = T - \mathrm{mom} - 1.799k_b^{-1}\sin k_b,$ $A_2 = \cos(1.799/k_b) - \mathrm{mom}^{-3} - k_b^{-1}A_3 + T^{-1},$ $A_1 = \cos\{\sin(e^{\sin A_2})\} + 1.799k_b^{-1} + T^{-1}$	0.532	0.544	0.757	0.747	22.552
3	$\mathrm{mom}\Big[\sin^2(e^{\cos T}) + \sin^2\Big(e^{\cos\{\sin^2(e^{\cos(k_b^{-1})})\}}\Big)\Big]\cos(H^{-1})$	0.556	0.581	0.745	0.709	21.557
4	$\frac{H}{T} + k_b^{-1}\Big[-\frac{0.588k_b}{\mathrm{mom} + \exp\{-2H\sin(k_b^{-1}) - \cos T\} - 0.635} + k_b^{-1}\Big\{\cos T + c^{-1} + T^{-1}(-0.480c + 3\mathrm{mom} - 1.905)\Big\} + c^{-1}\Big]$	0.444	0.557	0.836	0.756	21.991
5	$(H - T + \mathrm{mom})\sin\{\sin(k_b^{-1})\} + T^{-1}\Big\{\frac{H\,\mathrm{mom}}{Tc} + \frac{\mathrm{mom}}{c}\Big\}\sin(k_b^{-1})$	0.443	0.381	0.858	0.877	17.420
Noise level: $\sigma = 0.20$						
1	$\frac{\mathrm{mom}}{T}\Big[\frac{0.605\sin\{e^{1/k_b}\sin(M/k_b)\}}{k_b} + \frac{1}{c}\Big]e^{1/k_b}$	0.552	0.469	0.755	0.831	27.758
2	$c^{-1}\Big[k_b e^{-k_b}\Big\{e^{\mathrm{mom}/T} + \frac{\mathrm{mom}(M + 0.175)\sin(\epsilon^{-1})}{k_b^2}\Big\} + \frac{\mathrm{mom}\{-\cos H + c^{-1}\}\sin(\epsilon^{-1})}{k_b^2}\Big] + \frac{-\cos H + \mathrm{mom}/k_b}{T}$	0.385	0.346	0.873	0.901	22.410
3	$\frac{\mathrm{mom}/k_b + (0.446M - \cos\mathrm{mom})/k_b}{T} + \frac{H/k_b + \mathrm{mom}/k_b}{Tc}$	0.470	0.447	0.826	0.855	17.045
4	$\frac{H}{T k_b} + T^{-1}\Big[\frac{1.193\{\mathrm{mom} + c^{-2}(\mathrm{mom} + \cos T)\}}{k_b} + c^{-2}\Big]$	0.514	0.568	0.787	0.821	19.553
5	$\frac{H}{T}\Big[\cos(k_b) + \cos\{\cos^3[\cos(\cos\{\mathrm{mom} + \cos k_b + \cos(T^{-1}\{\cos(T^{-1}) + \cos k_b + 0.633\}) + 0.633 + T^{-3}\})]\} + 0.633\Big] - \cos(\mathrm{mom})$	0.516	0.540	0.854	0.849	19.143

Table 38: `operon` results for the Feynman equation II_36_38 (FMMM): $f = \frac{\mathrm{mom}H}{k_b T} + \frac{\mathrm{mom}\alpha}{\epsilon c^2 k_b T}H$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$0.038 - 1.793 \times \frac{\left\{\frac{(-1.402)\mathrm{mom}}{2.718k_b}\right\}(2.303H)}{1.302T\,[\sin\{(0.673\epsilon + 0.568c - 0.232\alpha - 0.172M)(0.318H)\} + 0.248c]}$	0.192	0.207	0.970	0.972	25.131
2	$-0.413 - 0.671\left[\left\{\frac{-1.390\mathrm{mom}}{-0.522c} + 1.389\mathrm{mom}\right\} \times \frac{1.599H + 0.608M}{\cos(0.236\alpha)(5.930T)(-0.597k_b)} - \frac{-0.639}{-0.568\epsilon}\right]$	0.184	0.285	0.971	0.917	24.610
3	$-0.749 + 0.013\left[\frac{0.312M}{(-0.120\epsilon)^2} - \left\{\frac{0.312M}{0.144c} + \frac{2.327H}{0.286T}\right\}\frac{0.298\mathrm{mom}}{-0.029k_b} + \frac{3.453\alpha}{0.104c}\right]$	0.248	0.193	0.948	0.968	29.759
4	$-0.074 - 0.127\left[\frac{\frac{1.443\alpha}{-2.307T}}{(0.026\epsilon)(2.551k_b)} - 0.318H\frac{\frac{1.678\mathrm{mom}}{-0.044T}}{2.194k_b\sin\{\sin(-0.438c)\}}\right]$	0.270	0.237	0.938	0.949	29.578
5	$0.028 - 0.087\frac{\left[\left\{\frac{1.188c}{e^{0.186\epsilon}} - \cos(0.619M) + 1.286H\right\}\frac{0.707\mathrm{mom}}{e^{-1.611\epsilon} - 0.828T}\right]}{(2.718c)(0.022k_b)}$	0.278	0.222	0.950	0.958	28.538
Noise level: $\sigma = 0.15$						
1	$0.134 - 4.891(0.495H) \times \frac{0.771\mathrm{mom} + 0.656c + 0.366\alpha + 0.729\mathrm{mom} - \cos(0.448M) - 0.628\epsilon}{(-1.588T)(2.398c)(0.424k_b)}$	0.292	0.338	0.936	0.919	25.042
2	$0.084 - 3.079\left[\frac{-0.060H}{(0.130c)(1.994k_b)}\frac{1.583\mathrm{mom}}{0.709T} - \frac{-1.278\alpha}{2.450k_b}\frac{0.521\mathrm{mom}}{(-0.686T)(1.559\epsilon)}\right]$	0.325	0.303	0.909	0.921	31.595
3	$\frac{A\,B\,(-0.900\alpha)(-0.821\mathrm{mom}) + 0.697\mathrm{mom}}{C}$ $A = \sin\{\sin(-0.367M)\}$, $B = -0.368H + 0.292c - 0.188$, $C = (0.330T)(0.435k_b) - 0.004 + 0.394$	0.324	0.357	0.913	0.890	25.952
4	$-0.001 + 63.119 \times \frac{(0.930H)(0.862\mathrm{mom})}{2.718k_b\left[2.008c + \frac{1.078H - 1.772\alpha}{1.367c} + 1.333\epsilon + 0.910H\right](1.772T)}$	0.262	0.299	0.943	0.930	24.575
5	$-0.027 + 0.083 \times \frac{\frac{3.628\alpha}{-0.661c} - 1.488M + (1.178\mathrm{mom})(-1.809H) + \frac{1.746\alpha}{0.738\mathrm{mom}} + 1.737\epsilon}{(1.443T)\sin(-0.120k_b)}$	0.278	0.260	0.944	0.943	25.120
Noise level: $\sigma = 0.20$						
1	$-0.160 + 0.069 \times \frac{\frac{2.349\mathrm{mom}}{0.130T}\left[0.465M + \frac{0.533\mathrm{mom}}{0.408c} - 0.481\mathrm{mom} + (0.558H)^2\right] + \frac{1.548\alpha}{0.130\epsilon}}{1.432k_b}$	0.316	0.288	0.920	0.936	37.939
2	$0.043 - 0.057\left[\frac{\frac{0.176\mathrm{mom}}{0.112\epsilon}}{\frac{-0.524T}{2.170M}} - \frac{\frac{-0.541\mathrm{mom}}{\left\{\frac{-0.536T}{1.417H}\right\}(-0.594k_b + 0.106\alpha)}}{-0.102c}\right]$	0.316	0.265	0.914	0.942	31.098

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Table 38: `operon` results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T}H$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
3	$-0.015 + 2.075 \frac{\frac{\sin(-0.413H)}{0.184T}}{\frac{1.862k_b}{0.434\text{mom}}\left[\sin\{\sin(-0.509\epsilon)\} - \frac{2.186M+2.135\alpha}{(1.507\epsilon)(2.350c)}\right]^{-1}}$	0.235	0.260	0.957	0.951	27.554
4	$-0.010 - 22.478 \times \frac{\left[\frac{(-1.210\text{mom})(1.414H)}{-0.690c\, e^{-0.237\alpha}} \frac{1}{1.790/(0.318M)+2.003\epsilon} + 0.460\text{mom}\right]}{(6.921T)(-1.302k_b)}$	0.277	0.366	0.942	0.898	23.024
5	$1.650 + 2.162\left[\sin\{\sin(-0.586c)\} + \frac{2.238\epsilon\sin(-0.514c) + 1.737M + 1.951\alpha + 5.167H}{(2.410T)\{3.182k_b/(0.679\text{mom})\}}\right]$	0.288	0.294	0.932	0.932	27.426

Table 39: `PySR` results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T}H$ across noise levels and repetitions. Train `RMSE` and R^2 are computed on the 90% training set, while test `RMSE` and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train `RMSE`	Test `RMSE`	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$\frac{\text{mom}}{1.174\,Tk_b}\left\{H + \frac{\alpha M}{\epsilon c}\right\}$	0.173	0.174	0.975	0.980	66.425
2	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{M}{c^2}\right\}$	0.249	0.306	0.946	0.904	97.007
3	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{\alpha}{\epsilon(c-0.431)}\right\}$	0.216	0.185	0.961	0.970	76.063
4	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{M\alpha}{\epsilon c^2}\right\}$	0.000	0.000	1.000	1.000	73.380
5	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{1}{\epsilon(c+\alpha^{-1}-1.262)}\right\}$	0.231	0.194	0.965	0.968	76.818
Noise level: $\sigma = 0.15$						
1	$\frac{\text{mom}}{Tk_b}\left\{H + M\alpha e^{-c}\right\}$	0.300	0.408	0.932	0.882	73.086
2	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{1}{c-0.493}\right\}$	0.360	0.330	0.889	0.907	133.422
3	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{M\{\cos(\epsilon)+1.423\}}{c^2}\right\}$	0.246	0.246	0.950	0.948	71.924
4	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{\alpha}{\epsilon(c-0.411)}\right\}$	0.261	0.298	0.943	0.930	89.639
5	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{M}{c^2}\right\}$	0.340	0.286	0.917	0.931	86.361
Noise level: $\sigma = 0.20$						
1	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{M}{c^2}\right\}$	0.331	0.298	0.912	0.932	208.305
2	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{\alpha+M-\epsilon}{c^2}\right\}$	0.226	0.228	0.956	0.957	110.782
3	$\frac{\text{mom}}{Tk_b}\left\{H + \frac{M}{c^2}\right\}$	0.331	0.356	0.914	0.909	62.105

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Table 39: PySR results for the Feynman equation II_36_38 (FMMM): $f = \frac{\mathrm{mom}H}{k_bT} + \frac{\mathrm{mom}\alpha}{\epsilon c^2 k_b T}H$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
4	$\frac{\mathrm{mom}}{Tk_b}\left\{H + \frac{\alpha}{c^2}\right\}$	0.367	0.441	0.898	0.852	70.073
5	$0.356\,\frac{\mathrm{mom}}{T}\{\sin(c) + H\}\{\cos(k_b) + 1.859\}$	0.430	0.464	0.849	0.831	141.019

Table 40: BMS results for the Feynman equation II_36_38 (FMMM): $f = \frac{\mathrm{mom}H}{k_bT} + \frac{\mathrm{mom}\alpha}{\epsilon c^2 k_b T}H$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set. Here, a_0 denotes a fitted constant learned by BMS.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$-\mathrm{mom}\, a_0\{a_0(H + k_b) + \cos T\} \times [-\{a_0 + a_0\}^2]^3 \exp\{\sin(\cos k_b)\}$	0.519	0.509	0.779	0.831	2414.000
2	$\mathrm{mom}\, H \exp\left[-\{ca_0\}^3 + Ta_0c(a_0M + k_b)\right]$	0.445	0.484	0.829	0.759	2980.680
3	$\mathrm{mom}\, H \exp[Tk_b(H + T)a_0(c + \epsilon)]$	0.457	0.446	0.824	0.828	2263.260
4	$\exp^3\left(a_0\{H + \cos a_0 + \sin(c + \epsilon) + \mathrm{mom} - Tk_b + \alpha^2 a_0\}\right)$	0.397	0.394	0.866	0.858	2485.010
5	$-2a_0\, \mathrm{mom}\, H \exp\{\cos(a_0)(c + T + k_b)\}$	0.561	0.416	0.797	0.855	2289.430
Noise level: $\sigma = 0.15$						
1	$\mathrm{mom}\, H \exp\left[(a_0 - T)k_b\, a_0^2 c\right] + a_0$	0.504	0.561	0.809	0.775	2309.130
2	$\mathrm{mom}\, [1 - \sin\{a_0(k_b + c + \cos(a_0 \cos H + \epsilon) + \epsilon + T)\}]$	0.560	0.546	0.731	0.746	2713.060
3	$\{\mathrm{mom}(\alpha + H) + \cos c + M\} \times \exp\{a_0(T + a_0 + k_b)\} + a_0$	0.422	0.399	0.852	0.862	2417.780
4	$\mathrm{mom}\, e^{-T}[a_0^3 + \epsilon + k_b + c + k_b - \{H + T + \alpha\} + k_b]$	0.501	0.569	0.790	0.745	2479.190
5	$a_0^2\{H + \cos c\}\{\mathrm{mom} + \cos T\} \exp(\cos k_b) + a_0^2$	0.513	0.454	0.810	0.826	2426.710
Noise level: $\sigma = 0.20$						
1	$\{\mathrm{mom}(H - a_0cM)\} \exp^2\left[-a_0(k_b + T)^3\right] + M^3 \sin(a_0\alpha) + 3a_0$	0.541	0.477	0.765	0.825	3558.890
2	$\mathrm{mom}\, H \exp\left[-a_0^2\{e^T + k_b^2 c\} + a_0\right] + a_0$	0.488	0.452	0.795	0.831	2786.200
3	$e^{a_0}\, 2a_0\{H + \cos T\}\{\cos k_b + \mathrm{mom}\} + a_0$	0.565	0.654	0.749	0.691	1977.010
4	$a_0\{-a_0\epsilon^2 + k_bc + k_bH\} + \mathrm{mom} + \cos T + H$	0.621	0.639	0.708	0.689	2035.710
5	$(H - c + \mathrm{mom}) \exp\left[\cos\{a_0^3(k_b + a_0 + T)\}\right]$	0.474	0.488	0.817	0.814	2647.510

Table 41: BSR results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom} H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} M$ across noise levels and repetitions. Train RMSE and R^2 are computed on the 90% training set, while test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	$1.319 - 0.130\exp\{\cos(\alpha)\} + 0.002\exp(\alpha) + 0.011\sin\{(c+\epsilon)^3\} + 0.068(\alpha + c\,\text{mom})$	1.093	1.237	0.019	0.001	2028.980
2	$10.235 - 9.025\cos\{\exp(-c)\} + 0.013\cos(M) + 0.262\cos(c) + 0.092M\alpha$	1.031	0.968	0.082	0.036	2046.020
3	$-0.048 + 0.026\exp(\alpha) - 0.016\exp(\epsilon) - 0.003\dfrac{\cos(\alpha)}{\sin(\alpha\,\text{mom})} + 0.802\,\text{mom}$	0.977	1.010	0.197	0.118	1910.030
4	$4.692 - 1.532\{k_b + \exp[\cos(k_b)]\} + 0.017\exp(\alpha) + 0.552H - 0.157\sin(\text{mom}^2)$	0.956	0.922	0.218	0.224	1912.210
5	$-2.822 + 0.388\{\cos(H^{-1}) + H + M\} + 1.020\,\epsilon^{-1} + 3.579\,T^{-1} - 6.204\times10^{-4}(T\epsilon)^3$	1.007	0.907	0.345	0.307	1932.020
Noise level: $\sigma = 0.15$						
1	$3.574 - 0.927T - 0.053\dfrac{\alpha}{M} + 0.516\sin(ck_b) + 0.010\sin^{-2}(\alpha)$	0.957	1.022	0.310	0.256	1923.980
2	$-1.097 - 0.117\cos(k_b^2) - 1.122\exp(-M) + 0.357\cos\{\sin(M)\} + 1.523\exp(T^{-1})$	0.947	0.917	0.229	0.280	1935.810
3	$0.387 + 0.340\exp\{\sin(c)\} - 0.249\cos(M) + 0.272\sin(k_b^2) - 0.979\cos(\text{mom})$	0.954	0.905	0.247	0.293	1921.380
4	$-0.293 + 0.028M\exp(H) + 1.805\sin\{\sin(T)\} - 0.054\sin^{-1}(\alpha) + 0.034\exp(\alpha)$	0.979	0.974	0.199	0.254	1960.230
5	$6.922 - 1.301\cos(H) - 2.606\,\text{mom}^{-1} - 0.467(H+c) - 0.243(T + \alpha + \epsilon k_b + M)$	0.855	0.707	0.473	0.577	1924.410
Noise level: $\sigma = 0.20$						
1	$2.398\times10^{-8} + 8.053\times10^{-6}\exp(HM) - 1.081\times10^{-8}(-k_b\alpha)^9 + 5.583\times10^{-7}\exp\left(\dfrac{k_b H}{M}\right) - 6.071\times10^{-8}\{\cos(\text{mom}) - M\}$	1.938	2.022	-2.020	-2.138	1931.390
2	$1.100\times10^{-8} + 8.718\times10^{-9}\sin(\epsilon) - 4.275\times10^{-9}\cos(\alpha) + 6.393\times10^{-9}\exp\{\exp(M)\} + 3.798\times10^{-8}cT$	1.881	1.826	-2.043	-1.753	1914.320
3	$4.287 - 0.692\,\alpha^{-1} - 0.185\sin(-c^3) - 2.741\cos\{\cos(\text{mom})\} + 0.130\sin\{\cos(\text{mom}\alpha)\}$	1.011	1.093	0.195	0.138	1921.150
4	$3.873 - 2.517\cos\{\cos(\text{mom})\} + 0.210\cos^2\{\sin(H^2)\} + 0.016\alpha^3 - 0.633\sin^2\{\cos(k_b) + H\}$	1.059	1.063	0.151	0.139	1959.312
5	$1.530 + 3.028\,T^{-1} - 0.907\,k_b^2 + 0.023\exp\{\cos(M\text{mom}) + M\} + 0.218\,k_b^3$	0.766	0.773	0.520	0.532	1933.380

D Results of HierBOSSS for Learning Feynman Equations

D.1 Experimental Settings of HierBOSSS

We evaluate HierBOSSS on the Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). Each experimental run uses a random subsample of $n = 2000$ observations out of 10^5 observations, followed by a 90/10 train-test split. We consider the noiseless setting (with the original target) and noisy regimes, where independent Gaussian noise with different standard deviations σ's is added to the target, as given below. Test RMSE and R^2 are computed on the held-out 10% test split. For each Feynman equation and noise level, we run 5 independent repetitions. RMSE and R^2 denotes the root mean square error and coefficient of determination, i.e., $\texttt{RMSE} = \sqrt{n^{-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ and $R^2 = 1 - \texttt{RMSE}^2/[n^{-1}\sum_{i=1}^{n}(y_i - \overline{y})^2]$ $(\overline{y} = n^{-1}\sum_{i=1}^{n} y_i)$, respectively.

HierBOSSS noise configurations and data splitting for Feynman equations

```
# Noiseless and noisy regimes
# 0.00 -> noiseless setting

feynman_equations = {
    "I_12_2": {      # CL
        "noise_levels": [0.00, 0.15, 0.20, 0.25],
    },
    "I_12_11": {     # FCE
        "noise_levels": [0.00, 0.25, 1.00, 2.00],
    },
    "I_24_6": {      # EHFO
        "noise_levels": [0.00, 0.15, 0.25, 1.00],
    },
    "I_50_26": {     # HOQN
        "noise_levels": [0.00, 0.15, 0.18, 0.20],
    },
    "II_36_38": {    # FMMM
        "noise_levels": [0.00, 0.15, 0.20, 0.25],
    },
}

# Data split settings
# y_clean -> original target
data_split_settings = {
    "base_seed": ...,
    "nreps": 5,
    "sample_n": 2000,
    "test_size": 0.10,
    "train_size": 0.90,
    "noise_model": "y = y_clean + Normal(0, sigma^2)",
    "split_seed_rule": "...",
}
```

The operator set $\mathbb{O}$ is kept common across every Feynman equation. The set consists of two binary operators and seven unary operators, as given below. In the numerical implementation, the inverse operator is protected using a threshold 10^{-8}, while the exponential operator is evaluated after clipping its input to $[-20, 20]$ for numerical stability.

HierBOSSS operator set

```
opset = [
    "add",  # (x, y) -> x + y
    "mul",  # (x, y) -> x*y
    "neg",  # x -> -x
    "inv",  # x -> 1/x, protected near zero
    "sin",  # x -> sin(x)
    "cos",  # x -> cos(x)
    "exp",  # x -> exp(x), with input clipped to [-20, 20]
    "sq",   # x -> x^2
    "cu",   # x -> x^3
]
```

The prior hyperparameters are set uniformly across all Feynman equations unless otherwise stated. The Dirichlet concentration parameters for both operator and feature assignments are set to one, i.e., $\boldsymbol{\alpha}_{\text{op}} = (1, \ldots, 1)$ and $\boldsymbol{\alpha}_{\text{ft}} = (1, \ldots, 1)$. The depth-dependent split probability is controlled by $(\alpha_0, \delta_0) = (0.95, 1.20)$. For the outer regression coefficient vector $\boldsymbol{\beta}$ with an intercept, we set the Gaussian prior mean vector to $\boldsymbol{\mu}_\beta = \mathbf{0}$, use Gaussian prior covariance matrix $\boldsymbol{\Sigma}_\beta = 10\boldsymbol{I}_{K+1}$, and assign the hyperparameters of the Inverse-Gamma prior on the model noise variance σ^2 to $\nu = 0.05 = \lambda$.

HierBOSSS prior configuration

```
HierBOSSS_prior = {
    # operator and feature Dirichlet weight prior concentration
    "alpha_op": "ones(len(opset))",
    "alpha_ft": "ones(number_of_features)",

    # split probability
    "alpha_0": 0.95,
    "delta_0": 1.20,

    # Gaussian prior parameters of outer regression coefficient vector
    "add_intercept": True,
    "beta0": "zeros(K + 1)",
    "V0": "10 * I_(K + 1)",

    # Inverse-Gamma prior parameters of model noise variance
    "a0": 0.05,
    "b0": 0.05,
    "sigma2_prior": "Inverse-Gamma(a0 / 2, b0 / 2)",
}
```

For each Feynman equation, the symbolic forest size K and number of MCMC iterations are chosen according to the symbolic complexity of the underlying target law, as given below. However, one can choose a suitable K balancing the expressiveness and computational efficiency for learning the underlying Feynman equations; see Section G.

HierBOSSS equation-specific K and MCMC iterations

```
# maxiter -> number of MCMC iterations within each parallel chain of HierBOSSS
# K -> symbolic forest size

HierBOSSS_feynman_settings = {
    "I_12_2": {             # CL
        "K": 4,
        "maxiter": 2000,
    },
    "I_12_11": {            # FCE
        "K": 3,
        "maxiter": 2000,
    },
    "I_24_6": {             # EHFO
        "K": 4,
        "maxiter": 2000,
    },
    "I_50_26": {            # HOQN
        "K": 4,
        "maxiter": 2000,
    },
    "II_36_38": {           # FMMM
        "K": 4,
        "maxiter": 20000,
    },
}
```

Each repetition is fit using 5 independent parallel MCMC chains of HierBOSSS. The final symbolic model selection is performed by pooling sampled forests across chains and ranking them by their joint marginal posterior over $\mathcal{T}$, i.e., $\mathrm{JMP}(\mathcal{T})$.

HierBOSSS parallel chain configuration

```
parallel_mcmc_settings = {
    "nchains": 5,               # number of parallel MCMC chains
    "burnin": 0,
    "thin": 1,
    "n_jobs": "nchains",
    "chain_seed_rule": "...",
    "target": "JMP(T)",
}
```

The MH kernel uses seven local symbolic tree proposal moves: grow (g), prune (p), change feature (cf), change operator (co), subtree replacement (st), delete node (del), and insert node (ins). Unless a move is unavailable for the current tree state, all moves are assigned equal proposal weight through the move weight vector $\boldsymbol{\pi}_{\text{move}}$. At each tree update, the sampler first identifies the set of valid moves and valid node addresses, normalizes the proposal weights over the available moves, samples a move type, and then samples a valid node address uniformly for that move.

HierBOSSS local symbolic tree proposal moves

```
# local symbolic tree proposal move weights

move_weights = {
    "grow": 1.0,                # g
    "prune": 1.0,               # p
    "change_feature": 1.0,      # cf
    "change_operator": 1.0,     # co
    "subtree_replace": 1.0,     # st
    "delete_node": 1.0,         # del
    "insert_node": 1.0,         # ins
}
```

The initial symbolic forests and proposal-generated subtrees use uniform operator and feature weights, i.e., uniform choices for $(\boldsymbol{w}_{\text{op,init}}, \boldsymbol{w}_{\text{op,prop}}, \boldsymbol{w}_{\text{ft,init}}, \boldsymbol{w}_{\text{ft,prop}})$. These weights are used only for initialization and proposal generation; the posterior target itself uses the Dirichlet-collapsed symbolic forest prior $\Pi_{\text{forest},K}(\mathcal{T} \mid \alpha_0, \delta_0, \boldsymbol{\alpha}_{\text{op}}, \boldsymbol{\alpha}_{\text{ft}})$. Thus, operator and feature preferences are learned through the marginal prior contribution in the joint marginal posterior rather than fixed at the proposal level.

HierBOSSS initialization and proposal weights

```
# initialization of operator and feature weight vectors with uniform choice
# proposal operator and feature weight vectors with uniform choice

initialization_and_proposal = {
    "wts_init_op": "ones(len(opset))",
    "wts_init_ft": "ones(number_of_features)",
    "wts_prop_op": "wts_init_op",
    "wts_prop_ft": "wts_init_ft",
    "proposal_normalization": "normalize over available moves and valid labels",
}
```

After sampling, all retained symbolic forests from the parallel MCMC chains are pooled and ranked by their log joint marginal posterior over $\mathcal{T}$, i.e., $\log \text{JMP}(\mathcal{T})$. For each repetition, the top $r = 10$ symbolic forests are retained as the representative posterior set. For each retained forest, the posterior mean of the outer regression coefficient vector is computed under the conjugate Gaussian layer, yielding the corresponding raw symbolic expression (***Raw***).

HierBOSSS posterior ranking configuration and obtaing *Raw* expressions

```
posterior_ranking = {
    "ranking_score": "log_JMP",
    "pooling": "all retained forests from all parallel chains",
    "top_r": 10,
    "raw_expression_coefficients": "posterior mean under NIG layer",
    "raw_expression_digits": 6, # rounding off decimals in raw expression
}
```

Finally, each top-ranked symbolic forest is passed through the post-MCMC symbolic model refinement step in HierBOSSS. The symbolic trees in a fixed forest are treated as candidate additive components, and a smaller subset is selected by minimizing BIC. After selecting the subset, the outer regression coefficient vector is re-estimated, and the retained expression is simplified using SymPy [Meurer et al., 2017]. The final reported expression corresponds to the ***Final*** expression after this refinement step with effective symbolic forest size $K_{\text{eff}} \leq K$.

HierBOSSS post-MCMC symbolic model refinement to obtain *Final* expressions

```
post_mcmc_refinement = {
    "candidate_models": "top_r forests ranked by log_JMP",
    "criterion": "BIC",
    "coefficient_refit": "posterior mean on selected symbolic design",
    "simplification": "SymPy",
    "final_expression_digits": 2, # rounding off decimals in final expression
    "coefficient_tolerance": 1e-4,
    "effective_forest_size": "K_eff = number of retained non-intercept trees",
}
```

D.2 Results for I_12_2 (CL)

D.2.1 Expressions Learned by HierBOSSS

Table 42: HierBOSSS results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	*Raw*: $-5.280\times10^{-6} + 0.080\,\frac{q_1 q_2}{\epsilon r^2} + 8.378\times10^{-6}\frac{1}{r} + 8.378\times10^{-6}\frac{1}{r} + 8.378\times10^{-6}\frac{1}{r}$	5.395×10^{-6}	5.504×10^{-6}	1.000	1.000	49.369
	Final: $0.080\,\frac{q_1 q_2}{\epsilon r^2}$	5.394×10^{-6}	5.503×10^{-6}	1.000	1.000	—
2	*Raw*: $-1.422\times10^{-6} + 2.295\times10^{-6}\frac{1}{r} + 2.295\times10^{-6}\frac{1}{r} + 2.295\times10^{-6}\frac{1}{r} + 0.040\,\frac{q_2(q_1+q_1)}{\epsilon r^2}$	1.441×10^{-6}	1.592×10^{-6}	1.000	1.000	47.098
	Final: $0.040\,\frac{2q_1 q_2}{\epsilon r^2}$	8.700×10^{-7}	9.738×10^{-7}	1.000	1.000	—
3	*Raw*: $5.501\times10^{-7} + 0.040\,\frac{q_2(q_1+q_1)}{\epsilon r^2} - 5.504\times10^{-8}\cos(q_2) - 5.504\times10^{-8}\cos(q_2) - 5.504\times10^{-8}\cos(q_2)$	9.758×10^{-7}	8.354×10^{-7}	1.000	1.000	47.762
	Final: $0.040\,\frac{2q_1 q_2}{\epsilon r^2}$	9.758×10^{-7}	8.354×10^{-7}	1.000	1.000	—
4	*Raw*: $-4.565\times10^{-6} + 7.093\times10^{-6}\frac{1}{r} + 0.080\,\frac{q_1 q_2}{\epsilon r^2} + 7.093\times10^{-6}\frac{1}{r} + 7.093\times10^{-6}\frac{1}{r}$	4.921×10^{-6}	3.756×10^{-6}	1.000	1.000	62.565
	Final: $0.080\,\frac{q_1 q_2}{\epsilon r^2}$	4.918×10^{-6}	3.753×10^{-6}	1.000	1.000	—
5	*Raw*: $1.220\times10^{-6} - 4.183\times10^{-7}\frac{1}{q_1} - 4.183\times10^{-7}\frac{1}{q_1} + 0.040\,\frac{q_1(q_2+q_2)}{\epsilon r^2} - 4.183\times10^{-7}\frac{1}{q_1}$	1.038×10^{-6}	9.866×10^{-7}	1.000	1.000	50.158
	Final: $0.040\,\frac{2q_1 q_2}{\epsilon r^2}$	1.038×10^{-6}	9.866×10^{-7}	1.000	1.000	—
Noise level: $\sigma = 0.15$						

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Table 42: HierBOSSS results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
1	***Raw***: $0.014 - 0.011\,\frac{1}{q_1} - 0.011\,\frac{1}{q_1} - 0.011\,\frac{1}{q_1} + 0.083\,\frac{q_1 q_2}{\epsilon r^2}$	0.149	0.143	0.281	0.238	48.198
	Final: $0.084\,\frac{q_1 q_2}{\epsilon r^2}$	0.149	0.144	0.279	0.235	—
2	***Raw***: $-0.007 + 0.019\,\frac{1}{\epsilon} + 0.019\,\frac{1}{\epsilon} + 0.074\,\frac{q_1 q_2}{\epsilon r^2} - 0.020\,\cos(r^3)$	0.152	0.147	0.229	0.135	46.409
	Final: $0.079\,\frac{q_1 q_2}{\epsilon r^2} - 0.020\,\cos(r^3)$	0.152	0.147	0.226	0.142	—
3	***Raw***: $0.006 - 0.003\,\frac{1}{q_1} - 0.003\,\frac{1}{q_1} - 0.003\,\frac{1}{q_1} + 0.081\,\frac{q_1 q_2}{\epsilon r^2}$	0.152	0.156	0.276	0.273	62.039
	Final: $0.082\,\frac{q_1 q_2}{\epsilon r^2}$	0.152	0.156	0.276	0.274	—
4	***Raw***: $-0.022 + 0.085\,\frac{q_1 q_2}{\epsilon r^2} + 0.014\,\frac{1}{q_2} + 0.014\,\frac{1}{q_2} + 0.014\,\frac{1}{q_2}$	0.152	0.148	0.231	0.274	63.390
	Final: $0.081\,\frac{q_1 q_2}{\epsilon r^2}$	0.152	0.148	0.228	0.275	—
5	***Raw***: $-0.133 + 0.075\,\frac{1}{r} + 0.075\,\frac{1}{r} + 0.083\,\frac{q_1 q_2}{\epsilon r^2} + 0.024\,r$	0.149	0.140	0.234	0.292	61.748
	Final: $0.084\,\frac{q_1 q_2}{\epsilon r^2}$	0.149	0.140	0.229	0.291	—
Noise level: $\sigma = 0.20$						
1	***Raw***: $-0.045 + 0.035\,\frac{1}{r} + 0.035\,\frac{1}{r} + 0.035\,\frac{1}{r} + 0.085\,\frac{q_1 q_2}{\epsilon r^2}$	0.198	0.198	0.127	0.205	56.024
	Final: $-0.045 + 0.110\,\frac{1}{r} + 0.085\,\frac{q_1 q_2}{\epsilon r^2}$	0.198	0.198	0.127	0.205	—

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Table 42: HierBOSSS results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
2	***Raw***: $0.003 - 0.017\,\frac{1}{\epsilon} + 0.083\,\frac{q_1 q_2}{\epsilon r^2} - 0.017\,\frac{1}{\epsilon} - 0.017\,\frac{1}{\epsilon}$	0.200	0.193	0.167	0.244	60.635
	Final: $-0.044\,\frac{1}{\epsilon} + 0.083\,\frac{q_1 q_2}{\epsilon r^2}$	0.200	0.193	0.167	0.244	—
3	***Raw***: $0.010 - 0.009\,\frac{1}{r} - 0.009\,\frac{1}{r} + 0.078\,\frac{q_1 q_2}{\epsilon r^2} - 0.009\,\frac{1}{r}$	0.201	0.199	0.148	0.289	58.254
	Final: $0.075\,\frac{q_1 q_2}{\epsilon r^2}$	0.201	0.199	0.148	0.285	—
4	***Raw***: $0.019 + 0.078\,\frac{q_1 q_2}{\epsilon r^2} - 0.006\,\frac{1}{r} - 0.006\,\frac{1}{r} - 0.006\,\frac{1}{r}$	0.197	0.211	0.107	0.134	55.818
	Final: $0.082\,\frac{q_1 q_2}{\epsilon r^2}$	0.197	0.211	0.104	0.129	—
5	***Raw***: $-0.063 + 0.081\,\frac{q_1 q_2}{\epsilon r^2} + 0.039\,\frac{1}{q_1} + 0.072\,\frac{1}{\epsilon} + 0.039\,\frac{1}{q_1}$	0.203	0.197	0.171	0.098	58.478
	Final: $0.080\,\frac{q_1 q_2}{\epsilon r^2}$	0.204	0.195	0.162	0.117	—
Noise level: $\sigma = 0.25$						
1	***Raw***: $-0.067 + 0.082\,\frac{q_1 q_2}{\epsilon r^2} + 0.060\,\frac{1}{r} + 0.060\,\frac{1}{r} + 0.060\,\frac{1}{r}$	0.252	0.236	0.097	0.108	55.903
	Final: $-0.067 + 0.082\,\frac{q_1 q_2}{\epsilon r^2} + 0.180\,\frac{1}{r}$	0.252	0.236	0.097	0.108	—
2	***Raw***: $0.026 + 0.088\,\frac{q_1 q_2}{\epsilon r^2} - 0.024\,\frac{1}{r} - 0.024\,\frac{1}{r} - 0.024\,\frac{1}{r}$	0.249	0.264	0.109	0.176	58.066
	Final: $0.081\,\frac{q_1 q_2}{\epsilon r^2}$	0.250	0.263	0.107	0.177	—

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Table 42: HierBOSSS results for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	***Raw***: $0.018 - 0.019\,\frac{1}{q_2} - 0.019\,\frac{1}{q_2} - 0.019\,\frac{1}{q_2} + 0.078\,\frac{q_1 q_2}{\epsilon r^2}$	0.254	0.242	0.123	0.049	58.756
	Final: $0.078\,\frac{q_1 q_2}{\epsilon r^2}$	0.254	0.241	0.121	0.063	—
4	***Raw***: $-0.014 + 0.082\,\frac{q_1 q_2}{\epsilon r^2} + 0.004\,\frac{1}{\epsilon} + 0.004\,\frac{1}{\epsilon} + 0.004\,\frac{1}{\epsilon}$	0.247	0.211	0.118	0.159	58.926
	Final: $0.079\,\frac{q_1 q_2}{\epsilon r^2}$	0.247	0.210	0.117	0.166	—
5	***Raw***: $-0.108 + 0.084\,\frac{q_1 q_2}{\epsilon r^2} + 0.075\,\frac{1}{r} + 0.075\,\frac{1}{r} + 0.048\,\frac{1}{q_1}$	0.246	0.245	0.134	-0.009	58.167
	Final: $-0.089 + 0.081\,\frac{q_1 q_2}{\epsilon r^2} + 0.160\,\frac{1}{r}$	0.246	0.245	0.133	-0.007	—

D.2.2 Top Expressions Ranked by JMP

Table 43: Top $r = 10$ expressions ranked by JMP learned by HierBOSSS for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. Test RMSE and R^2 are computed on the held-out 10% test set.

Rank	Learned expression	Test RMSE	Test R^2	log JMP
Noiseless setting, Repetition 1				
1–10	***Raw***: $-5.280\times10^{-6} + 0.080\,\frac{q_1 q_2}{\epsilon r^2} + 8.378\times10^{-6}\frac{1}{r} + 8.378\times10^{-6}\frac{1}{r} + 8.378\times10^{-6}\frac{1}{r}$	5.504×10^{-6}	1.000	8436.284
	Final: $0.080\,\frac{q_1 q_2}{\epsilon r^2}$	5.503×10^{-6}	1.000	—
Noise level: $\sigma = 0.15$, Repetition 1				

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Table 43: Top $r = 10$ expressions ranked by JMP learned by HierBOSSS for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. Test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rank	Learned expression	Test RMSE	Test R^2	log JMP
1–5	***Raw***: $0.014 - 0.011\,\frac{1}{q_1} - 0.011\,\frac{1}{q_1} - 0.011\,\frac{1}{q_1} + 0.083\,\frac{q_1 q_2}{\epsilon r^2}$	0.143	0.238	2432.541
	Final: $0.084\,\frac{q_1 q_2}{\epsilon r^2}$	0.144	0.235	—
6–10	***Raw***: $0.014 - 0.011\,\frac{1}{q_1} - 0.011\,\frac{1}{q_1} + 0.083\,\frac{q_1 q_2}{\epsilon r^2} - 0.011\,\frac{1}{q_1}$	0.143	0.238	2432.541
	Final: $0.084\,\frac{q_1 q_2}{\epsilon r^2}$	0.144	0.235	—
Noise level: $\sigma = 0.20$, Repetition 1				
1–9	***Raw***: $-0.045 + 0.035\,\frac{1}{r} + 0.035\,\frac{1}{r} + 0.035\,\frac{1}{r} + 0.085\,\frac{q_1 q_2}{\epsilon r^2}$	0.198	0.205	1922.118
	Final: $-0.045 + 0.110\,\frac{1}{r} + 0.085\,\frac{q_1 q_2}{\epsilon r^2}$	0.198	0.205	—
10	***Raw***: $-0.172 + 0.121\,\frac{1}{r} + 0.024\,r + 0.121\,\frac{1}{r} + 0.085\,\frac{q_1 q_2}{\epsilon r^2}$	0.198	0.202	1921.041
	Final: $-0.045 + 0.110\,\frac{1}{r} + 0.085\,\frac{q_1 q_2}{\epsilon r^2}$	0.198	0.205	—
Noise level: $\sigma = 0.25$, Repetition 1				
1	***Raw***: $-0.067 + 0.082\,\frac{q_1 q_2}{\epsilon r^2} + 0.060\,\frac{1}{r} + 0.060\,\frac{1}{r} + 0.060\,\frac{1}{r}$	0.236	0.108	1490.524
	Final: $-0.067 + 0.082\,\frac{q_1 q_2}{\epsilon r^2} + 0.180\,\frac{1}{r}$	0.236	0.108	—

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Table 43: Top $r = 10$ expressions ranked by JMP learned by `HierBOSSS` for the Feynman equation I_12_2 (CL): $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ across noise levels. ***Raw*** expression correspond to the symbolic forest learned by `HierBOSSS` followed by the ***Final*** expression after post-`MCMC` symbolic model refinement. Test `RMSE` and R^2 are computed on the held-out 10% test set *(continued)*.

Rank	Learned expression	Test `RMSE`	Test R^2	log JMP
2–10	***Raw***: $-0.064 - 0.033\sin(q_1) + 0.127\frac{q_2}{\epsilon r} + 0.088\frac{1}{r} + 0.088\frac{1}{r}$	0.240	0.074	1490.074
	Final: $-0.064 - 0.033\sin(q_1) + 0.130\frac{q_2}{\epsilon r} + 0.180\frac{1}{r}$	0.240	0.074	—

D.3 Results for I_12_11 (FCE)

D.3.1 Expressions Learned by HierBOSSS

Table 44: HierBOSSS results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	***Raw***: $E_f q + Bqv\sin(\theta) + 3.088\times10^{-7}$	1.705×10^{-7}	1.782×10^{-7}	1.000	1.000	15.484
	Final: $E_f q + Bqv\sin(\theta)$	8.627×10^{-8}	8.641×10^{-8}	1.000	1.000	—
2	***Raw***: $E_f q + Bqv\sin(\theta) + 7.311\times10^{-7}$	4.202×10^{-7}	4.186×10^{-7}	1.000	1.000	14.780
	Final: $E_f q + Bqv\sin(\theta) + 1.196\times10^{-6}$	6.817×10^{-7}	6.778×10^{-7}	1.000	1.000	—
3	***Raw***: $E_f q + Bqv\sin(\theta) + 2.381\times10^{-7}$	1.340×10^{-7}	1.308×10^{-7}	1.000	1.000	15.398
	Final: $E_f q + Bqv\sin(\theta) + 2.453\times10^{-7}$	1.381×10^{-7}	1.347×10^{-7}	1.000	1.000	—
4	***Raw***: $E_f q + Bqv\sin(\theta) + 5.151\times10^{-7}$	2.838×10^{-7}	2.975×10^{-7}	1.000	1.000	15.166
	Final: $E_f q + Bqv\sin(\theta)$	1.460×10^{-7}	1.446×10^{-7}	1.000	1.000	—
5	***Raw***: $E_f q + Bqv\sin(\theta) + 1.896\times10^{-7}$	1.063×10^{-7}	1.085×10^{-7}	1.000	1.000	14.862
	Final: $E_f q + Bqv\sin(\theta) + 1.907\times10^{-7}$	1.076×10^{-7}	1.097×10^{-7}	1.000	1.000	—
Noise level: $\sigma = 0.25$						
1	***Raw***: $E_f q + Bqv\sin(\theta) - 0.091\,\frac{1}{B} + 0.038$	0.248	0.240	1.000	1.000	15.472
	Final: $1.001\,E_f q + Bqv\sin(\theta)$	0.248	0.238	1.000	1.000	—
2	***Raw***: $1.002\,E_f q + Bqv\sin(\theta) + 0.057\,\frac{1}{E_f} - 0.033$	0.255	0.244	1.000	1.000	13.165
	Final: $1.001\,E_f q + Bqv\sin(\theta)$	0.255	0.244	1.000	1.000	—
3	***Raw***: $1.001\,E_f q + Bqv\sin(\theta) + 0.062\,\frac{1}{E_f} - 0.032$	0.254	0.261	1.000	1.000	13.696
	Final: $E_f q + Bqv\sin(\theta)$	0.254	0.260	1.000	1.000	—

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Table 44: HierBOSSS results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
4	***Raw***: $E_f q + 1.001\,Bqv\sin(\theta) - 0.041\,\frac{1}{B} + 0.009$	0.252	0.246	1.000	1.000	14.391
	Final: $E_f q + 1.001\,Bqv\sin(\theta)$	0.253	0.246	1.000	1.000	—
5	***Raw***: $E_f q + Bqv\sin(\theta) + 0.041\,\frac{1}{B} - 0.018$	0.249	0.231	1.000	1.000	13.686
	Final: $0.999\,E_f q + Bqv\sin(\theta)$	0.249	0.231	1.000	1.000	—
Noise level: $\sigma = 1.00$						
1	***Raw***: $1.002\,E_f q + 0.999\,Bqv\sin(\theta) + 0.342\,\frac{1}{B} - 0.125$	0.985	0.983	0.999	0.999	13.525
	Final: $1.003\,E_f q + 0.999\,Bqv\sin(\theta)$	0.987	0.978	0.999	0.999	—
2	***Raw***: $E_f q + 1.002\,Bqv\sin(\theta) - 0.174\,\frac{1}{q} - 0.001$	1.001	0.969	0.998	0.998	13.876
	Final: $0.995\,E_f q + 1.002\,Bqv\sin(\theta)$	1.003	0.968	0.998	0.998	—
3	***Raw***: $0.994\,E_f q + 1.001\,Bqv\sin(\theta) - 0.415\,\frac{1}{q} + 0.208$	1.004	0.990	0.999	0.999	14.962
	Final: $E_f q + 1.001\,Bqv\sin(\theta)$	1.007	0.996	0.999	0.999	—
4	***Raw***: $1.005\,E_f q + 0.999\,Bqv\sin(\theta) - 0.109\,\sin(e^q) + 0.008$	0.982	1.060	0.999	0.998	17.241
	Final: $1.005\,E_f q + 0.999\,Bqv\sin(\theta) - 0.110\,\sin(e^q)$	0.982	1.061	0.999	0.998	—
5	***Raw***: $0.996\,E_f q + Bqv\sin(\theta) + 0.022$	1.021	0.973	0.998	0.999	15.910
	Final: $0.998\,E_f q + Bqv\sin(\theta)$	1.021	0.974	0.998	0.999	—
Noise level: $\sigma = 2.00$						
1	***Raw***: $1.002\,E_f q + Bqv\sin(\theta) + 0.270\,\frac{1}{E_f} - 0.054$	2.014	1.849	0.994	0.995	15.336
	Final: $1.005\,E_f q + Bqv\sin(\theta)$	2.016	1.846	0.994	0.995	—
2	***Raw***: $1.012\,E_f q + Bqv\sin(\theta) + 0.247\,\frac{1}{q} - 0.174$	1.995	2.103	0.994	0.992	15.370
	Final: $1.004\,E_f q + Bqv\sin(\theta)$	1.996	2.107	0.994	0.992	—

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Table 44: HierBOSSS results for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	***Raw***: $1.008\,E_f q + 1.001\,Bqv\sin(\theta) + 0.235\,\cos(B^2) - 0.105$	2.027	1.927	0.994	0.995	14.927
	Final: $0.999\,E_f q + 1.001\,Bqv\sin(\theta) + 0.237\,\cos(B^2)$	2.027	1.926	0.994	0.995	—
4	***Raw***: $1.001\,E_f q + Bqv\sin(\theta) + 0.345\,\frac{1}{q} - 0.199$	1.977	1.691	0.994	0.996	15.981
	Final: $0.994\,E_f q + Bqv\sin(\theta)$	1.978	1.689	0.994	0.996	—
5	***Raw***: $E_f q + 1.004\,Bqv\sin(\theta) + 0.236\,\cos(B) + 0.036$	1.962	1.958	0.995	0.994	15.997
	Final: $0.994\,E_f q + 1.004\,Bqv\sin(\theta)$	1.965	1.958	0.995	0.994	—

D.3.2 Top Expressions Ranked by JMP

Table 45: Top $r = 10$ expressions ranked by JMP learned by HierBOSSS for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. Test RMSE and R^2 are computed on the held-out 10% test set.

Rank	Learned expression	Test RMSE	Test R^2	log JMP
Noiseless setting, Repetition 1				
1–10	***Raw***: $E_f q + Bqv\sin(\theta) + 3.088\times10^{-7}$	1.782×10^{-7}	1.000	8367.026
	Final: $E_f q + Bqv\sin(\theta)$	8.641×10^{-8}	1.000	—
Noise level: $\sigma = 0.25$, Repetition 1				
1–10	***Raw***: $E_f q + Bqv\sin(\theta) - 0.091\,\frac{1}{B} + 0.038$	0.240	1.000	1517.584
	Final: $1.001\,E_f q + Bqv\sin(\theta)$	0.238	1.000	—
Noise level: $\sigma = 1.00$, Repetition 1				
1–10	***Raw***: $1.002\,E_f q + 0.999\,Bqv\sin(\theta) + 0.342\,\frac{1}{B} - 0.125$	0.983	0.999	-964.878
	Final: $1.003\,E_f q + 0.999\,Bqv\sin(\theta)$	0.978	0.999	—
Noise level: $\sigma = 2.00$, Repetition 1				

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Table 45: Top $r = 10$ expressions ranked by JMP learned by HierBOSSS for the Feynman equation I_12_11 (FCE): $F = q(E_f + Bv\sin\theta)$ across noise levels. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. Test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rank	Learned expression	Test RMSE	Test R^2	log JMP
1–10	***Raw***: $1.002\, E_f q + Bqv\sin(\theta) + 0.270\,\frac{1}{E_f} - 0.054$	1.849	0.995	-2252.841
	Final: $1.005\, E_f q + Bqv\sin(\theta)$	1.846	0.995	—

D.4 Results for I_24_6 (EHFO)

D.4.1 Expressions Learned by HierBOSSS

Table 46: HierBOSSS results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	***Raw***: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2 - 4.500\times10^{-9} + 4.780\times10^{-8}\frac{1}{\omega}$	2.817×10^{-8}	2.595×10^{-8}	1.000	1.000	66.103
	Final: $0.248\,m\omega^2x^2 + 0.248\,m\omega_0^2x^2$	2.816×10^{-8}	2.595×10^{-8}	1.000	1.000	—
2	***Raw***: $0.250\,mx^2(\omega^2+\omega_0^2) - 3.780\times10^{-7}\sin(\omega) + 6.420\times10^{-7}$	2.551×10^{-7}	2.464×10^{-7}	1.000	1.000	70.628
	Final: $0.250\,mx^2(\omega^2+\omega_0^2)$	2.004×10^{-7}	2.021×10^{-7}	1.000	1.000	—
3	***Raw***: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2 + 8.120\times10^{-8} - 6.460\times10^{-8}\frac{1}{m}$	3.260×10^{-8}	3.451×10^{-8}	1.000	1.000	62.323
	Final: $0.248\,m\omega^2x^2 + 0.248\,m\omega_0^2x^2$	2.589×10^{-8}	2.687×10^{-8}	1.000	1.000	—
4	***Raw***: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2 + 6.090\times10^{-9}\sin(\omega_0) + 2.530\times10^{-8}$	2.645×10^{-8}	2.542×10^{-8}	1.000	1.000	63.981
	Final: $0.248\,m\omega^2x^2 + 0.252\,m\omega_0^2x^2$	2.594×10^{-8}	2.524×10^{-8}	1.000	1.000	—
5	***Raw***: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2 - 1.150\times10^{-7}\sin(\omega) + 1.250\times10^{-7}$	3.481×10^{-8}	3.788×10^{-8}	1.000	1.000	60.097
	Final: $0.248\,m\omega^2x^2 + 0.248\,m\omega_0^2x^2$	2.323×10^{-8}	2.547×10^{-8}	1.000	1.000	—
Noise level: $\sigma = 0.15$						
1	***Raw***: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2 - 0.034\,\sin(x) + 0.038$	0.149	0.143	1.000	1.000	61.153
	Final: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2$	0.149	0.143	1.000	1.000	—
2	***Raw***: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2 - 0.012 + 0.016\,\frac{1}{\omega_0}$	0.153	0.146	1.000	1.000	63.261
	Final: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2$	0.153	0.146	1.000	1.000	—
3	***Raw***: $0.129\,m^2x^2e^{-\omega_0^2} - 0.514\,m\omega x^2\cos(\omega) + 0.925\,m\omega_0\left(x^2+\frac{1}{\omega_0}\right) - 1.310\,mx\sin(\sin(\omega_0)) - 0.514\,\omega_0 + 0.929$	0.483	0.496	0.999	0.999	71.026

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Table 46: HierBOSSS results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
	Final: $0.130\, m^2x^2e^{-\omega_0^2} - 0.510\, m\omega x^2\cos(\omega) - 1.300\, mx\sin(\sin(\omega_0)) + 0.920\, m(\omega_0 x^2 + 1) - 0.510\, \omega_0 + 0.930$	0.483	0.496	0.999	0.999	—
4	***Raw***: $-0.360\, m^3 + 0.541\, m^2\omega_0 x + 0.251\, m\omega^2x^2 + 0.541\, m + 0.035\, \omega_0^3x^3 - 0.360\, \omega_0 - 0.360\, x + 0.494$	0.571	0.622	0.998	0.998	63.494
	Final: $-0.360\, m^3 + 0.250\, m\omega^2x^2 + 0.540\, m(m\omega_0 x + 1) + 0.035\, \omega_0^3x^3 - 0.360\, \omega_0 - 0.360\, x + 0.490$	0.571	0.622	0.998	0.998	—
5	***Raw***: $0.665\, m\omega^2x + 0.885\, m\omega_0 x^2 - 0.018\, x^4(m + 2\omega)\cos(\omega) - 0.032\, (m + \omega + x + \sin(\sin(\omega_0)))^3 + 0.821$	0.633	0.586	0.998	0.998	61.686
	Final: $0.660\, m\omega^2x + 0.880\, m\omega_0 x^2 - 0.018\, x^4(m + 2\omega)\cos(\omega) - 0.032\, (m + \omega + x + \sin(\sin(\omega_0)))^3 + 0.820$	0.633	0.586	0.998	0.998	—
Noise level: $\sigma = 0.25$						
1	***Raw***: $0.250\, m\omega^2x^2 + 0.250\, m\omega_0^2x^2 + 0.050 - 0.048\, \frac{1}{\omega_0}$	0.247	0.245	1.000	1.000	62.123
	Final: $0.250\, m\omega^2x^2 + 0.250\, m\omega_0^2x^2$	0.247	0.245	1.000	1.000	—
2	***Raw***: $-0.497\, m\omega\omega_0 x^2 + 0.249\, mx^2(\omega + \omega_0)^2 - 0.031 + 0.029\, \frac{1}{\omega}$	0.251	0.242	1.000	1.000	65.230
	Final: $-0.500\, m\omega\omega_0 x^2 + 0.250\, mx^2(\omega + \omega_0)^2$	0.251	0.242	1.000	1.000	—
3	***Raw***: $3.980\, m\omega x + 1.030\, m\omega_0 x^2 - 7.600\, mx + 3.720\, m\left(\cos(\omega) + \frac{1}{\omega\omega_0}\right) + 1.030\, \omega_0 - 2.290$	0.875	0.879	0.996	0.996	63.661
	Final: $4.000\, m\omega x - 7.600\, mx + 3.700\, m\frac{\omega\omega_0\cos(\omega) + 1}{\omega\omega_0} + \omega_0(mx^2 + 1) - 2.300$	0.875	0.879	0.996	0.996	—

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Table 46: HierBOSSS results for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
4	***Raw***: $0.250\,m\omega^2x^2 + 0.250\,mx\left(\omega_0^2x + \frac{1}{x}\right) - 0.426\,m - 0.209\,\cos(m) + 0.289$	0.246	0.263	1.000	1.000	64.655
	Final: $0.250\,m\omega^2x^2 + 0.250\,m(\omega_0^2x^2 + 1) - 0.250\,m$	0.246	0.264	1.000	1.000	—
5	***Raw***: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2 - 0.033 + 0.041\,\frac{1}{\omega_0}$	0.255	0.244	1.000	1.000	62.556
	Final: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2$	0.255	0.243	1.000	1.000	—
Noise level: $\sigma = 1.00$						
1	***Raw***: $0.250\,mx^2(\omega^2 + \omega_0^2) - 0.098 + 0.220\,\frac{1}{\omega_0}$	1.007	0.925	0.995	0.995	66.461
	Final: $0.250\,mx^2(\omega^2 + \omega_0^2)$	1.008	0.922	0.995	0.995	—
2	***Raw***: $0.250\,m\omega^2x^2 + 0.247\,m\omega_0^2x^2 - 0.386\,\sin(x) + 0.667 - 0.412\,\frac{1}{\omega_0}$	0.995	1.061	0.995	0.992	63.424
	Final: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2$	0.998	1.054	0.995	0.992	—
3	***Raw***: $0.251\,m\omega^2x^2 + 0.130\,m\omega_0^2e^x - 0.259\,m\omega_0^2\cos^6(x) - 0.177\,\omega_0^2 + 0.251\,\omega_0x^2\sin(x) - 0.687$	1.033	0.971	0.994	0.995	67.736
	Final: $0.130\,m\omega_0^2e^x - 0.260\,m\omega_0^2\cos^6(x) - 0.180\,\omega_0^2 + 0.250\,x^2(m\omega^2 + \omega_0\sin(x)) - 0.690$	1.033	0.971	0.994	0.995	—
4	***Raw***: $0.250\,m\omega^2x^2 + 0.049\,\omega_0^3x^2 + 0.066\,x^2(m\omega_0 + \sin(m))^2 + 0.570 - 0.991\,\frac{1}{\omega_0}$	1.001	0.878	0.995	0.996	67.552
	Final: $0.250\,m\omega^2x^2 + 0.049\,\omega_0^3x^2 + 0.066\,x^2(m\omega_0 + \sin(m))^2 + 0.570 - 0.990\,\frac{1}{\omega_0}$	1.001	0.878	0.995	0.996	—
5	***Raw***: $0.249\,m\omega^2x^2 + 0.249\,m\omega_0^2x^2 + 0.197 - 0.315\,\frac{1}{x}$	0.983	0.975	0.996	0.995	67.711
	Final: $0.250\,m\omega^2x^2 + 0.250\,m\omega_0^2x^2$	0.984	0.974	0.996	0.995	—

D.4.2 Top Expressions Ranked by JMP

Table 47: Top $r = 10$ expressions ranked by JMP learned by HierBOSSS for the Feynman equation I_24_6 (EHFO): $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$ across noise levels. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. Test RMSE and R^2 are computed on the held-out 10% test set.

Rank	Learned expression	Test RMSE	Test R^2	log JMP
Noiseless setting, Repetition 1				
1–7	***Raw***: $0.250\, m\omega^2 x^2 + 0.250\, m\omega_0^2 x^2 - 4.500\times10^{-9} + 4.780\times10^{-8}\frac{1}{\omega}$	2.595×10^{-8}	1.000	8391.666
	Final: $0.248\, m\omega^2 x^2 + 0.248\, m\omega_0^2 x^2$	2.595×10^{-8}	1.000	—
8–10	***Raw***: $0.250\, m\omega^2 x^2 + 0.250\, m\omega_0^2 x^2 - 1.160\times10^{-7}\sin(\omega_0) + 1.270\times10^{-7}$	3.259×10^{-8}	1.000	8391.327
	Final: $0.248\, m\omega^2 x^2 + 0.248\, m\omega_0^2 x^2$	3.258×10^{-8}	1.000	—
Noise level: $\sigma = 0.15$, Repetition 1				
1–10	***Raw***: $0.250\, m\omega^2 x^2 + 0.250\, m\omega_0^2 x^2 - 0.034\sin(x) + 0.038$	0.143	1.000	2414.742
	Final: $0.250\, m\omega^2 x^2 + 0.250\, m\omega_0^2 x^2$	0.143	1.000	—
Noise level: $\sigma = 0.25$, Repetition 1				
1–10	***Raw***: $0.250\, m\omega^2 x^2 + 0.250\, m\omega_0^2 x^2 + 0.050 - 0.048\frac{1}{\omega_0}$	0.245	1.000	1508.432
	Final: $0.250\, m\omega^2 x^2 + 0.250\, m\omega_0^2 x^2$	0.245	1.000	—
Noise level: $\sigma = 1.00$, Repetition 1				
1	***Raw***: $0.250\, mx^2(\omega^2 + \omega_0^2) - 0.098 + 0.220\frac{1}{\omega_0}$	0.925	0.995	-1017.126
	Final: $0.250\, mx^2(\omega^2 + \omega_0^2)$	0.922	0.995	—
2–6	***Raw***: $0.250\, mx^2(\omega^2 + \omega_0^2) + 0.916\,\omega + 1.110\cos(\omega) - 1.400$	0.920	0.995	-1017.169
	Final: $0.250\, mx^2(\omega^2 + \omega_0^2)$	0.922	0.995	—
7–10	***Raw***: $0.250\, mx^2(\omega^2 + \omega_0^2) - 0.165\sin(m) + 0.182$	0.921	0.995	-1017.247
	Final: $0.250\, mx^2(\omega^2 + \omega_0^2)$	0.922	0.995	—

D.5 Results for I_50_26 (HOQN)

D.5.1 Expressions Learned by HierBOSSS

Table 48: HierBOSSS results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	***Raw***: $0.999\,\alpha x_1\cos^2(\omega t) + 1.000\,x_1\cos(\omega t) + 2.458\times10^{-5} - 3.023\times10^{-5}\frac{1}{t}$	1.511×10^{-5}	1.556×10^{-5}	1.000	1.000	61.611
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	1.336×10^{-5}	1.354×10^{-5}	1.000	1.000	—
2	***Raw***: $0.999\,x_1\{\alpha + \cos(\omega t)\} - 1.000\,\alpha x_1\sin^2(\omega t) - 9.780\times10^{-6}\sin(\omega) + 1.710\times10^{-5}$	7.647×10^{-6}	7.587×10^{-6}	1.000	1.000	59.610
	Final: $x_1\{-0.996\,\alpha\sin^2(\omega t) + \alpha + \cos(\omega t)\}$	7.647×10^{-6}	7.586×10^{-6}	1.000	1.000	—
3	***Raw***: $0.999\,\alpha x_1\cos^2(\omega t) + 1.000\,x_1\cos(\omega t) - 6.102\times10^{-6}\sin(\omega) + 1.358\times10^{-5}$	8.878×10^{-6}	9.173×10^{-6}	1.000	1.000	58.390
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	7.959×10^{-6}	8.139×10^{-6}	1.000	1.000	—
4	***Raw***: $-1.000\,\alpha x_1\sin^2(\omega t) + 1.000\,\alpha x_1 + 1.000\,x_1\cos(\omega t) + 5.479\times10^{-5}\omega - 5.040\times10^{-5}$	8.840×10^{-5}	9.163×10^{-5}	1.000	1.000	62.377
	Final: $-\alpha x_1\sin^2(\omega t) + \alpha x_1 + x_1\cos(\omega t)$	8.840×10^{-5}	9.163×10^{-5}	1.000	1.000	—
5	***Raw***: $0.999\,\alpha x_1\cos^2(\omega t) + 1.000\,x_1\cos(\omega t) + 2.642\times10^{-5} - 3.337\times10^{-5}\frac{1}{t}$	1.494×10^{-5}	1.378×10^{-5}	1.000	1.000	60.502
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	1.319×10^{-5}	1.195×10^{-5}	1.000	1.000	—
Noise level: $\sigma = 0.15$						
1	***Raw***: $0.184\,\alpha\omega t x_1\cos^2(\omega t) + 0.184\,\alpha\{x_1 + \sin(t + x_1)\} + 0.167\,t\sin(\omega + x_1) - 1.374\,\cos\{\exp(\cos(\omega t))\} + 0.326$	0.413	0.405	0.954	0.962	58.388
	Final: $0.180\,\alpha\omega t x_1\cos^2(\omega t) + 0.180\,\alpha\{x_1 + \sin(t + x_1)\} + 0.170\,t\sin(\omega + x_1) - 1.400\,\cos\{\exp(\cos(\omega t))\} + 0.330$	0.413	0.405	0.954	0.962	—
2	***Raw***: $1.002\,\alpha x_1\cos^2(\omega t) + 1.000\,x_1\cos(\omega t) + 0.008\,t - 0.015$	0.153	0.146	0.994	0.993	63.595

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Table 48: HierBOSSS results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.154	0.146	0.994	0.993	—
3	*Raw*: $-0.095\,x_1^2 + 0.243\,(\alpha + x_1)^2\cos^2(\omega t) - 1.720\,\sin\{\cos(t)\}\cos(\omega)\cos(x_1) + 1.801\,\cos(\omega t) + 0.275$	0.326	0.333	0.974	0.978	66.467
	Final: $-0.095\,x_1^2 + 0.240\,(\alpha + x_1)^2\cos^2(\omega t) - 1.700\,\sin\{\cos(t)\}\cos(\omega)\cos(x_1) + 1.800\,\cos(\omega t) + 0.280$	0.326	0.333	0.974	0.978	—
4	*Raw*: $0.097\,\alpha x_1^2 + 0.494\,\alpha x_1\cos(2\omega t) + 0.588\,\alpha + 0.588\,\sin\{x_1 + \cos(\omega)\} + 1.979\,\cos(\omega t) - 0.473$	0.394	0.379	0.960	0.956	63.475
	Final: $0.097\,\alpha x_1^2 + 0.490\,\alpha x_1\cos(2\omega t) + 0.590\,\alpha + 0.590\,\sin\{x_1 + \cos(\omega)\} + 2.000\,\cos(\omega t) - 0.470$	0.394	0.379	0.960	0.956	—
5	*Raw*: $0.529\,\alpha x_1 + 0.819\,\sin^6(t)\cos(x_1) + 1.984\,\cos(\omega t) + 2.025\,\cos(2\omega t) + 0.819\,\dfrac{\sin^6(t)}{\alpha\omega} - 0.066$	0.637	0.613	0.897	0.906	65.792
	Final: $0.530\,\alpha x_1 + 0.820\,\sin^6(t)\cos(x_1) + 2.000\,\cos(\omega t) + 2.000\,\cos(2\omega t) + 0.820\,\dfrac{\sin^6(t)}{\alpha\omega}$	0.637	0.613	0.897	0.906	—
Noise level: $\sigma = 0.18$						
1	*Raw*: $0.118\,\alpha^2 x_1 + 0.294\,\exp\{x_1\cos(\omega t)\} + 1.095\,\sin\{\cos(\omega t)\} + 1.558\,\cos(2\omega t) + 0.242$	0.546	0.548	0.926	0.914	60.667
	Final: $0.120\,\alpha^2 x_1 + 0.290\,\exp\{x_1\cos(\omega t)\} + 1.100\,\sin\{\cos(\omega t)\} + 1.550\,\cos(2\omega t) + 0.250$	0.546	0.548	0.926	0.914	—
2	*Raw*: $1.000\,\alpha x_1\cos^2(\omega t) + 1.000\,x_1\cos(\omega t) - 0.001 - 0.022\,\dfrac{1}{\omega}$	0.181	0.174	0.992	0.993	65.954
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.181	0.172	0.992	0.993	—
3	*Raw*: $0.999\,\alpha x_1\cos^2(\omega t) + 1.002\,x_1\cos(\omega t) + 0.016\,\dfrac{1}{x_1} - 0.007$	0.181	0.179	0.992	0.993	64.002
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.181	0.179	0.992	0.993	—
4	*Raw*: $0.996\,\alpha x_1\cos^2(\omega t) + 0.998\,x_1\cos(\omega t) + 0.025\,\sin(\omega) - 0.002$	0.177	0.189	0.992	0.992	59.533

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Table 48: HierBOSSS results for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
	Final: $\alpha x_1\cos^2(\omega t) + x_1\cos(\omega t) + 0.023\sin(\omega)$	0.177	0.189	0.992	0.992	—
5	***Raw***: $-1.592\,\alpha\sin^2(\omega t) + 1.829\,\alpha + 0.884\,x_1\exp\{\cos(\omega t)\} - 0.098\,x_1 - 2.114$	0.422	0.447	0.952	0.948	61.797
	Final: $-1.600\,\alpha\sin^2(\omega t) + 1.800\,\alpha + 0.880\,x_1\exp\{\cos(\omega t)\} - 0.098\,x_1 - 2.100$	0.422	0.447	0.952	0.948	—
Noise level: $\sigma = 0.20$						
1	***Raw***: $-0.013\,\alpha^2\omega\left(t + \frac{1}{t}\right) + 0.204\,\alpha\sin(x_1) + 0.204\,\alpha + 0.204\,x_1 + 0.054\,(\alpha x_1 + \omega + t)^2\cos^2(\omega t) + 0.203\left(t + x_1^2 + x_1 + \frac{1}{t}\right)\sin\{\cos(\omega t)\} - 0.816$	0.255	0.264	0.984	0.984	61.204
	Final: $-0.013\,\alpha^2\omega\left(t + \frac{1}{t}\right) + 0.200\,\alpha\sin(x_1) + 0.200\,\alpha + 0.200\,x_1 + 0.054\,(\alpha x_1 + \omega + t)^2\cos^2(\omega t) + 0.200\left(t + x_1^2 + x_1 + \frac{1}{t}\right)\sin\{\cos(\omega t)\} - 0.820$	0.255	0.264	0.984	0.984	—
2	***Raw***: $0.501\,\alpha x_1\cos(2\omega t) + 0.503\,\alpha x_1 + 1.002\,x_1\cos(\omega t) - 0.007$	0.200	0.211	0.990	0.984	62.652
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.201	0.211	0.990	0.984	—
3	***Raw***: $1.001\,\alpha x_1\cos^2(\omega t) + 1.005\,x_1\cos(\omega t) + 0.039\,\frac{1}{x_1} - 0.025$	0.203	0.192	0.990	0.991	62.807
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.203	0.192	0.990	0.991	—
4	***Raw***: $0.999\,\alpha x_1\cos^2(\omega t) + 0.996\,x_1\cos(\omega t) + 0.009\,t - 0.023$	0.198	0.169	0.989	0.993	64.214
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.198	0.169	0.989	0.993	—
5	***Raw***: $1.000\,\alpha x_1\cos^2(\omega t) + 0.999\,x_1\cos(\omega t) - 0.010\,t + 0.013$	0.197	0.195	0.990	0.990	68.162
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.197	0.195	0.990	0.990	—

D.5.2 Top Expressions Ranked by JMP

Table 49: Top $r = 10$ expressions ranked by JMP learned by HierBOSSS for the Feynman equation I_50_26 (HOQN): $x = x_1[\cos(\omega t) + \alpha\cos^2(\omega t)]$ across noise levels. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. Test RMSE and R^2 are computed on the held-out 10% test set.

Rank	Learned expression	Test RMSE	Test R^2	log JMP
Noiseless setting, Repetition 1				
1–10	***Raw***: $0.999\,\alpha x_1\cos^2(\omega t) + 1.000\,x_1\cos(\omega t) + 2.458\times10^{-5} - 3.023\times10^{-5}\frac{1}{t}$	1.556×10^{-5}	1.000	7792.650
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	1.354×10^{-5}	1.000	—
Noise level: $\sigma = 0.15$, Repetition 2				
1–8	***Raw***: $1.002\,\alpha x_1\cos^2(\omega t) + 1.000\,x_1\cos(\omega t) + 0.004\,t - 0.004\,(-t) - 0.015$	0.146	0.993	2364.333
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.146	0.993	—
9	***Raw***: $1.002\,\alpha x_1\cos^2(\omega t) + 1.000\,x_1\cos(\omega t) + 0.004\,t + 0.004\,t - 0.015$	0.146	0.993	2364.119
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.146	0.993	—
10	***Raw***: $1.002\,\alpha x_1\cos^2(\omega t) + 1.001\,x_1\cos(\omega t) - 0.002\,\cos(x_1) - 0.002\,\cos(x_1) + 8.634\times10^{-4}$	0.146	0.993	2364.089
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.146	0.993	—
Noise level: $\sigma = 0.18$, Repetition 2				
1–10	***Raw***: $1.000\,\alpha x_1\cos^2(\omega t) + 1.000\,x_1\cos(\omega t) - 0.011\,\frac{1}{\omega} - 0.011\,\frac{1}{\omega} - 8.373\times10^{-4}$	0.174	0.993	2067.444
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.172	0.994	—
Noise level: $\sigma = 0.20$, Repetition 2				
1–10	***Raw***: $0.501\,\alpha x_1\cos(2\omega t) + 1.002\,x_1\cos(\omega t) + 0.251\,\alpha x_1 + 0.251\,\alpha x_1 - 0.007$	0.211	0.984	1880.029
	Final: $x_1\{\alpha\cos^2(\omega t) + \cos(\omega t)\}$	0.211	0.983	—

D.6 Results for II_36_38 (FMMM)

D.6.1 Expressions Learned by HierBOSSS

Table 50: HierBOSSS results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} M$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
Noiseless setting						
1	*Raw*: $\frac{1.000 H \text{mom}}{T k_b} + \frac{1.000 M \alpha \text{mom}}{T c^2 \epsilon k_b} + 8.3\times10^{-7} - \frac{3.92\times10^{-7}}{H}$	4.840×10^{-7}	4.782×10^{-7}	1.000	1.000	1116.153
	Final: $\frac{1.0 H \text{mom}}{T k_b} + \frac{1.0 M \alpha \text{mom}}{T c^2 \epsilon k_b}$	4.840×10^{-7}	4.782×10^{-7}	1.000	1.000	—
2	*Raw*: $\frac{1.000 H \text{mom}}{T k_b} + \frac{1.000 M \alpha \text{mom}}{T c^2 \epsilon k_b} - 1.45\times10^{-6} + \frac{4.18\times10^{-6}}{c}$	9.941×10^{-7}	1.161×10^{-6}	1.000	1.000	1166.543
	Final: $\frac{1.0 H \text{mom}}{T k_b} + \frac{0.996 M \alpha \text{mom}}{T c^2 \epsilon k_b}$	9.936×10^{-7}	1.160×10^{-6}	1.000	1.000	—
3	*Raw*: $-1.9\times10^{-7} + 0.990\frac{H \text{mom}}{T k_b} + 0.190\frac{M^2 \text{mom}}{T \epsilon k_b c^2} + 0.0036\frac{M^2 \alpha \text{mom}}{T \epsilon k_b} + 0.00025\,\epsilon\left\{T\epsilon - \alpha^2 - \alpha k_b + c k_b + k_b - \epsilon \text{mom}\right\}$	0.035	0.030	0.999	0.999	1217.604
	Final: $\frac{0.990 H \text{mom}}{T k_b} + \frac{0.187 M^2 \text{mom}}{T \epsilon k_b c^2} + \frac{0.0033 M^2 \alpha \text{mom}}{T \epsilon k_b}$	0.035	0.030	0.999	0.999	—
4	*Raw*: $\frac{1.000 H \text{mom}}{T k_b} + \frac{1.000 M \alpha \text{mom}}{T c^2 \epsilon k_b} + 3.15\times10^{-7} + \frac{2.31\times10^{-6}}{c}$	1.043×10^{-6}	1.013×10^{-6}	1.000	1.000	1097.652
	Final: $\frac{0.990 H \text{mom}}{T k_b} + \frac{0.996 M \alpha \text{mom}}{T c^2 \epsilon k_b}$	1.043×10^{-6}	1.013×10^{-6}	1.000	1.000	—
5	*Raw*: $-1.7\times10^{-5} + 0.995\frac{H \text{mom}}{T k_b} + 0.156\frac{M^2 \text{mom}}{T \epsilon c^2 k_b} + 0.0025\frac{(M^2+\alpha)\alpha \text{mom}}{T \epsilon k_b} + 0.00006\frac{\sin(\cos(\epsilon)) + \alpha}{T + c + \epsilon}$	0.049	0.039	0.998	0.999	1113.009
	Final: $\frac{0.995 H \text{mom}}{T k_b} + \frac{0.154 M^2 \text{mom}}{T \epsilon c^2 k_b} + \frac{0.0025 (M^2+\alpha)\alpha \text{mom}}{T \epsilon k_b}$	0.049	0.039	0.998	0.999	—
Noise level: $\sigma = 0.15$						
1	*Raw*: $\frac{1.000 H \text{mom}}{T k_b} + \frac{0.995 M \alpha \text{mom}}{T c^2 \epsilon k_b} + 0.0488\sin(M) - 0.0361$	0.148	0.143	1.000	1.000	1039.296
	Final: $\frac{1.000 H \text{mom}}{T k_b} + \frac{0.990 M \alpha \text{mom}}{T c^2 \epsilon k_b}$	0.149	0.144	1.000	1.000	—
2	*Raw*: $-0.0173 + \frac{1.000 \text{mom}\left(H + \frac{M\alpha}{c^2 \epsilon}\right)}{T k_b} + \frac{0.0271}{M}$	0.153	0.146	1.000	1.000	1049.078
	Final: $\frac{1.000 \text{mom}\left(H c^2 \epsilon + M\alpha\right)}{T c^2 \epsilon k_b}$	0.153	0.146	1.000	1.000	—

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Table 50: HierBOSSS results for the Feynman equation II_36_38 (FMMM): $f = \frac{\mathrm{mom} H}{k_b T} + \frac{\mathrm{mom}\alpha}{\epsilon c^2 k_b T} M$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
3	*Raw*: $\frac{0.995M\alpha\mathrm{mom}}{Tc^2\epsilon k_b} + \frac{0.153}{k_b^3} + 0.159H\mathrm{mom} + 0.159T + 0.159k_b\cos(T) - 0.256 + 5.84H\mathrm{mom}e^{-T}e^{-k_b}$	0.160	0.162	0.999	0.999	1201.265
	Final: $\frac{0.990M\alpha\mathrm{mom}}{Tc^2\epsilon k_b} + \frac{0.15}{k_b^3} + 0.16H\mathrm{mom} + 0.16T + 0.16k_b\cos(T) - 0.26 + 5.7H\mathrm{mom}e^{-T}e^{-k_b}$	0.160	0.162	0.999	0.999	—
4	*Raw*: $\frac{0.970H\mathrm{mom}}{Tk_b} - 2.16 + \frac{0.538M\alpha}{Tc^2\epsilon k_b} + \frac{0.538M\mathrm{mom}}{Tc^2\epsilon k_b}$	0.156	0.157	0.999	0.999	1222.962
	Final: $\frac{0.97H\mathrm{mom}}{Tk_b} - 2.2 + \frac{0.54M(\alpha + \mathrm{mom})}{Tc^2\epsilon k_b}$	0.156	0.157	0.999	0.999	—
5	*Raw*: $\frac{1.000H\mathrm{mom}}{Tk_b} + \frac{1.000M\alpha\mathrm{mom}}{Tc^2\epsilon k_b} + 0.0349\sin(M) - 0.0342$	0.149	0.140	1.000	1.000	1041.973
	Final: $\frac{1.000H\mathrm{mom}}{Tk_b} + \frac{1.000M\alpha\mathrm{mom}}{Tc^2\epsilon k_b}$	0.149	0.139	1.000	1.000	—
Noise level: $\sigma = 0.20$						
1	*Raw*: $\frac{1.010H\mathrm{mom}}{Tk_b} + \frac{1.27M\mathrm{mom}(2\alpha + \epsilon)}{c^2k_b(T\epsilon + 1)^2} + \frac{0.00429\alpha^2(M\mathrm{mom} + M\cos(k_b) + 2)^2}{c^2\epsilon} - 0.0444 + \frac{0.0773}{c}$	0.199	0.204	1.000	1.000	1177.871
	Final: $\frac{1.000H\mathrm{mom}}{Tk_b} + \frac{1.3M\mathrm{mom}(2\alpha + \epsilon)}{c^2k_b(T\epsilon + 1)^2} + \frac{0.0043\alpha^2(M\mathrm{mom} + M\cos(k_b) + 2)^2}{c^2\epsilon} + \frac{0.077}{c}$	0.199	0.202	1.000	1.000	—
2	*Raw*: $\frac{1.000H\mathrm{mom}}{Tk_b} + \frac{1.01M\alpha\mathrm{mom}}{Tc^2\epsilon k_b} + 0.00215 - \frac{0.0429}{k_b}$	0.200	0.193	1.000	1.000	1082.862
	Final: $\frac{0.990H\mathrm{mom}}{Tk_b} + \frac{1.000M\alpha\mathrm{mom}}{Tc^2\epsilon k_b}$	0.201	0.193	1.000	1.000	—
3	*Raw*: $\frac{1.020H\mathrm{mom}}{Tk_b} + \frac{0.999M\alpha\mathrm{mom}}{Tc^2\epsilon k_b} + 2.48\times10^{-5} - \frac{0.0365}{\alpha}$	0.201	0.200	1.000	1.000	1003.359
	Final: $\frac{1.000H\mathrm{mom}}{Tk_b} + \frac{1.000M\alpha\mathrm{mom}}{Tc^2\epsilon k_b} - \frac{0.036}{\alpha}$	0.201	0.200	1.000	1.000	—
4	*Raw*: $\frac{1.000M\alpha\mathrm{mom}}{Tc^2\epsilon k_b} + 0.972 + \frac{0.415H(k_b\cos(T) + \mathrm{mom})}{k_b} + 0.296H + 0.296T - 1.74\cos\left(\frac{\mathrm{mom}\sin(T)}{T^2k_b}\right)$	0.214	0.222	1.000	1.000	1149.019
	Final: $\frac{1.000M\alpha\mathrm{mom}}{Tc^2\epsilon k_b} + 0.97 + \frac{0.41H(k_b\cos(T) + \mathrm{mom})}{k_b} + 0.3H + 0.3T - 1.7\cos\left(\frac{\mathrm{mom}\sin(T)}{T^2k_b}\right)$	0.214	0.222	1.000	1.000	—

Continued on next page

Table 50: HierBOSSS results for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom}H}{k_bT} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T}M$ across noise levels and repetitions. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and R^2 are computed on the 90% training set, while the test RMSE and R^2 are computed on the held-out 10% test set *(continued)*.

Rep.	Learned expression	Train RMSE	Test RMSE	Train R^2	Test R^2	Runtime (s)
5	*Raw*: $\frac{1.010H\text{mom}}{Tk_b} + \frac{1.22(M+\alpha)\cos(\alpha+\text{mom})}{Tc^2\epsilon k_b} - \frac{1.68\left(M^2+c\right)}{T\alpha^2c^2\epsilon k_b\text{mom}} - 0.00638 + \frac{5.73M}{Tc^2\epsilon k_b}$	0.205	0.198	1.000	1.000	1173.493
	Final: $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.2(M+\alpha)\cos(\alpha+\text{mom})}{Tc^2\epsilon k_b} - \frac{1.7\left(M^2+c\right)}{T\alpha^2c^2\epsilon k_b\text{mom}} + \frac{5.8M}{Tc^2\epsilon k_b}$	0.205	0.197	1.000	1.000	—
Noise level: $\sigma = 0.25$						
1	*Raw*: $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b} - 0.0288 + \frac{0.0723}{M}$	0.251	0.230	1.000	1.000	1018.236
	Final: $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$	0.252	0.231	1.000	1.000	—
2	*Raw*: $\frac{1.010H\text{mom}}{Tk_b} + \frac{0.979M\alpha\text{mom}}{Tc^2\epsilon k_b} + 0.0707\sin(\epsilon) - 0.0536$	0.249	0.266	1.000	1.000	1086.514
	Final: $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$	0.249	0.263	1.000	1.000	—
3	*Raw*: $\frac{0.988H\text{mom}}{Tk_b} + \frac{0.977M\alpha\text{mom}}{Tc^2\epsilon k_b} - 0.0183 + \frac{0.0663}{k_b}$	0.254	0.243	1.000	1.000	1073.973
	Final: $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.980M\alpha\text{mom}}{Tc^2\epsilon k_b}$	0.254	0.242	1.000	1.000	—
4	*Raw*: $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} + 0.0587\sin(\alpha) - 0.0482$	0.247	0.212	1.000	1.000	1078.759
	Final: $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$	0.247	0.211	1.000	1.000	—
5	*Raw*: $\frac{1.020H\text{mom}}{Tk_b} + \frac{4.6M\alpha\text{mom}}{c\epsilon\left(Tc^2+2Tck_b+Tk_b^2+c\right)} + 0.00862 - \frac{0.0695}{M}$	0.247	0.249	1.000	1.000	1042.153
	Final: $\frac{1.000H\text{mom}}{Tk_b} - \frac{0.058}{M} + \frac{4.6M\alpha\text{mom}}{c\epsilon\left(Tc^2+2Tck_b+Tk_b^2+c\right)}$	0.247	0.249	1.000	1.000	—

D.6.2 Top Expressions Ranked by JMP

Table 51: Top $r = 10$ expressions ranked by JMP learned by HierBOSSS for the Feynman equation II_36_38 (FMMM): $f = \frac{\text{mom} H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} M$ across noise levels. ***Raw*** expression correspond to the symbolic forest learned by HierBOSSS followed by the ***Final*** expression after post-MCMC symbolic model refinement. Test RMSE and R^2 are computed on the held-out 10% test set.

Rank	Learned expression	Test RMSE	Test R^2	log JMP
Noiseless setting, Repetition 1				
1–2	***Raw***: $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} + 8.300\times10^{-7} - 3.920\times10^{-7}\frac{1}{H}$	4.782×10^{-7}	1.000	8296.095
	Final: $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$	4.782×10^{-7}	1.000	—
3–10	***Raw***: $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} + 1.051\times10^{-6} - 5.877\times10^{-7}\sin(\text{mom})$	4.729×10^{-7}	1.000	8295.793
	Final: $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.005M\alpha\text{mom}}{Tc^2\epsilon k_b}$	3.377×10^{-7}	1.000	—
Noise level: $\sigma = 0.15$, Repetition 1				
1–10	***Raw***: $\frac{1.004H\text{mom}}{Tk_b} + \frac{0.995M\alpha\text{mom}}{Tc^2\epsilon k_b} + 0.049\sin(M) - 0.036$	0.143	0.985	2358.727
	Final: $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$	0.144	0.985	—
Noise level: $\sigma = 0.20$, Repetition 2				
1–10	***Raw***: $\frac{1.003H\text{mom}}{Tk_b} + \frac{1.006M\alpha\text{mom}}{Tc^2\epsilon k_b} + 0.002 - 0.043\frac{1}{k_b}$	0.193	0.969	1816.438
	Final: $\frac{0.990H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$	0.193	0.969	—
Noise level: $\sigma = 0.25$, Repetition 1				
1–10	***Raw***: $\frac{1.002H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b} - 0.029 + 0.072\frac{1}{M}$	0.230	0.951	1419.573
	Final: $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$	0.231	0.951	—

E Effective Symbolic Forest Size

Figure 1 displays the effective symbolic forest size K_{eff} selected by the post-MCMC symbolic model refinement in HierBOSSS. Since K_{eff} is determined by the BIC-based subset selection step, it measures how many symbolic tree components are retained in the ***Final*** expression after removing collinear and weakly contributing tree-structured symbolic expressions. The results show that HierBOSSS often selects effective forests smaller than the nominal value of K, particularly for the simpler Feynman equations such as I_12_2 (CL) and I_12_11 (FCE). In contrast, structurally richer equations such as I_24_6 (EHFO), II_36_38 (FMMM), and I_50_26 (HOQN) sometimes retain 3 or 4 components (closer to the nominal K used), especially under increased noise levels. This illustrates that the post-MCMC symbolic model refinement provides a data-adaptive notion of symbolic complexity, balancing parsimony with the need to preserve multiple meaningful additive components capturing the symbolic structure underlying the data.

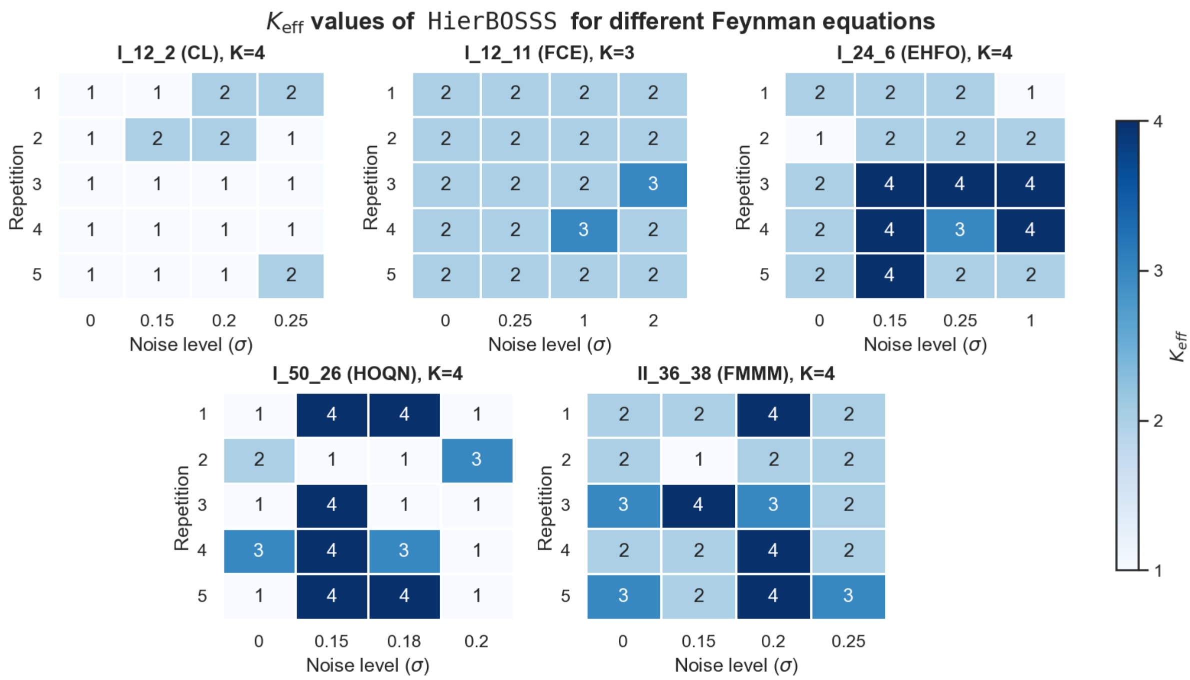


Figure 1: Effective symbolic forest size (K_{eff}) of HierBOSSS for the Feynman equations in I_12_2 (CL)-II_36_38 (FMMM), across 5 independent repetitions and different noise levels.

F Expression Catching, Complexity, Accuracy, and Runtime Summaries for Learning Feynman Equations

F.1 Structural Recovery

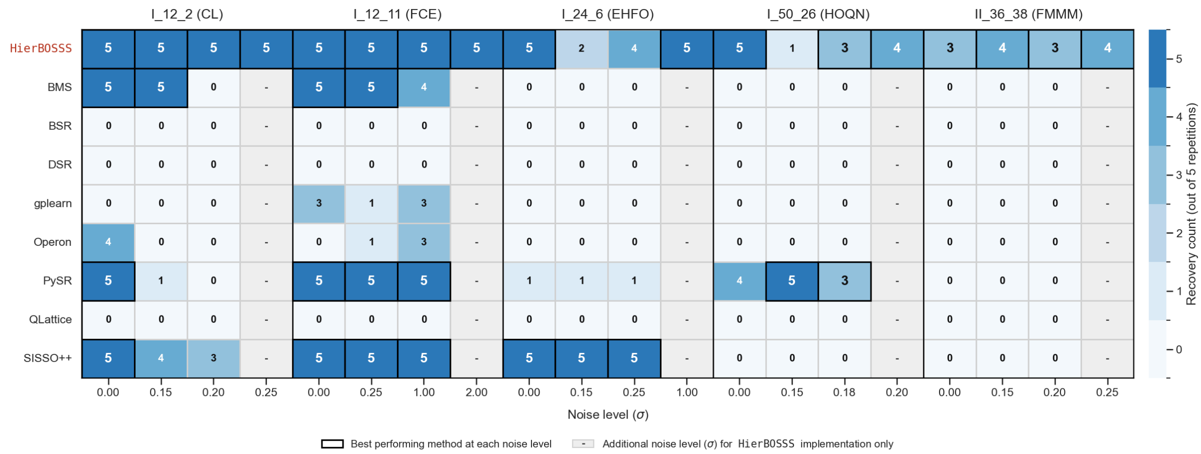
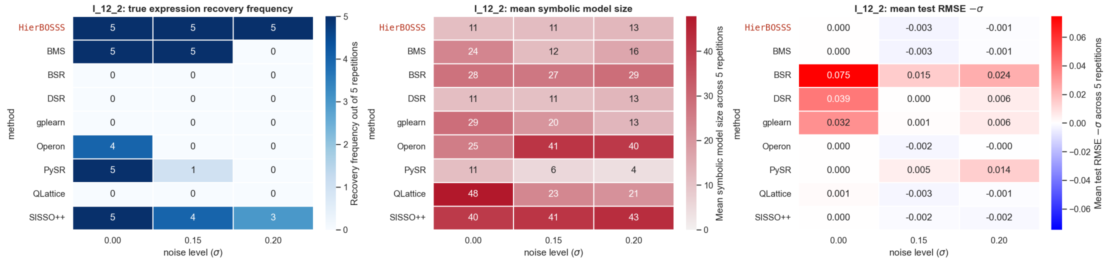


Figure 2: Structural recovery performance (out of 5 independent repetitions) of HierBOSSS and competitors across Feynman equations in I_12_2 (CL)-II_36_38 (FMMM) and different noise levels.

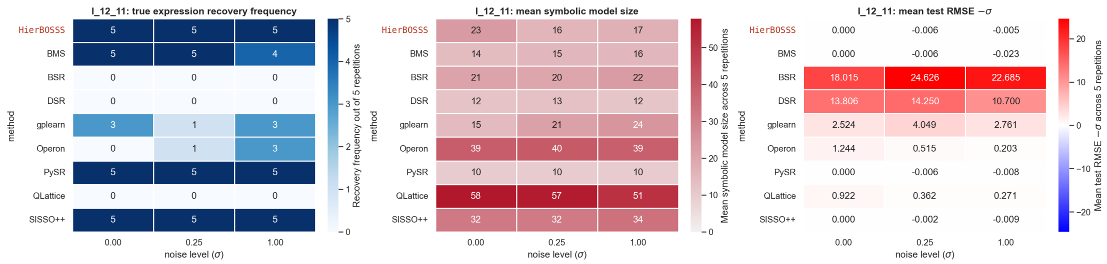
F.2 Expression Complexity

We evaluate the complexity of the symbolic expressions learned by HierBOSSS and the competing SR methods, for the Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). Predictive accuracy alone does not fully characterize the quality of a learned scientific expression, since an accurate expression may still be difficult to interpret if it contains redundant operators, unnecessary constants, deeply nested transformation, or unwieldy algebraic structure. This behavior is observed for several competitors (see Section C), where good numerical fit is sometimes achieved through complex nested output expressions rather than compact, interpretable scientific laws. We therefore examine expression complexity jointly with structural recovery and held-out predictive error. Experimental configurations for HierBOSSS and the competing methods are kept the same as those described in Sections D.1 and B.

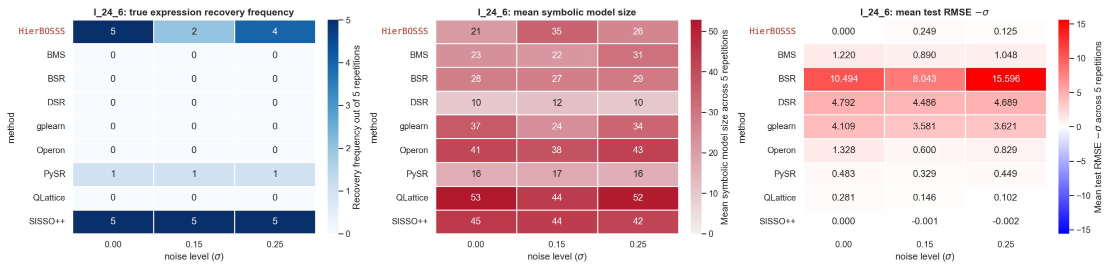
For a learned expression, we calculate its symbolic model size as the total number of nodes appearing in its expression tree [Imai Aldeia et al., 2025, La Cava et al., 2021]. Nodes include numerical constants, symbolic constants, variables or features, and mathematical operators. This gives a common symbolic model size measure across heterogeneous SR methods. For the competing methods, symbolic model size is computed from the expression returned (learned) by the corresponding method. For HierBOSSS, symbolic model size is computed from the ***Final*** expression. For each Feynman equation in I_12_2 (CL)-II_36_38 (FMMM), noise level, and method, we average the symbolic model size across the 5 independent repetitions. This average is reported together with the structural recovery frequency (out of the same 5 repetitions) and the mean (across the same 5 repetitions) test error gap, i.e., mean test $\texttt{RMSE} - \sigma$.



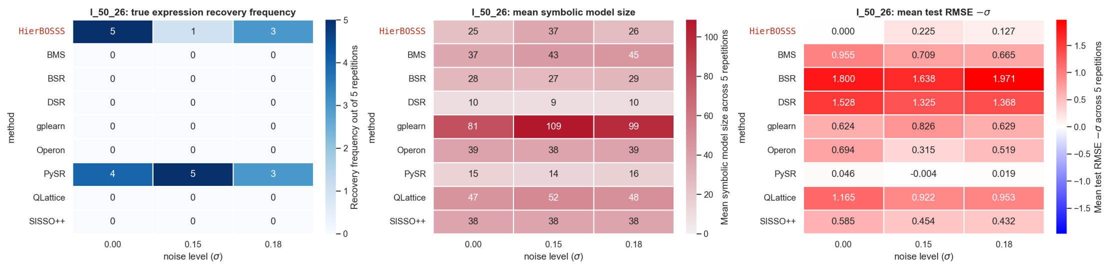
(a) I_12_2 (CL)



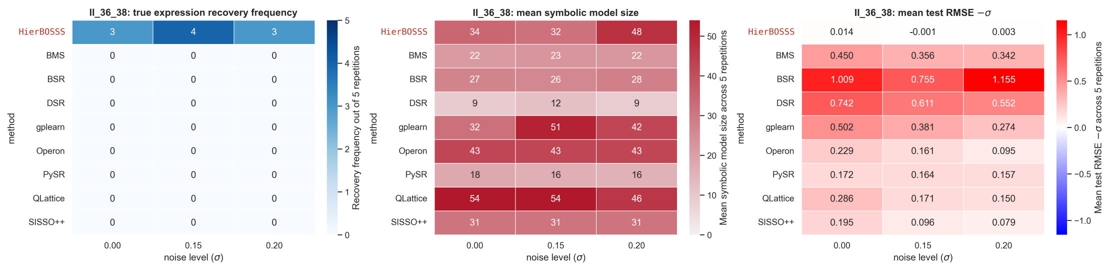
(b) I_12_11 (FCE)



(c) I_24_6 (EHFO)

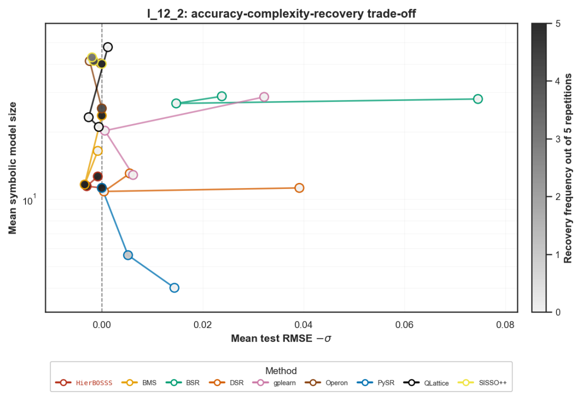


(d) I_50_26 (HOQN)

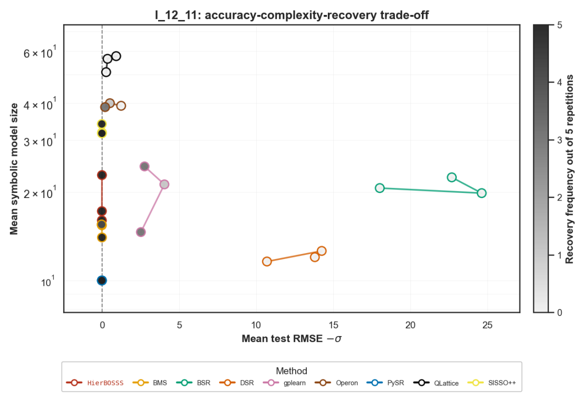


(e) II_36_38 (FMMM)

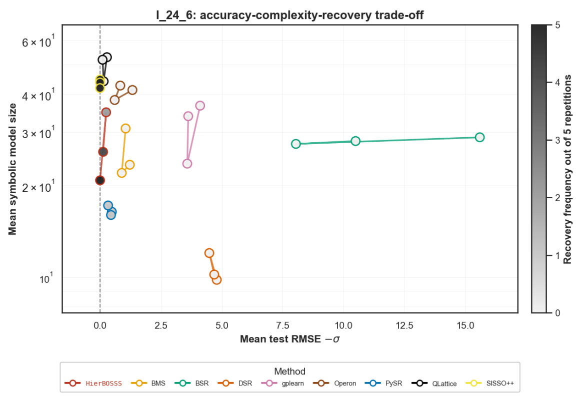
Figure 3: Expression recovery, symbolic model size, and predictive error gap. For each Feynman equation in I_12_2 (CL)-II_36_38 (FMMM), the three panels report structurally correct recovery frequency, mean symbolic model size, and mean test RMSE$-\sigma$, respectively, across 5 independent repetitions.



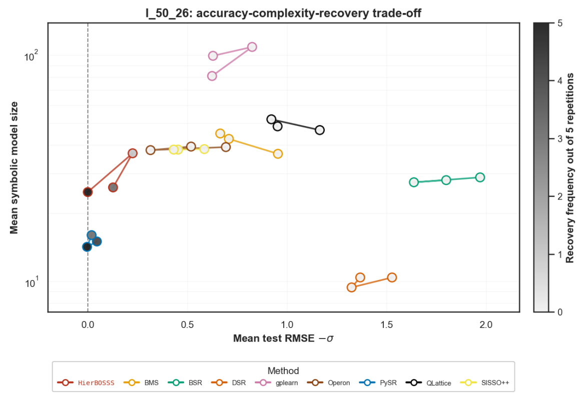
(a) I_12_2 (CL)



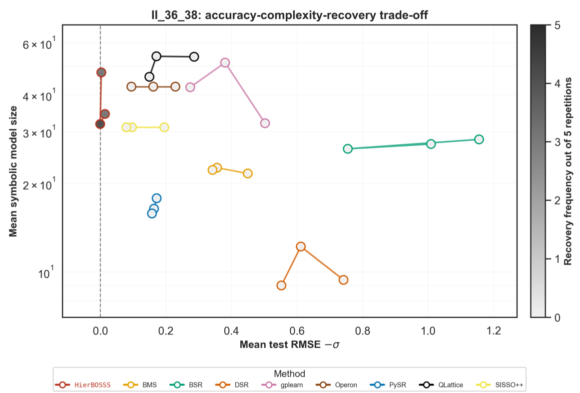
(b) I_12_11 (FCE)



(c) I_24_6 (EHFO)



(d) I_50_26 (HOQN)



(e) II_36_38 (FMMM)

Figure 4: Accuracy–complexity–recovery trade-off plots for the Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). Each point corresponds to a method at a given noise level. The horizontal axis reports mean test $\texttt{RMSE} - \sigma$, the vertical axis reports mean symbolic model size, and point shading indicates structurally correct recovery frequency, across 5 independent repetitions.

Figure 3 summarize these three diagnostics for each of the Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). These heatmaps distinguish methods that recover the true symbolic structure of the underlying Feynman equation from methods that achieve reasonable prediction through larger or only partially correct expressions.

To complement the preceding diagnostic heatmaps, we also provide accuracy-complexity-recovery trade-off plots in Figure 4. Each point represents one method at one noise level, with the horizontal axis measuring the mean test-error gap viz., mean test RMSE $-\ \sigma$, the vertical axis measuring the mean symbolic model size, and the point shading indicating the structurally correct recovery frequency, across 5 independent repetitions. Hence, the most desirable region is near the vertical zero-error reference line, with small model size and darker point shading. Such points correspond to expressions that are compact, predictive near the noise floor, and structurally correct.

Across the Feynman equations in I_12_2 (CL)-II_36_38 (FMMM), HierBOSSS generally exhibits a favorable balance of the accuracy-complexity-recovery trade-off. For the simpler equations I_12_2 (CL) and I_12_11 (FCE), HierBOSSS recovers the true symbolic structure across all displayed noise levels while maintaining compact model sizes and test errors close to the noise level. Some competitors, such as BMS, PySR, and SISSO + +, can also perform well for selected equations and noise levels, but their behavior is less uniform across the Feynman benchmark. Other methods often either fail to recover the true structure, produce substantially larger expressions, or exhibit larger test error gaps.

For more challenging equations, such as I_24_6 (EHFO), I_50_26 (HOQN), and II_36_38 (FMMM), the distinction between prediction and structural discovery becomes more pronounced. Several competing SR methods (e.g., DSR, BMS, PySR, operon, and SISSO + +) obtain reasonable prediction errors or compact expressions in some settings, but this does not necessarily imply recovery of the underlying Feynman equation. In particular, a method may lie close to the zero-error line while still having low recovery frequency, indicating that it has found a predictive approximation rather than the correct symbolic structure. Conversely, some methods return large expressions with many redundant or nested components, making the resulting formulas less interpretable even when predictive performance is acceptable.

The behavior of HierBOSSS is consistent. The depth-dependent splitting rule favors parsimonious symbolic tree structures, encoding the Occam's razor principle [Jefferys and Berger, 1992] directly into the symbolic forest prior over $\mathcal{T}$. In addition, post-MCMC symbolic model refinement applies an Occam's window-based [Madigan and Raftery, 1994] symbolic model selection using the joint marginal posterior of the symbolic forest, JMP$(\mathcal{T})$, retaining high-posterior expressions while discarding unnecessary symbolic components. This complete probabilistic SR framework of HierBOSSS allows it balance expressiveness, predictive accuracy, structural recovery, and symbolic parsimony.

F.3 Test RMSE and R^2

Table 52: Summary of test RMSE (computed on a held-out 10% test set) for HierBOSSS and competitors across Feynman equations in I_12_2 (CL)-II_36_38 (FMMM) and different noise levels. Each entry reports the mean ± standard deviation over 5 repetitions. — denotes additional noise level (σ) for HierBOSSS implementation only.

Noise (σ)	HierBOSSS	BMS	BSR	DSR	gplearn	operon	PySR	QLattice	SISSO + +
I_12_2 (CL)									
0.00	0.000 ± 0.000	0.000 ± 0.000	0.075 ± 0.005	0.039 ± 0.011	0.032 ± 0.013	0.000 ± 0.000	0.000 ± 0.000	0.001 ± 0.001	0.000 ± 0.000

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Table 52: Summary of test RMSE (computed on a held-out 10% test set) for HierBOSSS and competitors across Feynman equations in I_12_2 (CL)-II_36_38 (FMMM) and different noise levels. Each entry reports the mean ± standard deviation over 5 repetitions. — denotes additional noise level (σ) for HierBOSSS implementation only *(continued)*.

Noise (σ)	HierBOSSS	BMS	BSR	DSR	gplearn	operon	PySR	QLattice	SISSO++
0.15	0.147 ± 0.006	0.147 ± 0.006	0.165 ± 0.008	0.150 ± 0.005	0.151 ± 0.005	0.148 ± 0.007	0.155 ± 0.003	0.147 ± 0.006	0.148 ± 0.002
0.20	0.199 ± 0.007	0.199 ± 0.007	0.224 ± 0.015	0.205 ± 0.006	0.206 ± 0.008	0.200 ± 0.007	0.214 ± 0.013	0.199 ± 0.005	0.198 ± 0.003
0.25	0.239 ± 0.019	—	—	—	—	—	—	—	—
I_12_11 (FCE)									
0.00	0.000 ± 0.000	0.000 ± 0.000	18.015 ± 1.724	13.805 ± 4.072	2.524 ± 3.543	1.244 ± 0.868	0.000 ± 0.000	0.922 ± 0.451	0.000 ± 0.000
0.25	0.244 ± 0.011	0.244 ± 0.011	24.876 ± 3.416	14.500 ± 1.855	4.298 ± 3.260	0.765 ± 0.435	0.244 ± 0.011	0.612 ± 0.285	0.248 ± 0.003
1.00	0.995 ± 0.038	0.977 ± 0.014	23.685 ± 4.119	11.700 ± 4.319	3.760 ± 3.785	1.203 ± 0.304	0.992 ± 0.038	1.271 ± 0.236	0.991 ± 0.011
2.00	1.905 ± 0.153	—	—	—	—	—	—	—	—
I_24_6 (EHFO)									
0.00	0.000 ± 0.000	1.219 ± 0.090	10.494 ± 2.528	4.792 ± 0.087	4.109 ± 1.872	1.328 ± 0.536	0.482 ± 0.377	0.281 ± 0.012	0.000 ± 0.000
0.15	0.399 ± 0.236	1.040 ± 0.325	8.193 ± 1.748	4.637 ± 0.227	3.731 ± 1.025	0.750 ± 0.613	0.479 ± 0.261	0.296 ± 0.036	0.149 ± 0.002
0.25	0.375 ± 0.282	1.298 ± 0.370	15.846 ± 5.789	4.939 ± 0.258	3.871 ± 0.612	1.080 ± 0.387	0.699 ± 0.274	0.352 ± 0.066	0.248 ± 0.003
1.00	0.960 ± 0.066	—	—	—	—	—	—	—	—
I_50_26 (HOQN)									
0.00	0.000 ± 0.000	0.955 ± 0.139	1.800 ± 0.093	1.528 ± 0.151	0.624 ± 0.391	0.694 ± 0.217	0.046 ± 0.102	1.165 ± 0.134	0.585 ± 0.015
0.15	0.375 ± 0.167	0.859 ± 0.239	1.788 ± 0.162	1.475 ± 0.122	0.975 ± 0.585	0.465 ± 0.297	0.146 ± 0.006	1.072 ± 0.097	0.603 ± 0.012
0.18	0.307 ± 0.178	0.845 ± 0.277	2.151 ± 0.316	1.548 ± 0.170	0.809 ± 0.198	0.699 ± 0.357	0.199 ± 0.029	1.133 ± 0.139	0.612 ± 0.012
0.20	0.206 ± 0.036	—	—	—	—	—	—	—	—
II_36_38 (FMMM)									
0.00	0.014 ± 0.019	0.450 ± 0.047	1.009 ± 0.134	0.742 ± 0.082	0.502 ± 0.113	0.229 ± 0.035	0.172 ± 0.110	0.286 ± 0.035	0.195 ± 0.023
0.15	0.150 ± 0.010	0.506 ± 0.075	0.905 ± 0.120	0.761 ± 0.104	0.531 ± 0.086	0.311 ± 0.038	0.314 ± 0.061	0.321 ± 0.023	0.246 ± 0.015
0.20	0.203 ± 0.011	0.542 ± 0.096	1.355 ± 0.538	0.752 ± 0.069	0.474 ± 0.087	0.295 ± 0.042	0.357 ± 0.098	0.350 ± 0.048	0.279 ± 0.013
0.25	0.239 ± 0.020	—	—	—	—	—	—	—	—

Table 53: Summary of test R^2 (computed on a held-out 10% test set) for HierBOSSS and competitors across Feynman equations in I_12_2 (CL)-II_36_38 (FMMM) and different noise levels. Each entry reports the mean ± standard deviation over 5 repetitions. — denotes additional noise level (σ) for HierBOSSS implementation only.

Noise (σ)	HierBOSSS	BMS	BSR	DSR	gplearn	operon	PySR	QLattice	SISSO++
I_12_2 (CL)									
0.00	1.000 ± 0.000	1.000 ± 0.000	0.210 ± 0.096	0.776 ± 0.092	0.827 ± 0.113	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
0.15	0.243 ± 0.060	0.246 ± 0.063	0.053 ± 0.081	0.206 ± 0.060	0.202 ± 0.080	0.235 ± 0.059	0.152 ± 0.085	0.234 ± 0.068	0.249 ± 0.030
0.20	0.196 ± 0.073	0.196 ± 0.073	−0.011 ± 0.078	0.145 ± 0.058	0.140 ± 0.047	0.190 ± 0.074	0.073 ± 0.033	0.194 ± 0.077	0.155 ± 0.024
0.25	0.101 ± 0.076	—	—	—	—	—	—	—	—
I_12_11 (FCE)									
0.00	1.000 ± 0.000	1.000 ± 0.000	0.499 ± 0.047	0.691 ± 0.166	0.976 ± 0.037	0.997 ± 0.004	1.000 ± 0.000	0.999 ± 0.001	1.000 ± 0.000
0.25	1.000 ± 0.000	1.000 ± 0.000	0.090 ± 0.223	0.687 ± 0.092	0.961 ± 0.041	0.999 ± 0.001	1.000 ± 0.000	1.000 ± 0.001	1.000 ± 0.000
1.00	0.999 ± 0.001	0.999 ± 0.001	0.111 ± 0.329	0.774 ± 0.139	0.962 ± 0.050	0.998 ± 0.001	0.999 ± 0.001	0.998 ± 0.001	0.998 ± 0.000
2.00	0.994 ± 0.002	—	—	—	—	—	—	—	—
I_24_6 (EHFO)									
0.00	1.000 ± 0.000	0.993 ± 0.001	0.459 ± 0.240	0.892 ± 0.007	0.909 ± 0.080	0.991 ± 0.007	0.998 ± 0.002	1.000 ± 0.000	1.000 ± 0.000
0.15	0.999 ± 0.001	0.994 ± 0.003	0.657 ± 0.147	0.894 ± 0.013	0.927 ± 0.040	0.996 ± 0.005	0.999 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
0.25	0.999 ± 0.002	0.991 ± 0.005	−0.397 ± 0.980	0.880 ± 0.013	0.925 ± 0.024	0.994 ± 0.005	0.997 ± 0.002	0.999 ± 0.001	1.000 ± 0.000
1.00	0.995 ± 0.002	—	—	—	—	—	—	—	—
I_50_26 (HOQN)									
0.00	1.000 ± 0.000	0.758 ± 0.078	0.159 ± 0.063	0.390 ± 0.120	0.869 ± 0.162	0.867 ± 0.088	0.997 ± 0.007	0.643 ± 0.087	0.908 ± 0.003
0.15	0.959 ± 0.033	0.792 ± 0.151	0.180 ± 0.106	0.434 ± 0.122	0.716 ± 0.258	0.923 ± 0.074	0.994 ± 0.001	0.704 ± 0.050	0.903 ± 0.004
0.18	0.968 ± 0.036	0.804 ± 0.150	−0.112 ± 0.320	0.433 ± 0.085	0.838 ± 0.083	0.865 ± 0.140	0.990 ± 0.004	0.691 ± 0.083	0.900 ± 0.005
0.20	0.988 ± 0.004	—	—	—	—	—	—	—	—
II_36_38 (FMMM)									
0.00	1.000 ± 0.001	0.826 ± 0.040	0.137 ± 0.128	0.530 ± 0.081	0.776 ± 0.096	0.953 ± 0.022	0.964 ± 0.036	0.927 ± 0.029	0.968 ± 0.005
0.15	1.000 ± 0.001	0.791 ± 0.052	0.332 ± 0.138	0.529 ± 0.099	0.768 ± 0.064	0.921 ± 0.020	0.920 ± 0.026	0.917 ± 0.007	0.951 ± 0.005
0.20	1.000 ± 0.000	0.770 ± 0.073	−0.616 ± 1.231	0.560 ± 0.085	0.851 ± 0.031	0.932 ± 0.020	0.896 ± 0.053	0.905 ± 0.026	0.937 ± 0.006
0.25	1.000 ± 0.000	—	—	—	—	—	—	—	—

F.4 Compute Runtime

Table 54: Summary of runtime in seconds for HierBOSSS and competitors across Feynman equations in I_12_2 (CL)-II_36_38 (FMMM) and different noise levels. Each entry reports the mean ± standard deviation over 5 repetitions. — denotes additional noise level (σ) for HierBOSSS implementation only.

Noise (σ)	HierBOSSS	BMS	BSR	DSR	gplearn	operon	PySR	QLattice	SISSO ++
I_12_2 (CL)									
0.00	51.390 ± 6.365	210.905 ± 14.077	188.939 ± 0.420	76.420 ± 15.339	17.349 ± 1.267	19.513 ± 0.976	69.953 ± 4.908	93.850 ± 1.890	16.639 ± 1.111
0.15	56.357 ± 8.312	361.300 ± 6.939	189.356 ± 1.034	107.400 ± 7.225	15.700 ± 0.688	24.173 ± 1.647	72.777 ± 6.282	83.086 ± 2.549	17.552 ± 0.298
0.20	57.842 ± 1.986	399.125 ± 15.148	189.955 ± 2.828	123.560 ± 24.557	15.695 ± 0.783	25.032 ± 2.637	81.711 ± 15.020	73.609 ± 4.531	17.585 ± 0.505
0.25	57.964 ± 1.210	—	—	—	—	—	—	—	—
I_12_11 (FCE)									
0.00	15.138 ± 0.313	272.269 ± 30.810	165.651 ± 1.424	48.720 ± 8.014	16.342 ± 1.109	26.616 ± 2.265	127.246 ± 8.985	93.075 ± 2.206	43.668 ± 1.667
0.25	14.082 ± 0.891	293.312 ± 17.077	165.657 ± 1.107	42.600 ± 0.911	16.877 ± 0.777	24.694 ± 1.669	121.428 ± 9.605	92.105 ± 1.032	42.753 ± 0.650
1.00	15.103 ± 1.520	295.094 ± 40.632	165.410 ± 1.454	42.500 ± 0.283	16.751 ± 1.640	24.779 ± 1.212	123.313 ± 13.838	93.531 ± 4.214	43.789 ± 1.669
2.00	15.522 ± 0.460	—	—	—	—	—	—	—	—
I_24_6 (EHFO)									
0.00	64.626 ± 4.014	235.951 ± 19.032	196.386 ± 4.105	31.800 ± 1.131	16.797 ± 2.877	26.912 ± 2.841	69.940 ± 10.127	95.689 ± 2.282	19.535 ± 0.596
0.15	64.124 ± 3.986	232.742 ± 17.890	192.569 ± 3.252	32.720 ± 1.410	14.813 ± 0.947	23.552 ± 3.040	76.673 ± 8.929	94.665 ± 1.604	19.015 ± 0.130
0.25	63.645 ± 1.326	242.652 ± 6.945	194.237 ± 6.542	31.640 ± 2.182	15.715 ± 1.215	23.836 ± 1.224	67.912 ± 9.139	94.928 ± 2.089	19.163 ± 0.330
1.00	66.577 ± 1.839	—	—	—	—	—	—	—	—
I_50_26 (HOQN)									
0.00	60.498 ± 1.581	274.056 ± 27.935	191.599 ± 1.395	44.880 ± 1.018	19.290 ± 1.877	24.758 ± 3.164	108.801 ± 15.951	85.273 ± 1.281	26.372 ± 0.552
0.15	63.543 ± 3.170	294.648 ± 29.925	191.651 ± 1.057	45.760 ± 1.031	20.649 ± 5.276	26.250 ± 1.756	103.283 ± 3.208	86.189 ± 7.616	26.119 ± 0.638
0.18	62.391 ± 2.588	300.061 ± 24.232	193.930 ± 5.809	46.000 ± 1.304	20.935 ± 2.589	28.398 ± 1.562	100.307 ± 11.418	85.358 ± 5.604	25.999 ± 0.704
0.20	63.808 ± 2.657	—	—	—	—	—	—	—	—
II_36_38 (FMMM)									
0.00	1142.192 ± 49.481	2486.476 ± 290.751	1965.850 ± 66.240	41.420 ± 0.638	19.391 ± 1.055	27.523 ± 2.473	77.939 ± 11.421	113.685 ± 16.482	17.127 ± 0.111
0.15	1110.915 ± 92.768	2469.174 ± 149.699	1933.160 ± 16.120	59.320 ± 14.941	20.437 ± 2.257	26.457 ± 2.915	90.886 ± 25.039	110.192 ± 5.903	16.914 ± 0.131
0.20	1117.321 ± 74.164	2601.064 ± 644.774	1931.850 ± 17.040	65.980 ± 11.217	21.182 ± 4.144	29.408 ± 5.562	118.457 ± 59.492	120.602 ± 28.463	16.825 ± 0.169

Continued on next page

Table 54: Summary of runtime in seconds for HierBOSSS and competitors across Feynman equations in I_12_2 (CL)-II_36_38 (FMMM) and different noise levels. Each entry reports the mean $\pm$ standard deviation over 5 repetitions. — denotes additional noise level (σ) for HierBOSSS implementation only *(continued)*.

Noise (σ)	HierBOSSS	BMS	BSR	DSR	gplearn	operon	PySR	QLattice	SISSO++
0.25	1059.927 $\pm$ 28.779	—	—	—	—	—	—	—	—

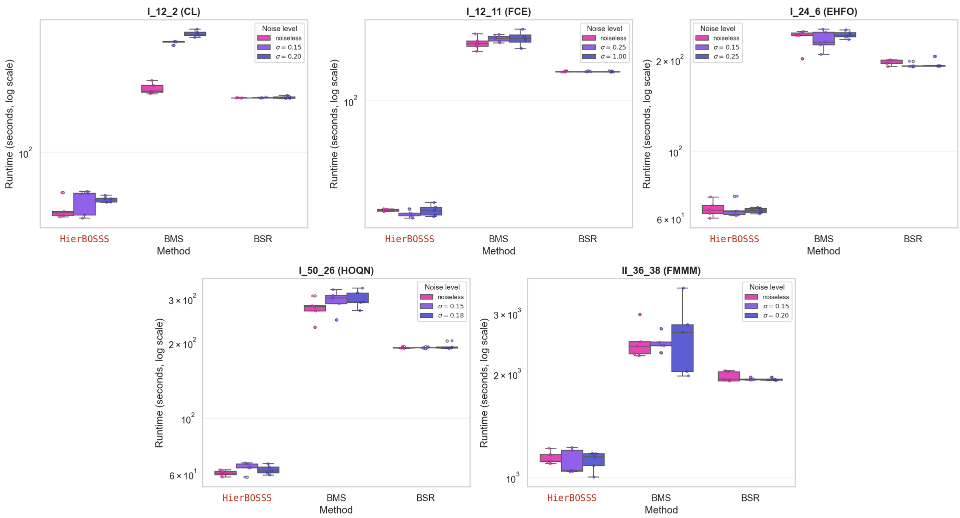


Figure 5: Compute runtimes (in log-scale) of HierBOSSS, BMS (5 parallel MCMC chains), and BSR (5 restarts) for the Feynman equations in I_12_2 (CL)-II_36_39 (FMMM), across different noise levels and 5 independent repetitions. Implementation details of HierBOSSS and the Bayesian competitors for each Feynman equation follow from Sections D.1, B.7, and B.8.

G Ablation Study for HierBOSSS and Practical Diagnostic for Choosing the Symbolic Forest Size

We conduct an ablation study to assess the sensitivity of HierBOSSS to the nominal number of symbolic trees K in the symbolic forest component $\mathcal{T}$. At the same time, this study provides a practical diagnostic for choosing K. The study is performed on two representative Feynman equations, I_12_2 (CL) and I_12_11 (FCE), under a fixed noise level $\sigma = 0.25$. For each equation, we vary K over $\{2, 3, 4, 5, 6, 8, 10, 12\}$ and run HierBOSSS with 5 independent repetitions under the experimental settings described in Section D.1.

Figures 6 and 7 summarize the effect of K on the predictive accuracy of HierBOSSS. Across the considered values of K, both the ***Raw*** and ***Final*** expressions exhibit stable average test RMSE and test R^2 over 5 repetitions. For I_12_2 (CL), increasing K beyond small to moderate values yields little change in predictive performance. A similar pattern is observed for I_12_11 (FCE), where the test RMSE and test R^2 remain highly stable across the full range of K. These results indicate that, once K is large enough to represent the main additive symbolic components, additional trees do not materially improve predictive performance.

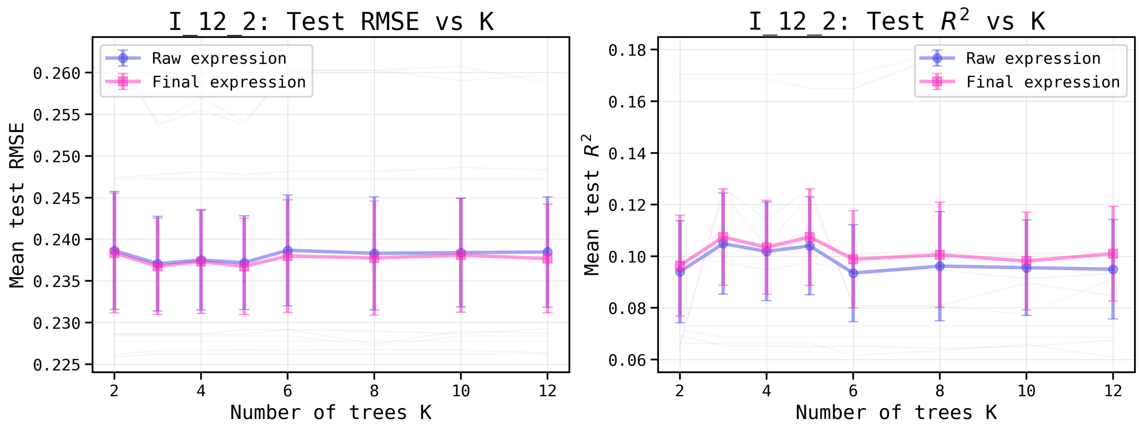


Figure 6: Average (over 5 independent repetitions) test RMSE and R^2 of HierBOSSS across different values of K for learning I_12_2 (CL) under noise level $\sigma = 0.25$.

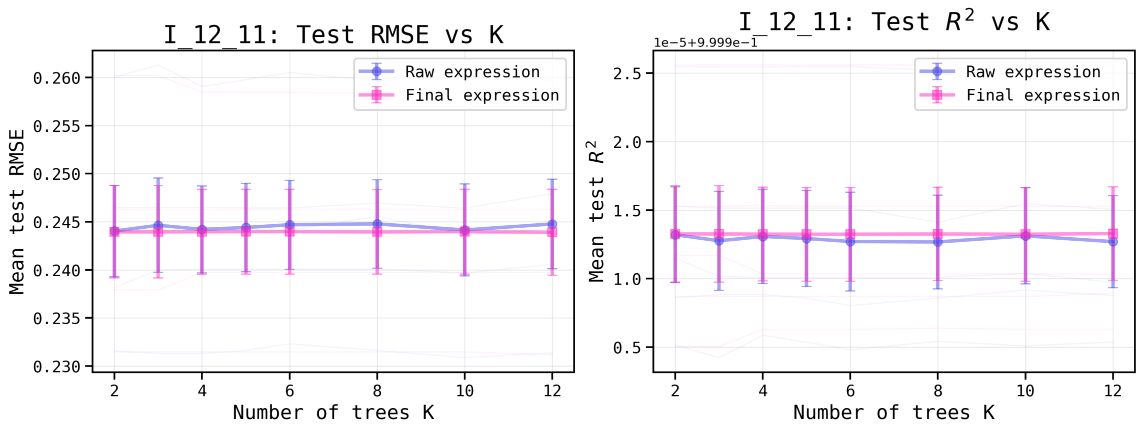
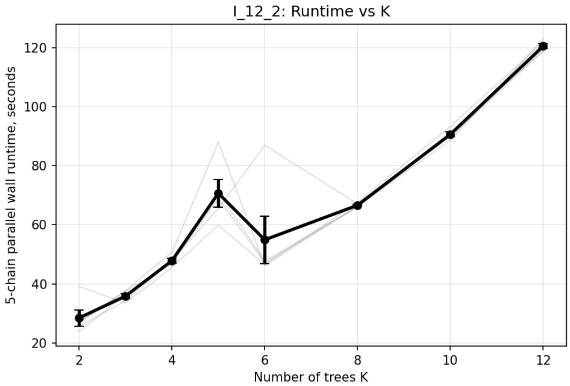
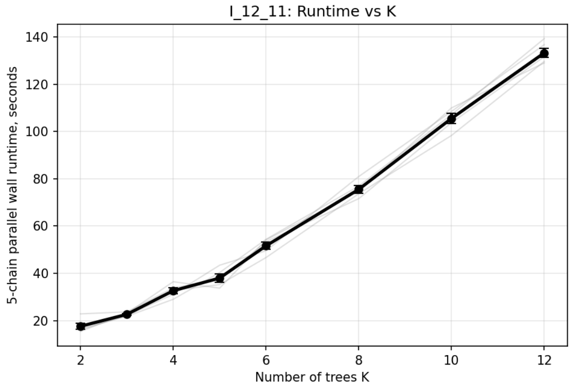


Figure 7: Average (over 5 independent repetitions) test RMSE and R^2 of HierBOSSS across different values of K for learning I_12_11 (FCE) under noise level $\sigma = 0.25$.

The runtime behavior is shown in Figure 8. As expected, the computational cost increases with K. This increase is moderate for smaller values of K, but becomes more pronounced for larger symbolic forest sizes. Thus, although larger K expands the symbolic expression search space, it also introduces additional computational overhead without commensurate gains in predictive accuracy for these Feynman equations.



(a) I_12_2 (CL)



(b) I_12_11 (FCE)

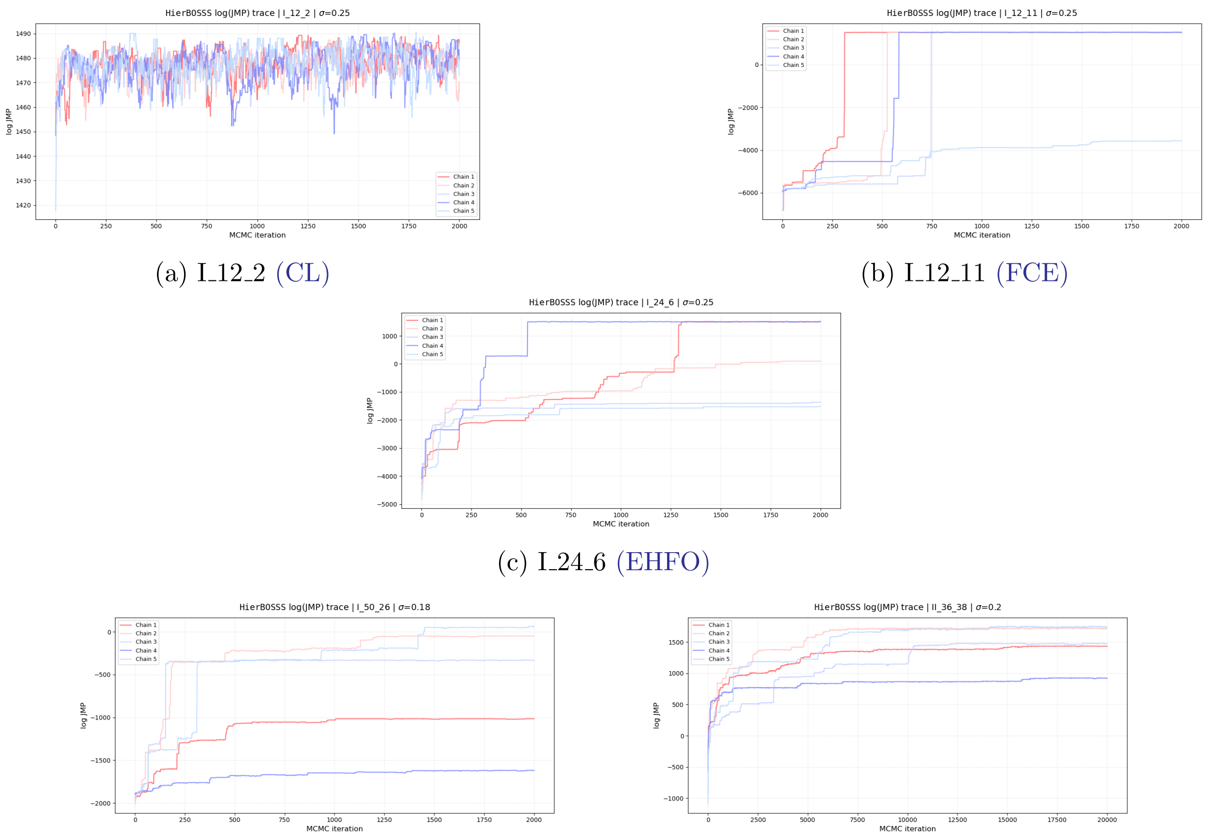
Figure 8: Runtime behavior of HierBOSSS across different values of K under noise level $\sigma = 0.25$ for I_12_2 (CL) and I_12_11 (FCE). Each panel reports the wall-clock runtime for 5 parallel chains across 5 independent repetitions, with the mean trend shown in black and repetition-level variability shown in gray.

Taken together, the preceding ablation study suggests a principled strategy for choosing K: select the smallest or a moderate value of K after which predictive accuracy stabilizes, while avoiding unnecessarily large forests that increase runtime. In this sense, K serves as a nominal expressivity parameter controlling the breadth of symbolic exploration, rather than the final number of symbolic components retained in the reported expression. The post-MCMC symbolic model refinement step of HierBOSSS subsequently determines the effective forest size K_{eff} in a data-adaptive manner by removing redundant or collinear symbolic tree structures. Therefore, a moderate nominal K provides sufficient exploratory flexibility, while the ***Final*** expression remains parsimonious through the refinement step.

H Posterior Diagnostics for the Bayesian SR Methods

We provide trace plots to examine the stochastic search and posterior convergence behavior of HierBOSSS, BMS, and BSR, while learning the Feynman equations in I_12_2 (CL)-II_36_38 (FMMM). For each equation, we show one representative repetition at the corresponding fixed noise level (σ), where HierBOSSS and BMS have been implemented with 5 parallel chains and BSR is executed for 5 restarts. All other experimental configurations follow from Sections D.1, B.7, and B.8. These traces visualize the stability of the posterior search and exploration of the Bayesian SR methods over the symbolic expression space. Since the plotted quantity differs across methods, the traces should be interpreted method-wise rather than as direct numerical comparisons across methods.

For HierBOSSS, the traces in Figure 9 show the evolution of the log joint marginal posterior value of $\mathcal{T}$, denoted by log(JMP), across MCMC iterations. These traces indicate how the sampler explores the posterior distribution over symbolic forests and whether the chains move toward regions of high posterior support. Stable traces across chains suggest proper mixing and that the posterior search repeatedly identifies comparable high-scoring symbolic structures.



(a) I_12_2 (CL)

(b) I_12_11 (FCE)

(c) I_24_6 (EHFO)

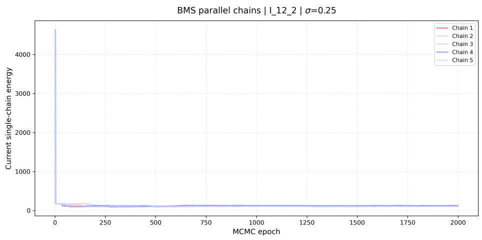
(d) I_50_26 (HOQN)

(e) II_36_38 (FMMM)

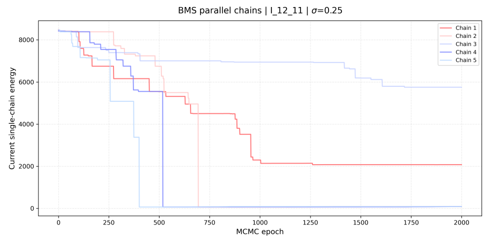
Figure 9: Trace plots for HierBOSSS across the Feynman equations in I_12_2 (CL)–II_36_38 (FMMM).

For BMS, the traces in Figure 10 report the current single-chain energy, or the description length [Guimerà et al., 2020], over MCMC epochs. This energy combines the data-fit term and the prior or symbolic complexity penalty. Lower values indicate models with better posterior support. Therefore, downward movements correspond to improvements in fit, parsimony, or both, while flat regions indicate that the chain remains in similar symbolic states.

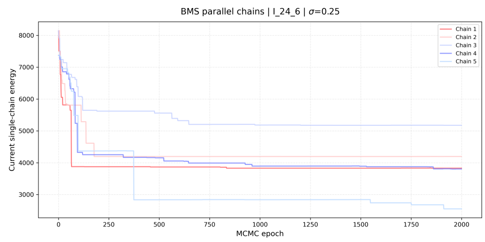
For BSR, the traces show the accepted-state log likelihood across proposal iterations [Jin et al., 2020]. Since the trace updates only when a proposed symbolic modification is accepted, flat segments correspond to rejected proposals, while jumps indicate accepted moves. Upward jumps typically reflect improved data fit, although occasional downward moves may occur due to the exploratory nature of the underlying MCMC routine.



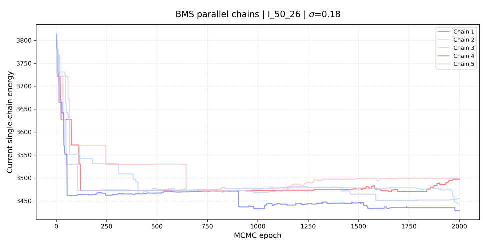
(a) I_12_2 (CL)



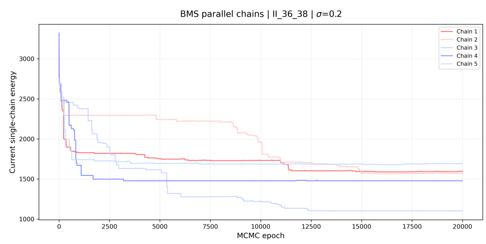
(b) I_12_11 (FCE)



(c) I_24_6 (EHFO)

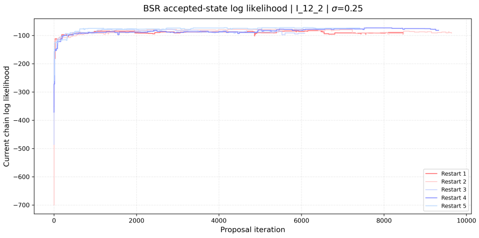


(d) I_50_26 (HOQN)

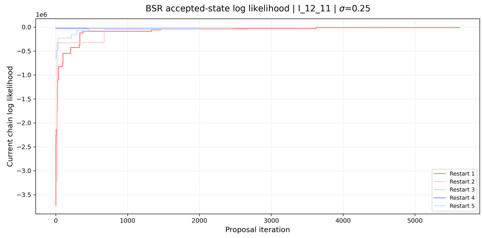


(e) II_36_38 (FMMM)

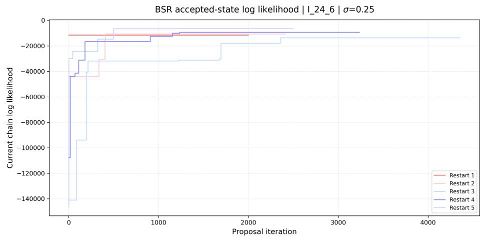
Figure 10: Trace plots for BMS across the Feynman equations in I_12_2 (CL)–II_36_38 (FMMM).



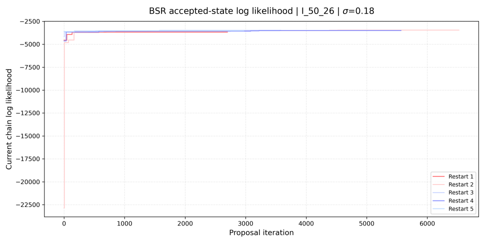
(a) I_12_2 (CL)



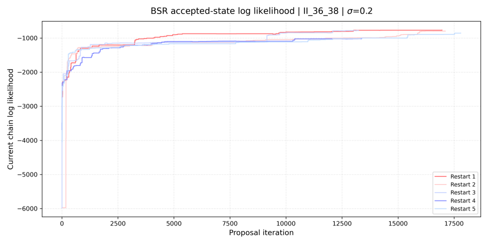
(b) I_12_11 (FCE)



(c) I_24_6 (EHFO)



(d) I_50_26 (HOQN)



(e) II_36_38 (FMMM)

Figure 11: Trace plots for `BSR` across the Feynman equations in I_12_2 (CL)–II_36_38 (FMMM).

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